HSPICE IN VLSI EXPERT

HSPICE PROGRAMMING AND TOOL USED

*RC step response circuit - case1

.OPTION WDF=1

R1 IN OUT 1k

C1 OUT 0 1u

V1 IN 0 pulse(0 5 0 .0001n .0001n 50m 100m)

.probe v(*) I(*)

.print v(*) I(*)

.tran .01m 50m

.end

[b23n_sarif_727@prac3 hspice_work]$ ls

assignment_hspice1 experiment3 experiment6 sxcmd.log.1 test_.lis test__wdf.sw1 VD.st0 VD_wdf.sw2

cc_work experiment4 rc_step_response sxcmd.log.2 test_.st0 test__wdf.sw2 VD_wdf.sw0

experiment2 experiment5 sxcmd.log sxcmd.log.3 test__wdf.sw0 VD.sp VD_wdf.sw1

```
*simple circuit

.OPTION WDF =1
.TEMP 25 125 -40
R1 1 2 4
R2 2 3 5
R3 3 0 1
V1 1 0 5
.DC V1 0 5 1
.print v(*) i(*)
.probe v(*) i(*)
.END
```

[b23n_sarif_727@prac3 hspice_work]$ hspice -i VD.sp

[b23n_sarif_727@prac3 hspice_work]$ which wv

/usr/synopsys/wv/W-2024.09-1/bin/wv

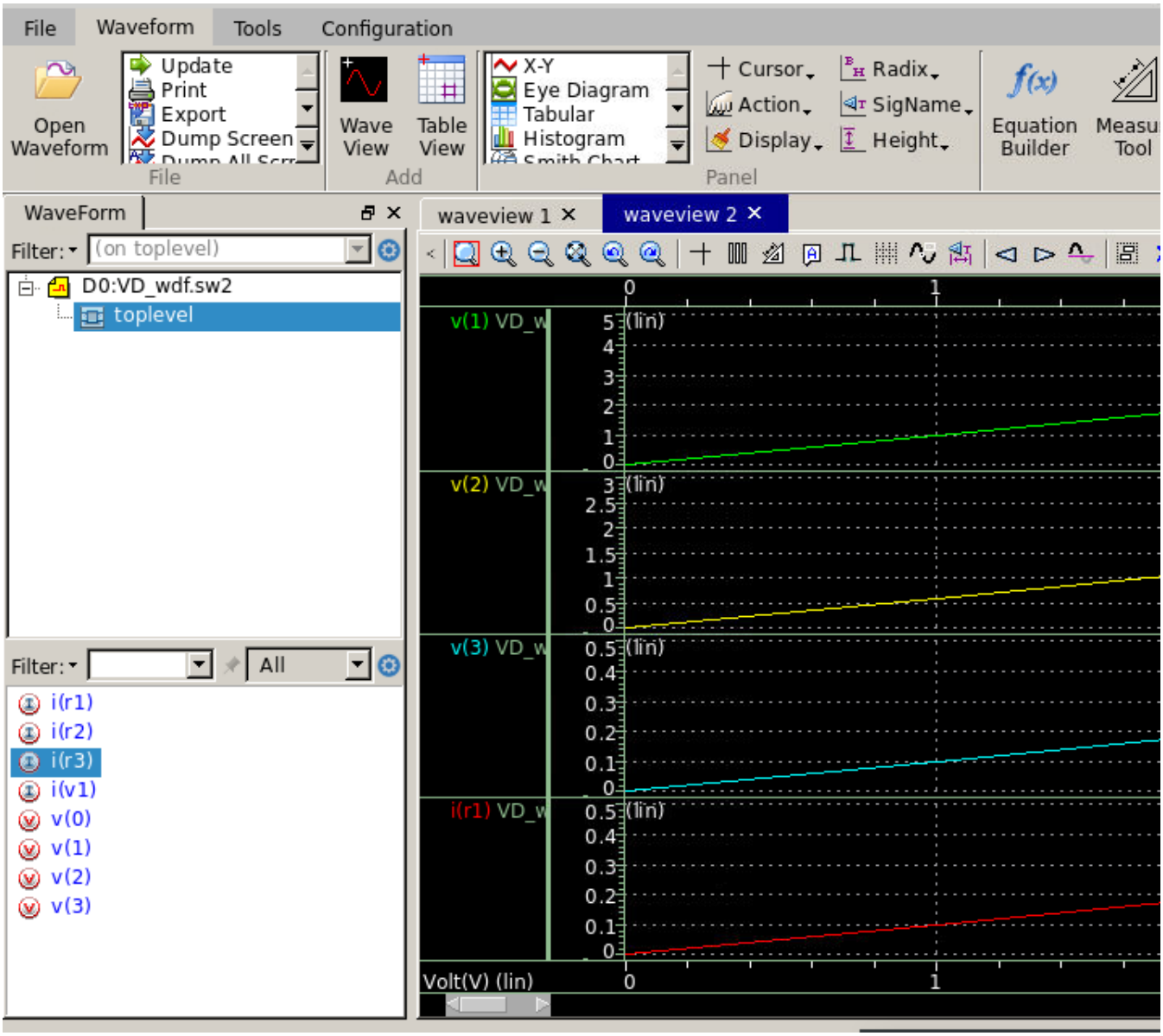
>info: ***** hspice job concluded

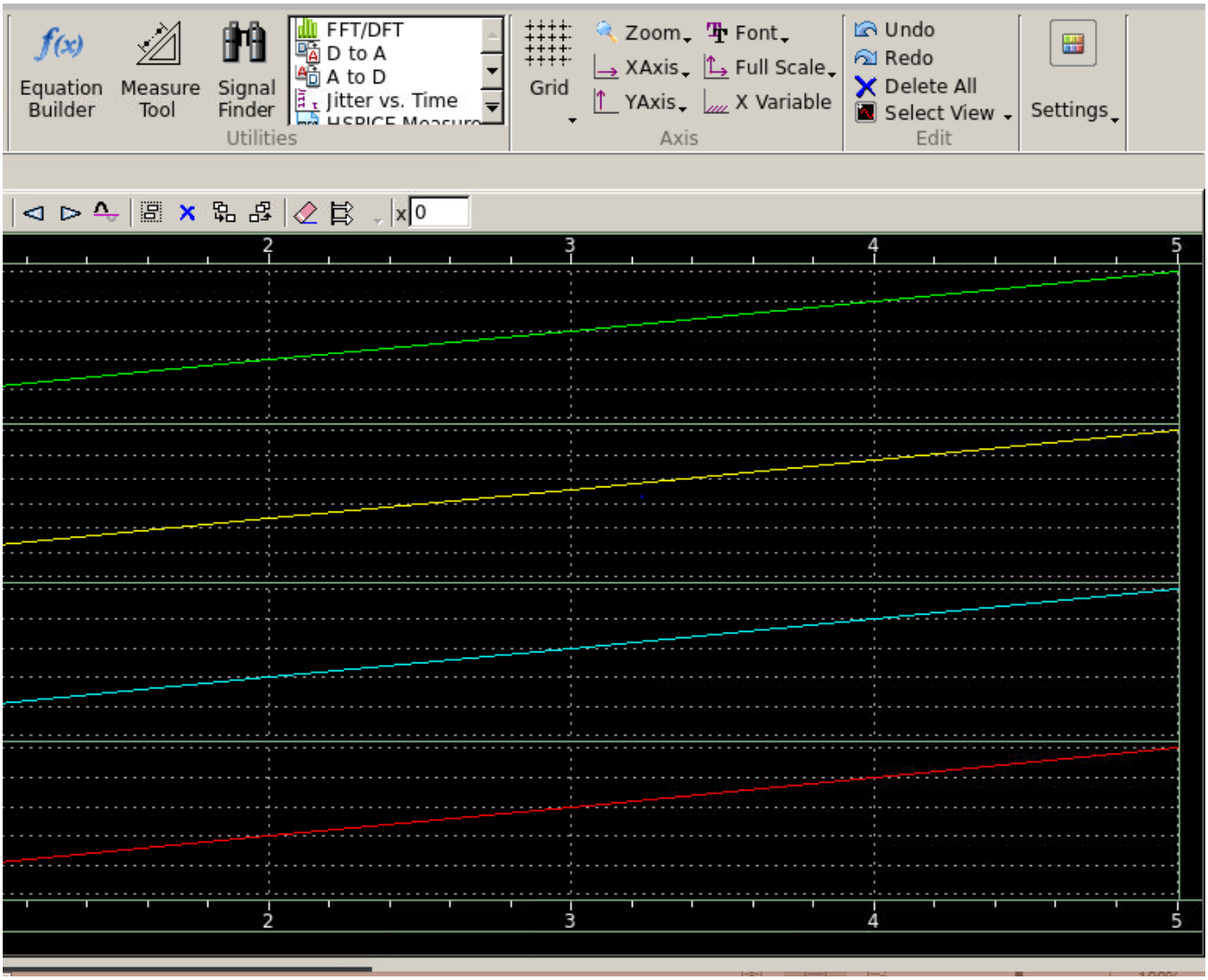
job total runtime 1.95 seconds

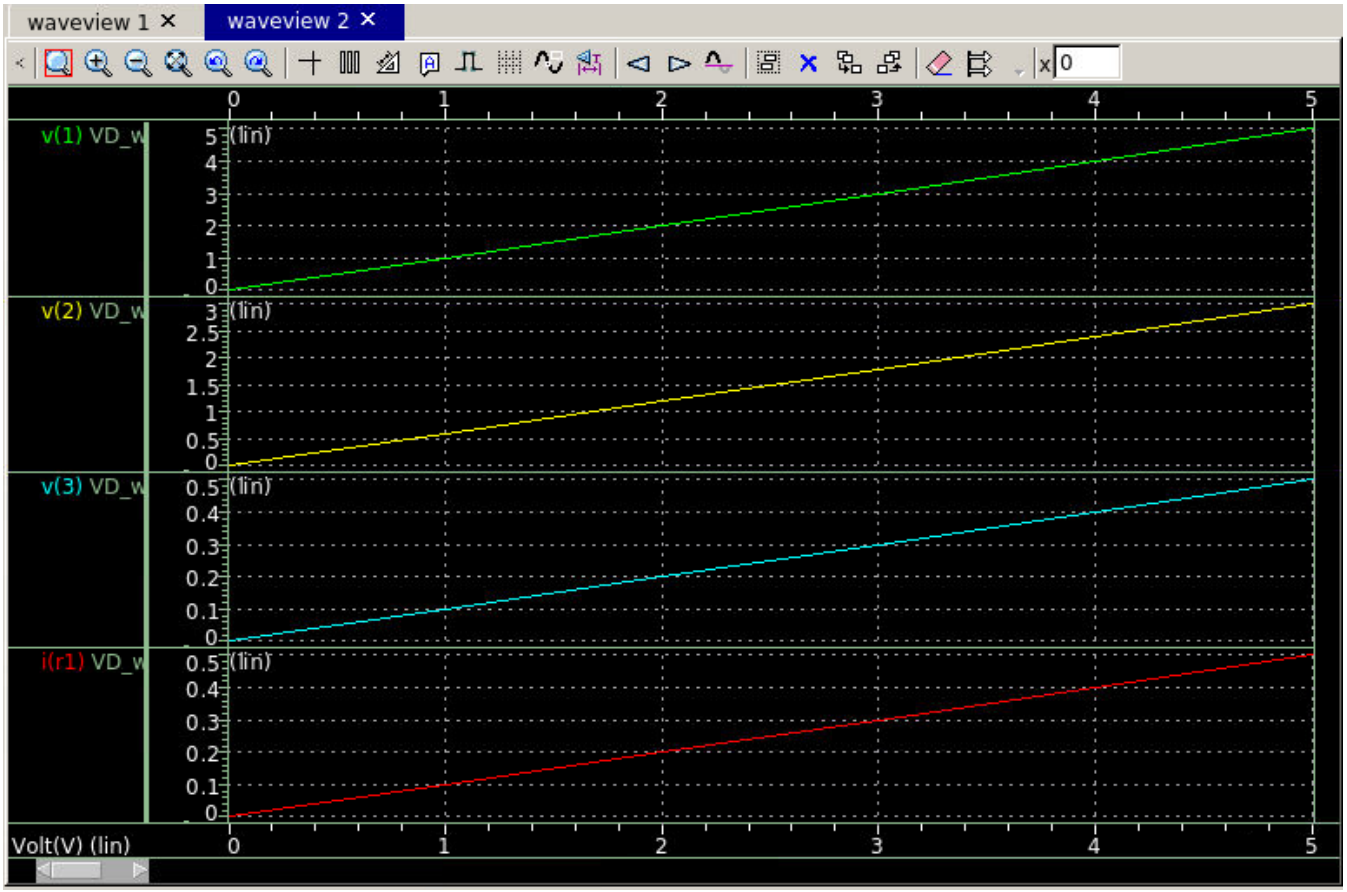
For the waveform analysis command:

[b23n_sarif_727@prac3 hspice_work]$ wv VD_wdf.sw2 &

[1] 519162


File
Waveform
Tools
Configuration
Open Waveform
Update
Print
Export
Dump Screen
File
Wave View
Table View
Add
X-Y
Eye Diagram
Tabular
Histogram
Panel
Cursor
Action
Display
Radix
SigName
Height
Equation Builder
Measu Tool
WaveForm
Filter:
(on toplevel)
D0:VD_wdf.sw2
toplevel
Filter:
All
i(r1)
i(r2)
i(r3)
i(v1)
v(0)
v(1)
v(2)
v(3)
waveview 1
waveview 2
v(1) VD_w
v(2) VD_w
v(3) VD_w
i(r1) VD_w
(lin)
Volt(V) (lin)


Equation Builder
Measure Tool
Signal Finder
FFT/DFT
D to A
A to D
Jitter vs. Time
Utilities
Grid
Zoom
Font
XAxis
Full Scale
YAxis
X Variable
Axis
Undo
Redo
Delete All
Select View
Edit
Settings
x 0
2
3
4
5



[b23n_sarif_727@prac3 hspice_work]$ cd rc_step_response/

[b23n_sarif_727@prac3 rc_step_response]$ ls

rc_circuit1.ic0 rc_circuit3.sp rc_circuit5.st0 rc_circuit_mesur_mod.mt0 sxcmd.log.2

rc_circuit1.sp rc_circuit3.st0 rc_circuit5_wdf.tr0 rc_circuit_mesur_mod.sp sxcmd.log.3

rc_circuit1.st0 rc_circuit3_wdf.tr0 rc_circuit.ic0 rc_circuit_mesur_mod.st0 sx_crash.log

rc_circuit1_wdf.tr0 rc_circuit4.ic0 rc_circuit_measuretplh.ic0 rc_circuit_mesur_mod_wdf.tr0 sx_crash.sx

rc_circuit2.ic0 rc_circuit4.sp rc_circuit_measuretplh.mt0 rc_circuit.sp testing.ic0

rc_circuit2.sp rc_circuit4.st0 rc_circuit_measuretplh.sp rc_circuit.st0 testing.lis

rc_circuit2.st0 rc_circuit4_wdf.tr0 rc_circuit_measuretplh.st0 rc_circuit_wdf.tr0 testing.mt0

rc_circuit2_wdf.tr0 rc_circuit5.ic0 rc_circuit_measuretplh_wdf.tr0 sxcmd.log testing.st0

rc_circuit3.ic0 rc_circuit5.sp rc_circuit_mesur_mod.ic0 sxcmd.log.1 testing_wdf.tr0

[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit1.sp

*RC step response circuit - case2

.OPTION WDF=1

R1 IN OUT 1k

C1 OUT 0 1u

V1 IN 0 pulse(0 5 0 .0001n .0001n 1m 2m)

.probe v(*) I(*)

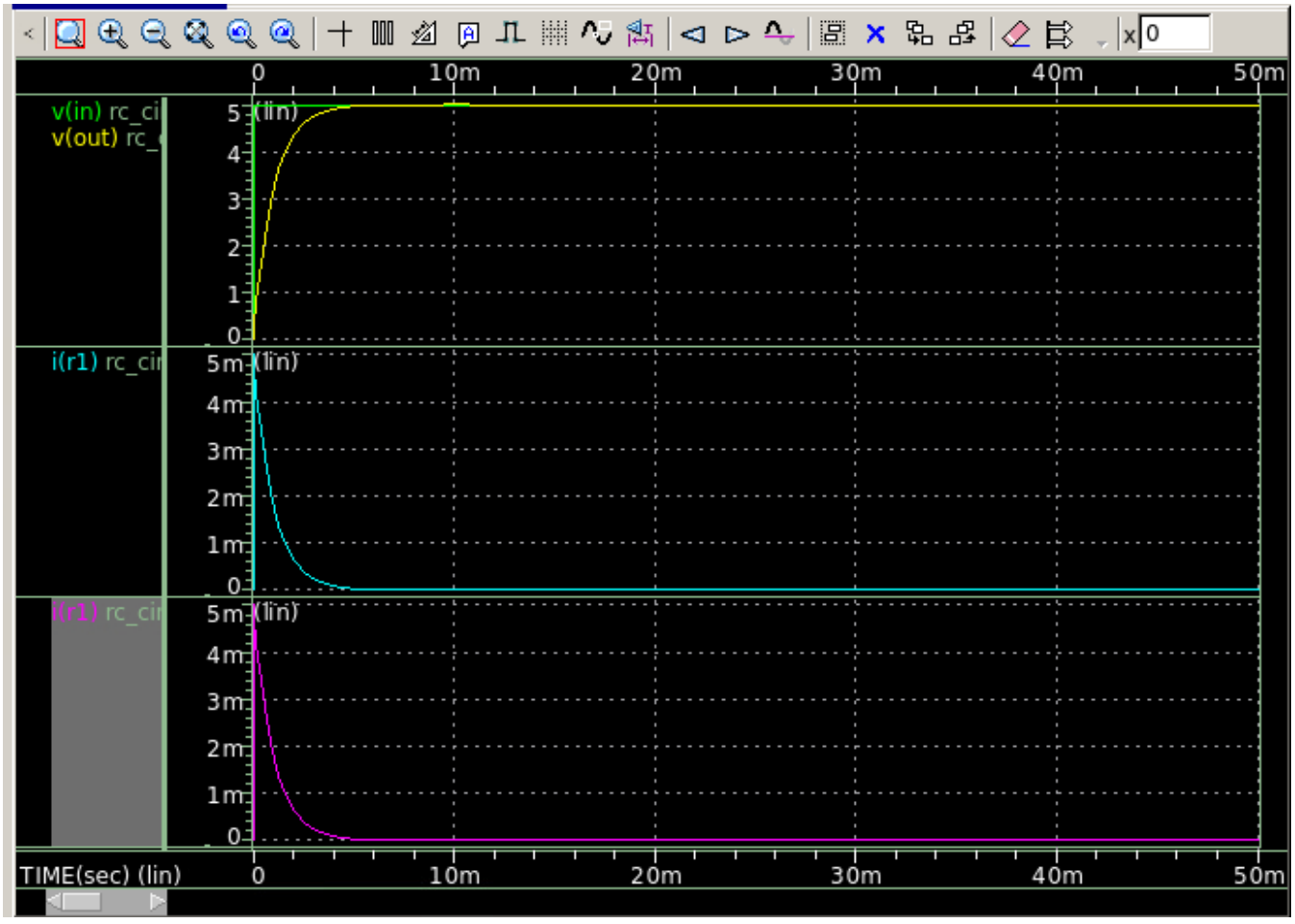
.print v(*) I(*)

.tran .01m 4m

.end

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit_wdf.tr0 &

[1] 540924



*RC step response circuit - case3

.OPTION WDF=1

R1 IN OUT 1k

C1 OUT 0 1u

V1 IN 0 pulse(0 5 0 .0001n .0001n 5m 10m)

.probe v(*) I(*)

.print v(*) I(*)

.tran .01m 20m

.end

[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit2.sp

```
*RC step response  circuit - case3

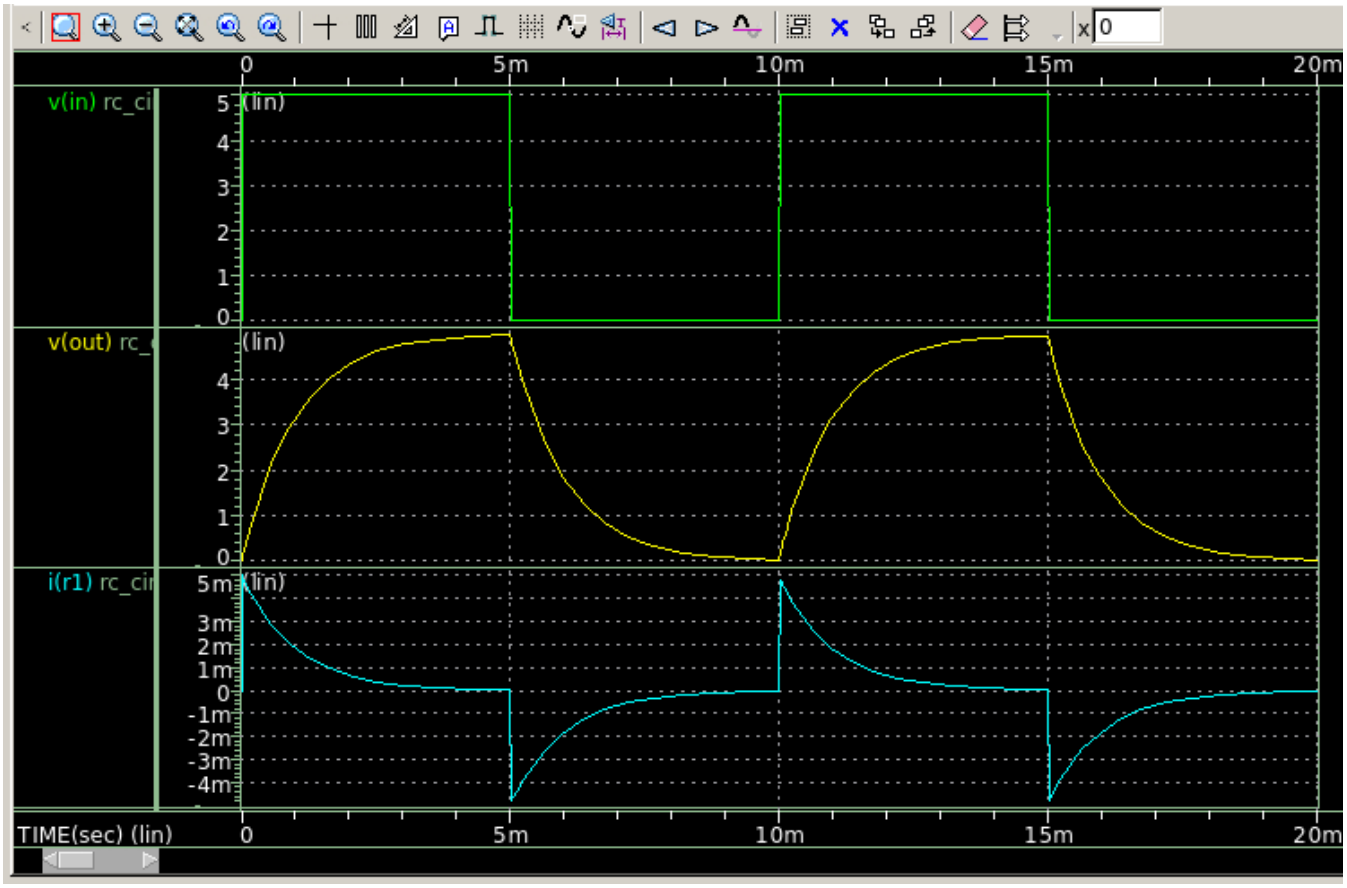

.OPTION WDF=1

R1 IN OUT 1k
C1 OUT 0 1u
V1 IN 0 pulse(0 5 0 .0001n .0001n 5m 10m)

.probe v(*) I(*)

.print v(*) I(*)
.tran .01m 20m
.end
```

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit2_wdf.tr0 &



[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit3.sp

```
*RC step response  circuit - case4


.OPTION WDF=1

R1 IN OUT 1k
C1 OUT 0 10p
V1 IN 0 pulse(0 5 0 .0001n .0001n 50m 100m)

.probe v(*) I(*)

.print v(*) I(*)
.tran .01m 50m
.end
```

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit3_wdf.tr0 &




```
*RC step response  circuit - case1


.OPTION WDF=1

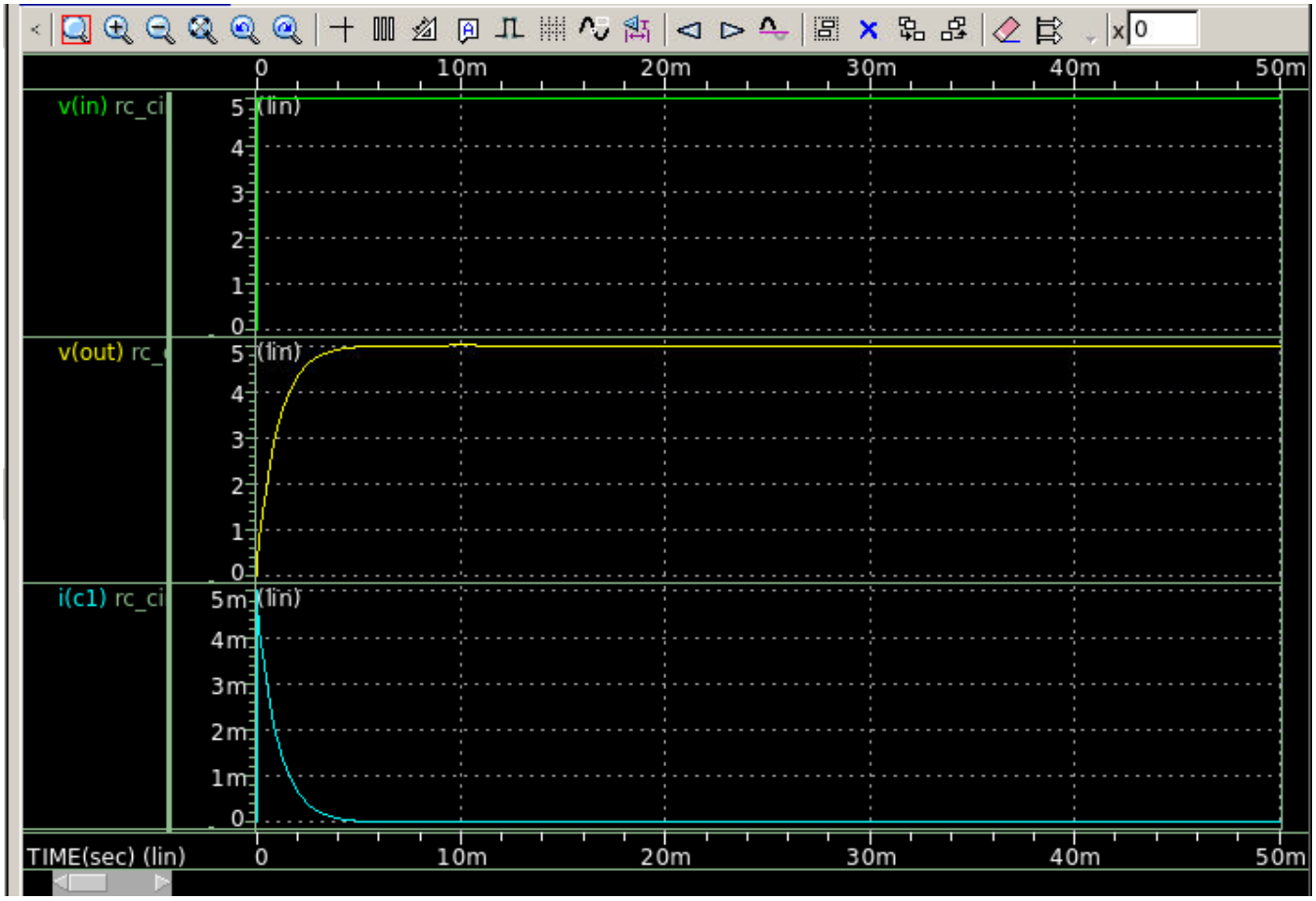
R1 IN OUT 1k
C1 OUT 0 1u
V1 IN 0 pulse(0 5 0 .0001n .0001n 50m 100m)

.probe v(*) I(*)

.print v(*) I(*)
.tran .01m 50m
.end
```

[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit.sp

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit_wdf.tr0 &



[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit4.sp

```
*RC step response  circuit - case5

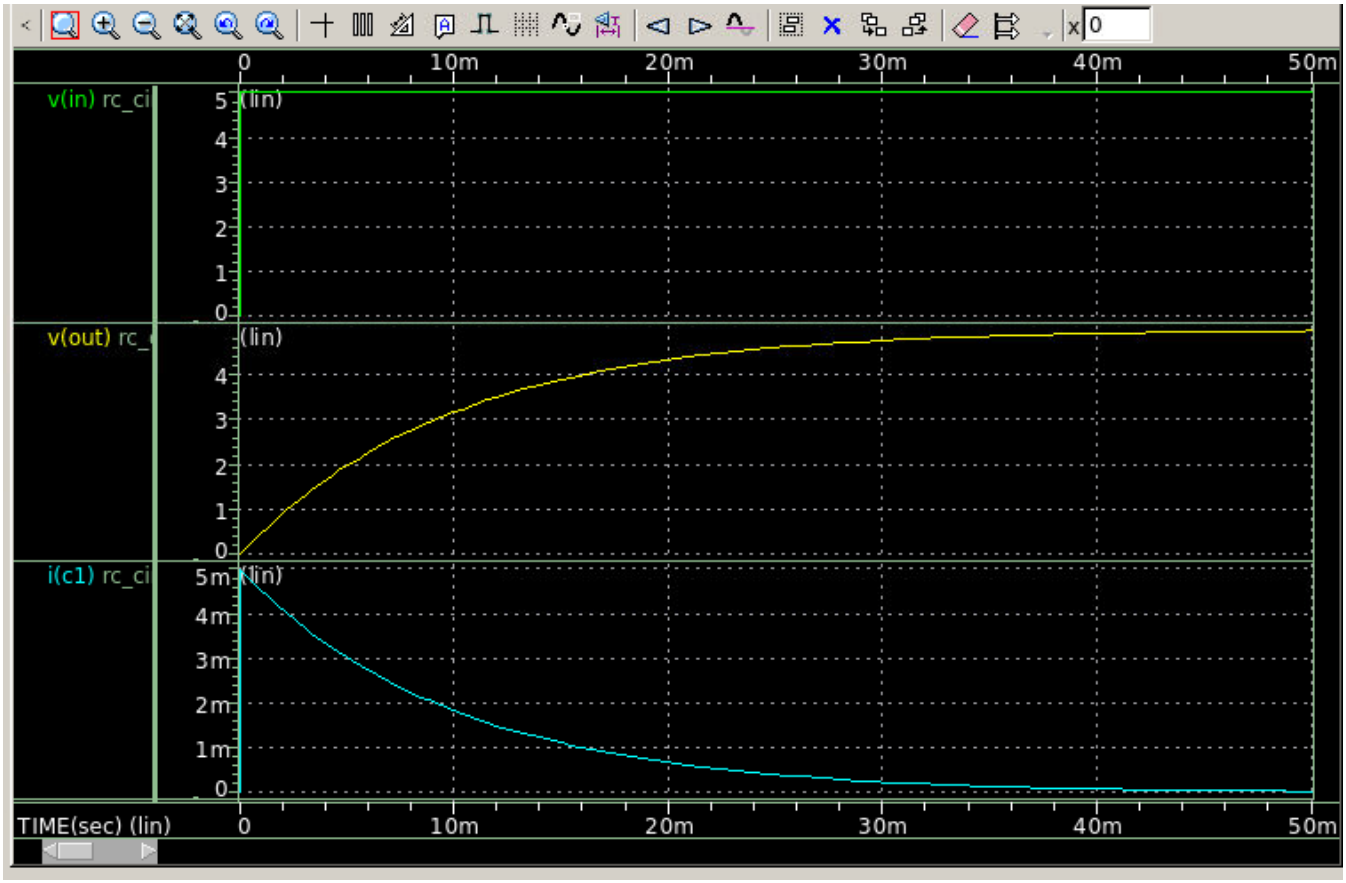

.OPTION WDF=1

R1 IN OUT 1k
C1 OUT 0 10u
V1 IN 0 pulse(0 5 0 .0001n .0001n 50m 100m)

.probe v(*) I(*)

.print v(*) I(*)
.tran .01m 50m
.end
```

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit4_wdf.tr0 &



[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit5.sp

```
*RC step response  circuit - case5

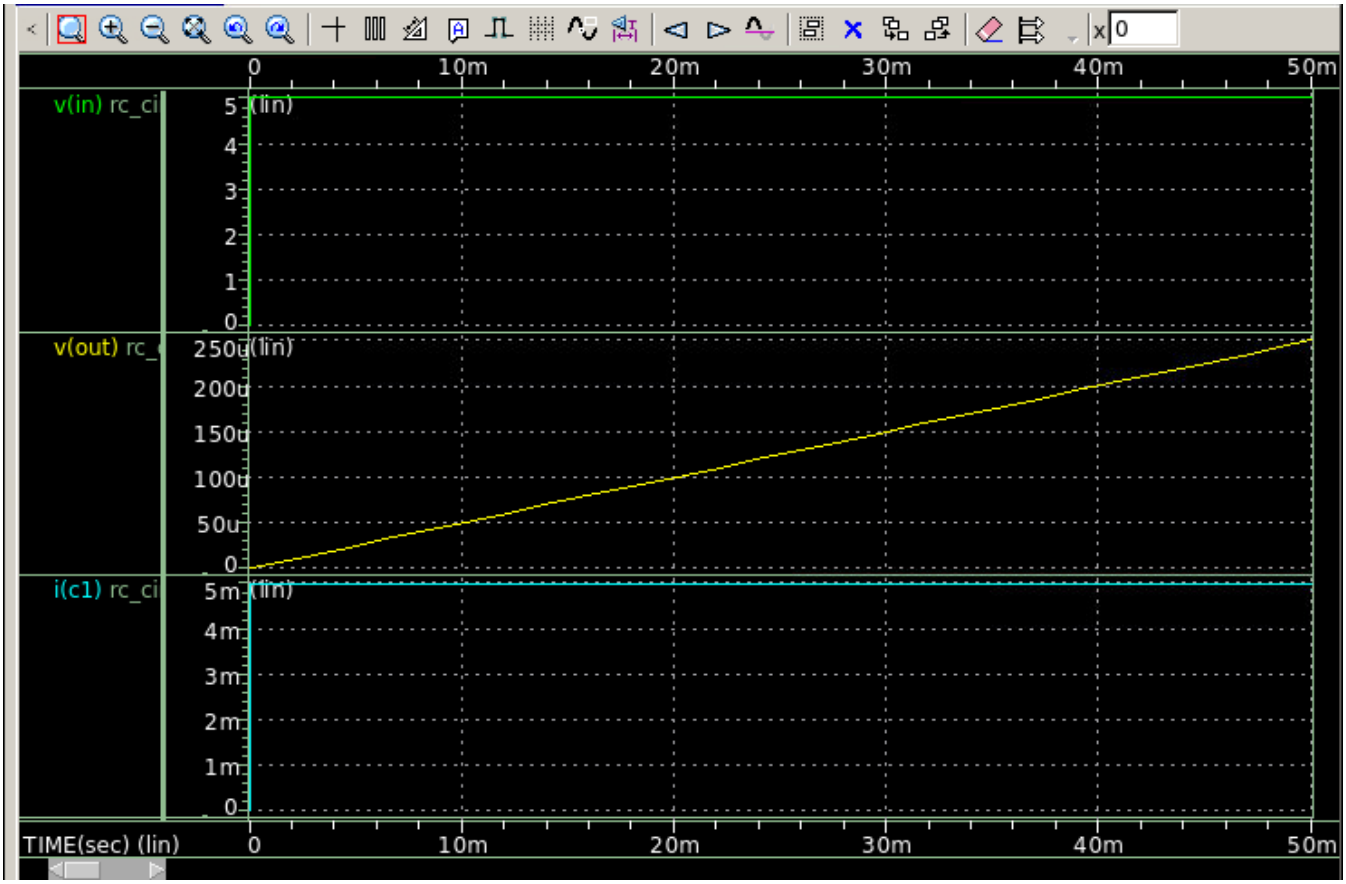

.OPTION WDF=1

R1 IN OUT 1k
C1 OUT 0 1
V1 IN 0 pulse(0 5 0 .0001n .0001n 50m 100m)

.probe v(*) I(*)

.print v(*) I(*)
.tran .01m 50m
.end
```

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit5_wdf.tr0 &



[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit_measuretplh.sp

```
*RC step response  circuit - case_measure tplh and tphl


.OPTION WDF=1
.TEMP 25
.param load =1u

R1 IN OUT 1k
C1 OUT 0 load
V1 IN 0 pulse(0 5 0 .0001n .0001n 2m 4m)

.TRAN .01m 50m sweep load 2u 8u 2u
.probe v(*) i(*)
.MEASURE TRAN TPLH TRIG V(IN) VAL=2.5 RISE=3
+ TARG V(OUT) VAL=2.5 RISE=3
.MEASURE TRAN TPHL TRIG V(IN) VAL=2.5 FALL=3  TARG V(OUT) VAL=2.5 FALL=3

.end
```

[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit_measuretplh.mt0

```
$DATA1 SOURCE='PrimeSim HSPICE' VERSION='V-2023.12-SP2 linux64' PARAM_COUNT=1
.TITLE '*rc step response  circuit - case_measure tplh and tphl'
 load              tplh              tphl              temper
                   alter#
  2.000e-06         7.925e-04         7.823e-04         25.0000
                  1
  4.000e-06         4.978e-03         4.785e-03         25.0000
                  1
  6.000e-06         9.194e-03         8.674e-03         25.0000
                  1
  8.000e-06         1.724e-02         1.668e-02         25.0000
                  1
```

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit_measuretplh_wdf.tr0 &



[b23n_sarif_727@prac3 rc_step_response]$ vi rc_circuit_mesur_mod.sp

[b23n_sarif_727@prac3 rc_step_response]$ wv rc_circuit_mesur_mod_wdf.tr0 &

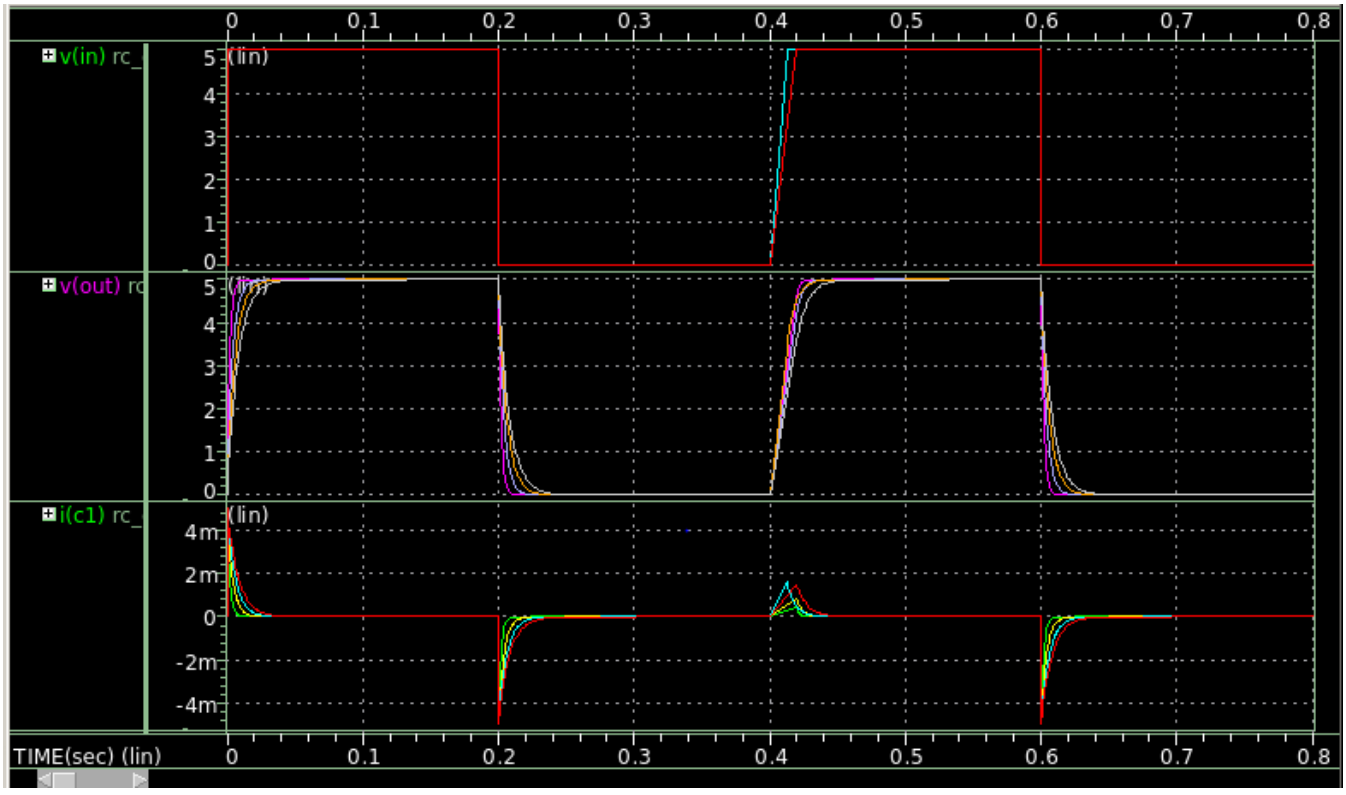
```
*RC step response  circuit - case_measure tplh and tphl


.OPTION WDF=1
.TEMP 25
.param load =1u

R1 IN OUT 1k
C1 OUT 0 load
V1 IN 0 pulse(0 5 0 .0001n .0001n 200m 400m)

.TRAN .01m 800m sweep load 2u 8u 2u
.probe v(*) i(*)
.MEASURE TRAN TPLH TRIG V(IN) VAL=2.5 RISE=3
+ TARG V(OUT) VAL=2.5 RISE=3
.MEASURE TRAN TPHL TRIG V(IN) VAL=2.5 FALL=3  TARG V(OUT) VAL=2.5 FALL=3

.end
```



Experiment-2

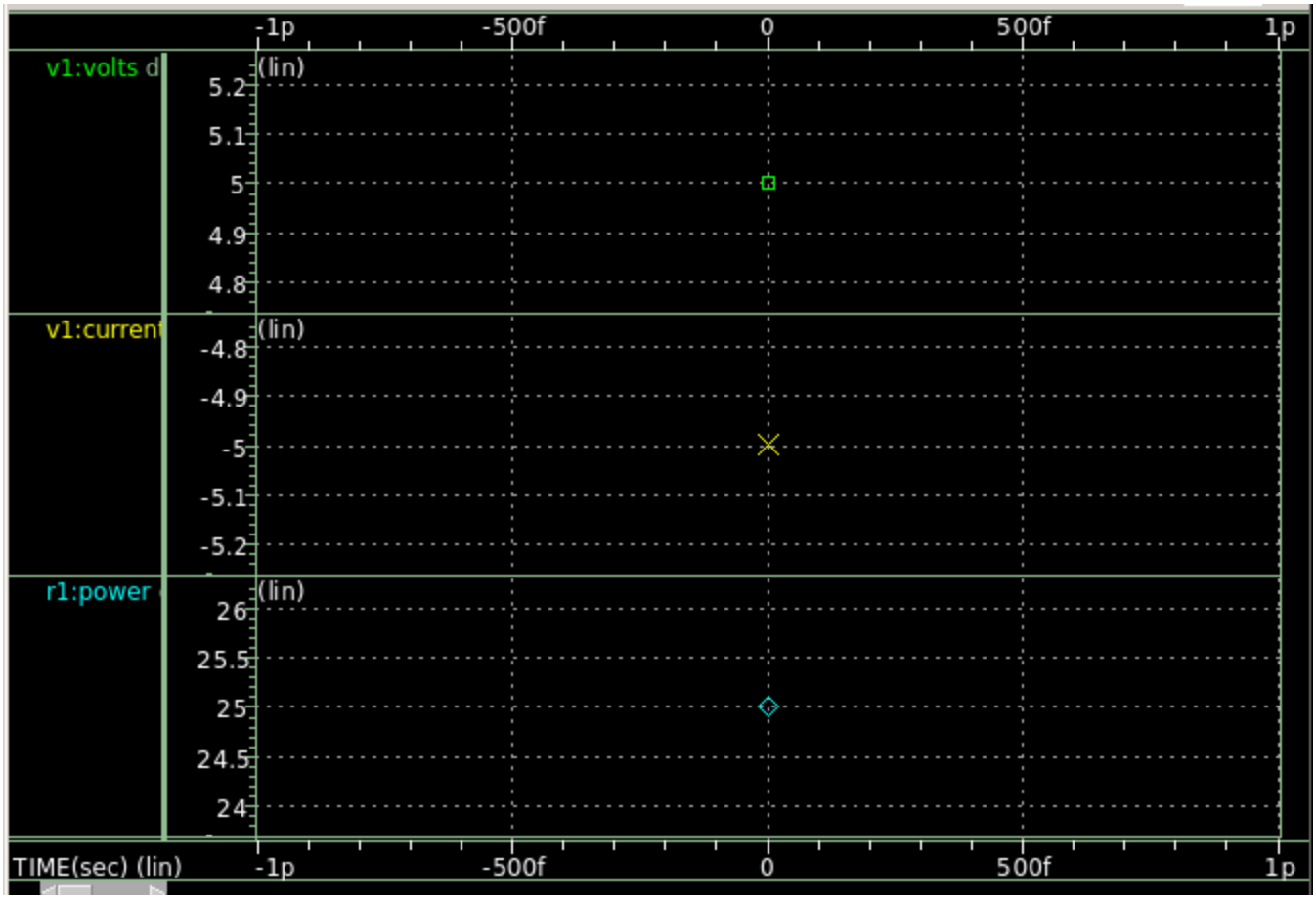
[b23n_sarif_727@prac3 experiment2]$ vi dangling.sp

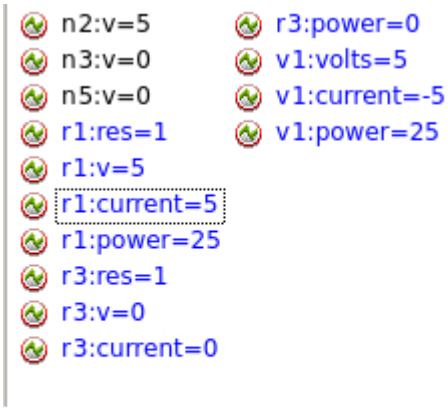
```
*dangling circuit


.OPTION WDF=1

V1 N2 0 5
R1 N2 N3 1
R2 N3 0
R3 N3 N5 1
.OP
.probe v(*) I(*)
.print v(*) I(*)
.end
```

[b23n_sarif_727@prac3 experiment2]$ wv dangling_wdf.op0 &





[b23n_sarif_727@prac3 experiment2]$ wv dangling_wdf.op0 &

Experiment3

[b23n_sarif_727@prac3 experiment3]$ vi cmos_inverter.sp

[b23n_sarif_727@prac3 experiment3]$ wv cmos_inverter_wdf.tr0 &

```
* cmos inverter of transient analyiss
.option search=/data/pdk/pdk_cc/hspice

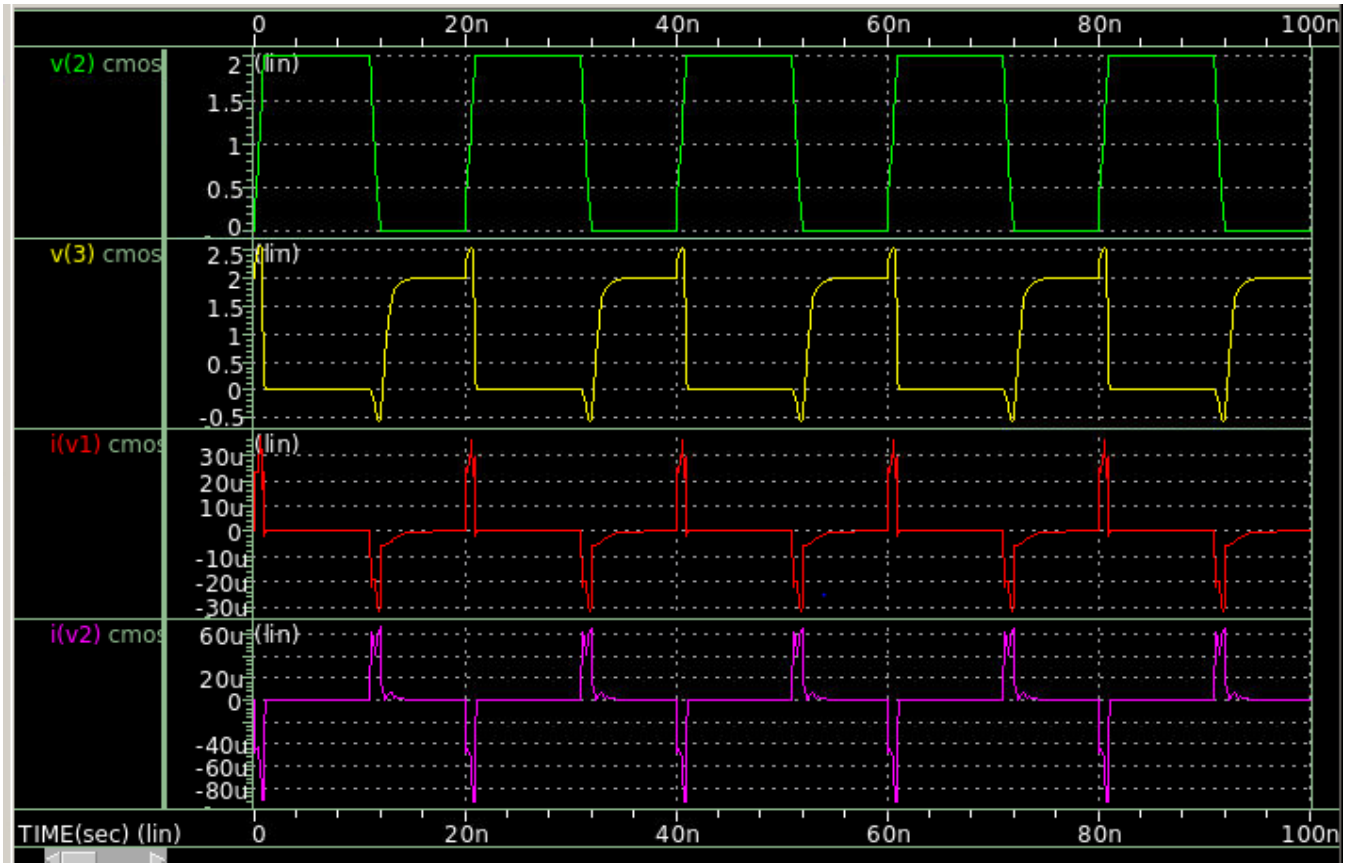

.lib reference40_models.lib TT

.option WDF=1
*.model  P25  pmos  level = 54
.TEMP 25
V1 1 0 2
V2 2 0 pulse(0 2 0 1n 1n 10n 20n)

M1 3 2 1 1 P25 W=2u L=1u
M2 3 2 0 0 N25 W=2u L=1u

.TRAN 0.01n 100n
*.IC V(3)=5
*.OP
.probe V(2) V(3)
.print TRAN V(2) V(3)

.end
```



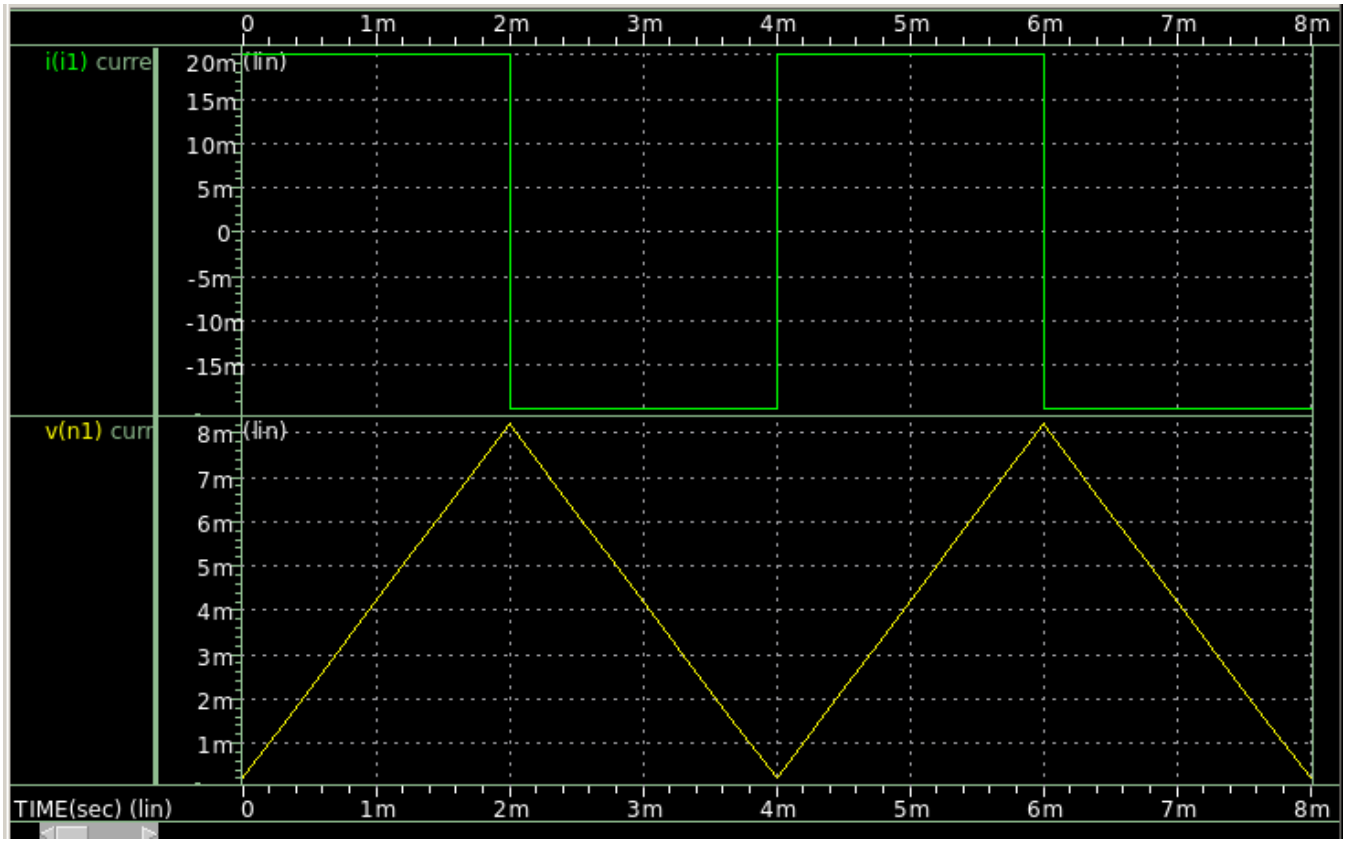

```
*input is squar waveform with current source.

.OPTION WDF=1

C1 N1 0 5m
I1 0 N1 PULSE(20m -20m 2m 1p 1p 2m 4m)
.IC V(N1)=0
.OP
.TRAN 1u 8m
.probe V(N1) I(I1)
.print TRAN V(N1) I(I1)
.end
```

[b23n_sarif_727@prac3 experiment3]$ vi current_square.sp

[b23n_sarif_727@prac3 experiment3]$ wv current_square_wdf.tr0 &




```
*input is triangular  waveform with current source.

.OPTION WDF=1

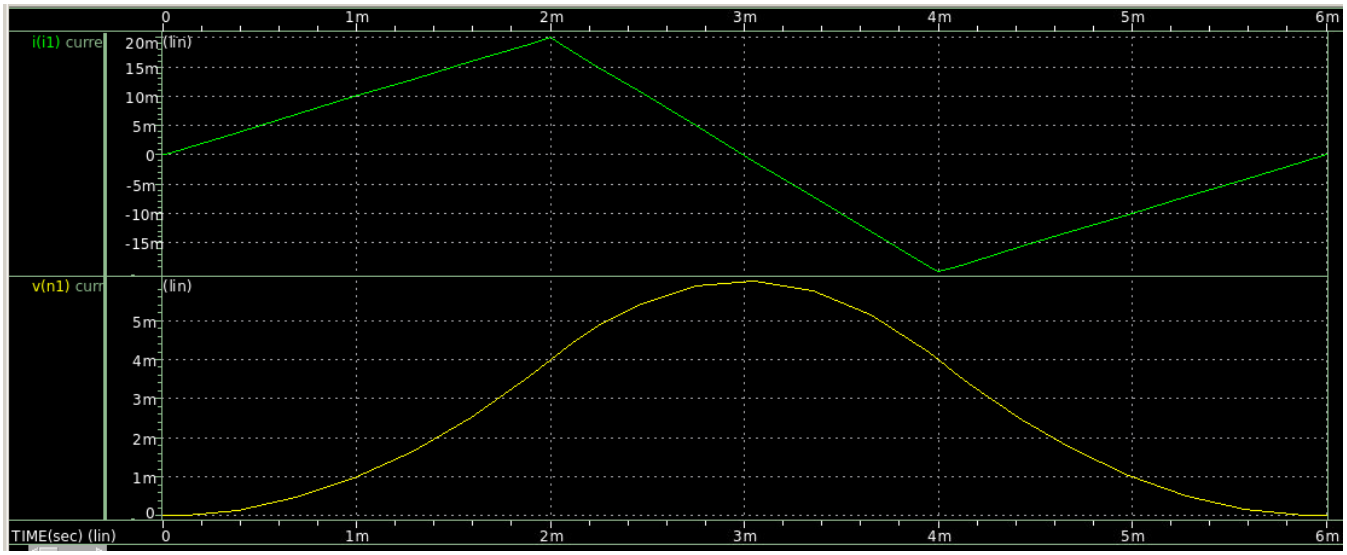
C1 N1 0 5m
I1 0 N1 PWL(0 0m 2m 20m 4m -20m 6m 0m)

.IC V(N1)=0

.TRAN 1u 6m
.probe V(N1) I(I1)
.print V(N1) I(I1)
.end
```

[b23n_sarif_727@prac3 experiment3]$ vi current_triangular.sp

[b23n_sarif_727@prac3 experiment3]$ wv current_triangular_wdf.tr0 &



[b23n_sarif_727@prac3 experiment3]$ vi mosfet.sp

[b23n_sarif_727@prac3 experiment3]$ wv mosfet_wdf.op0 &

```
* mosfetoperating point analysis
.option search=/data/pdk/pdk_cc/hspice

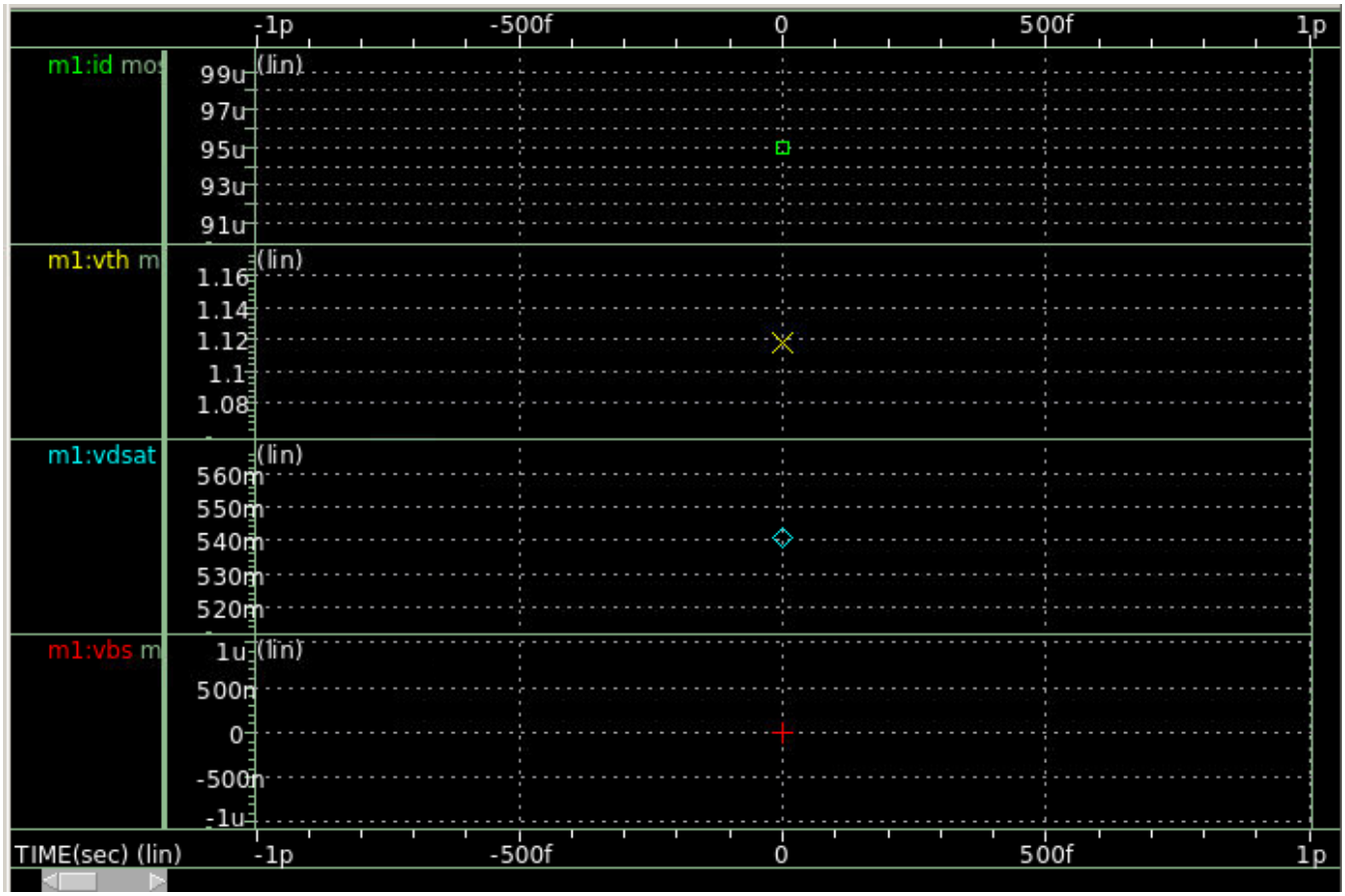

.lib reference40_models.lib TT

.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 0 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

1:v=2.5	m1:vdsat=541m	m1:cbtot=12.4f
2:v=2	m1:beta=407u	m1:cgs=16.1f
m1:region=2	m1:gam eff=1.17	m1:cgd=0.52f
m1:id=95u	m1:gm=178u	v1:volts=2
m1:ibs=6.11E-18	m1:gds=1.27u	v1:current=-891p
m1:ibd=-2.51p	m1:gmb=134u	v1:power=1.78n
m1:vgs=2	m1:vod=882m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-98.2u
m1:vbs=0	m1:cgtot=18.1f	v2:power=246u
m1:vth=1.12	m1:cstot=15.7f	



```
* mosfetoperating point analysis
.option search=/data/pdk/pdk_cc/hspice

.temp 25 125 -25
.lib reference40_models.lib TT

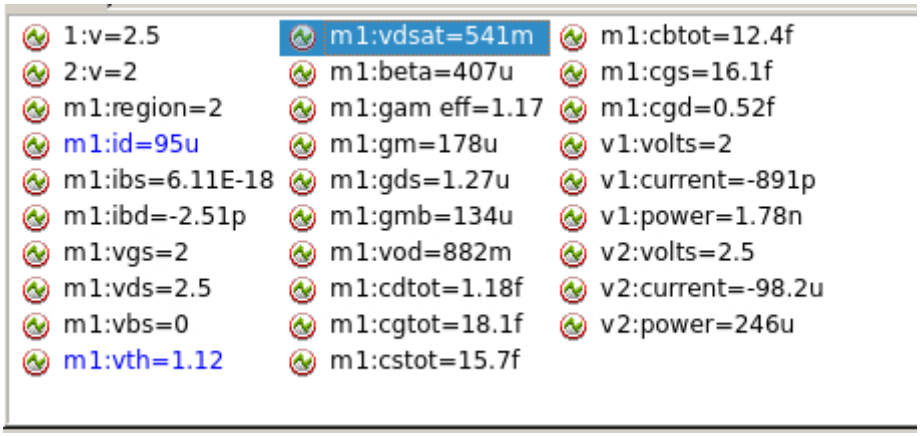
.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 0 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5

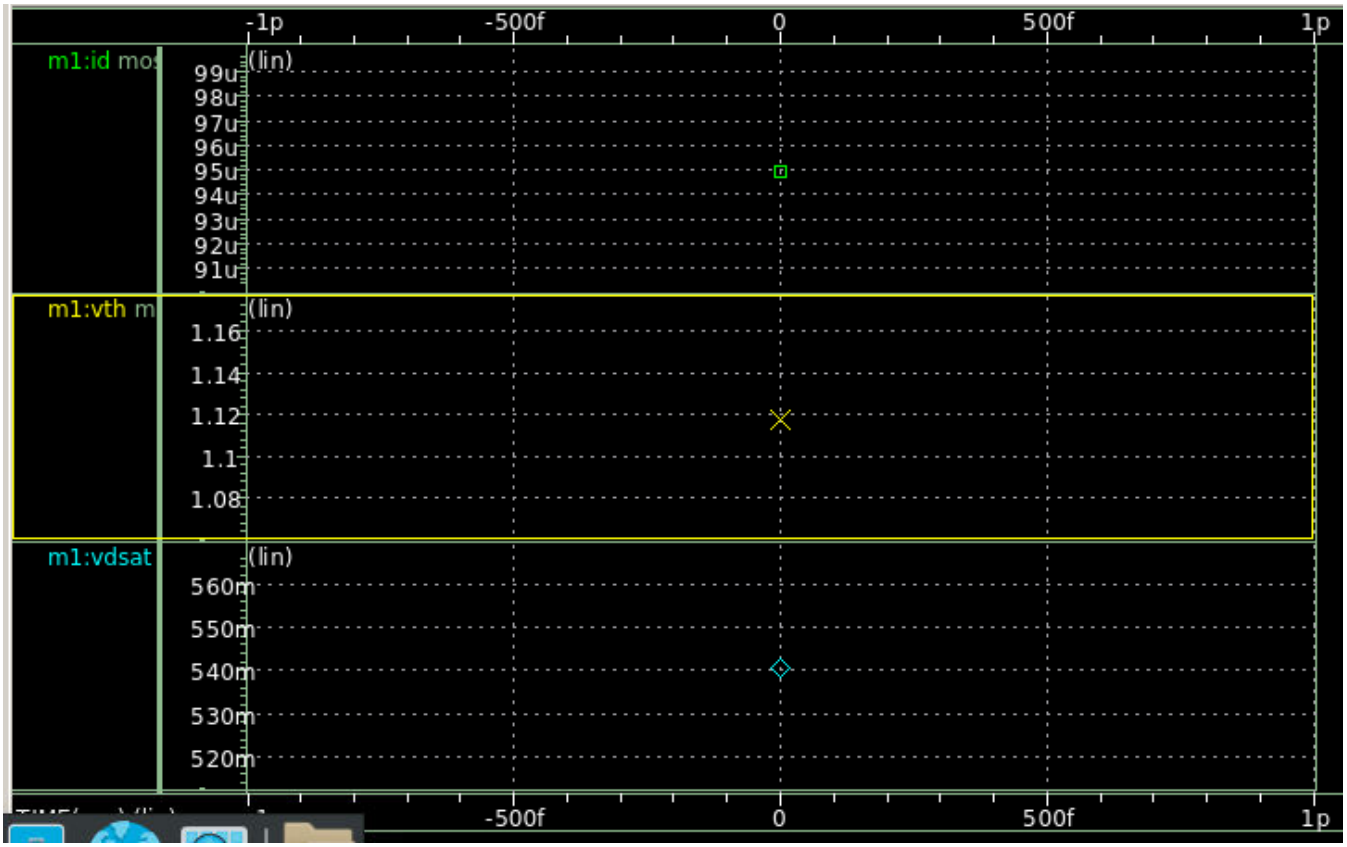
.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

[b23n_sarif_727@prac3 experiment3]$ vi mosfet_temp.sp

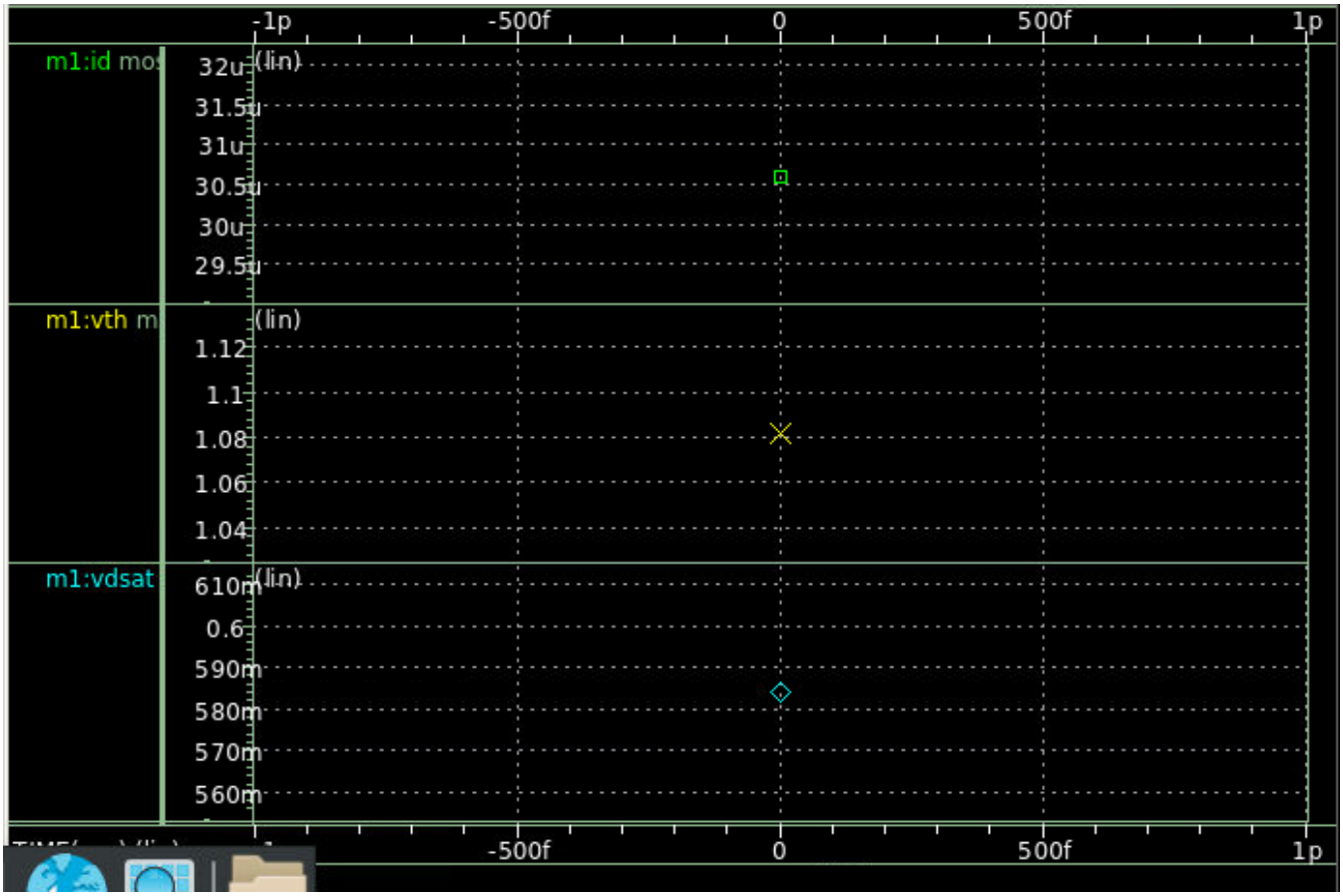
[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_wdf.op0 &





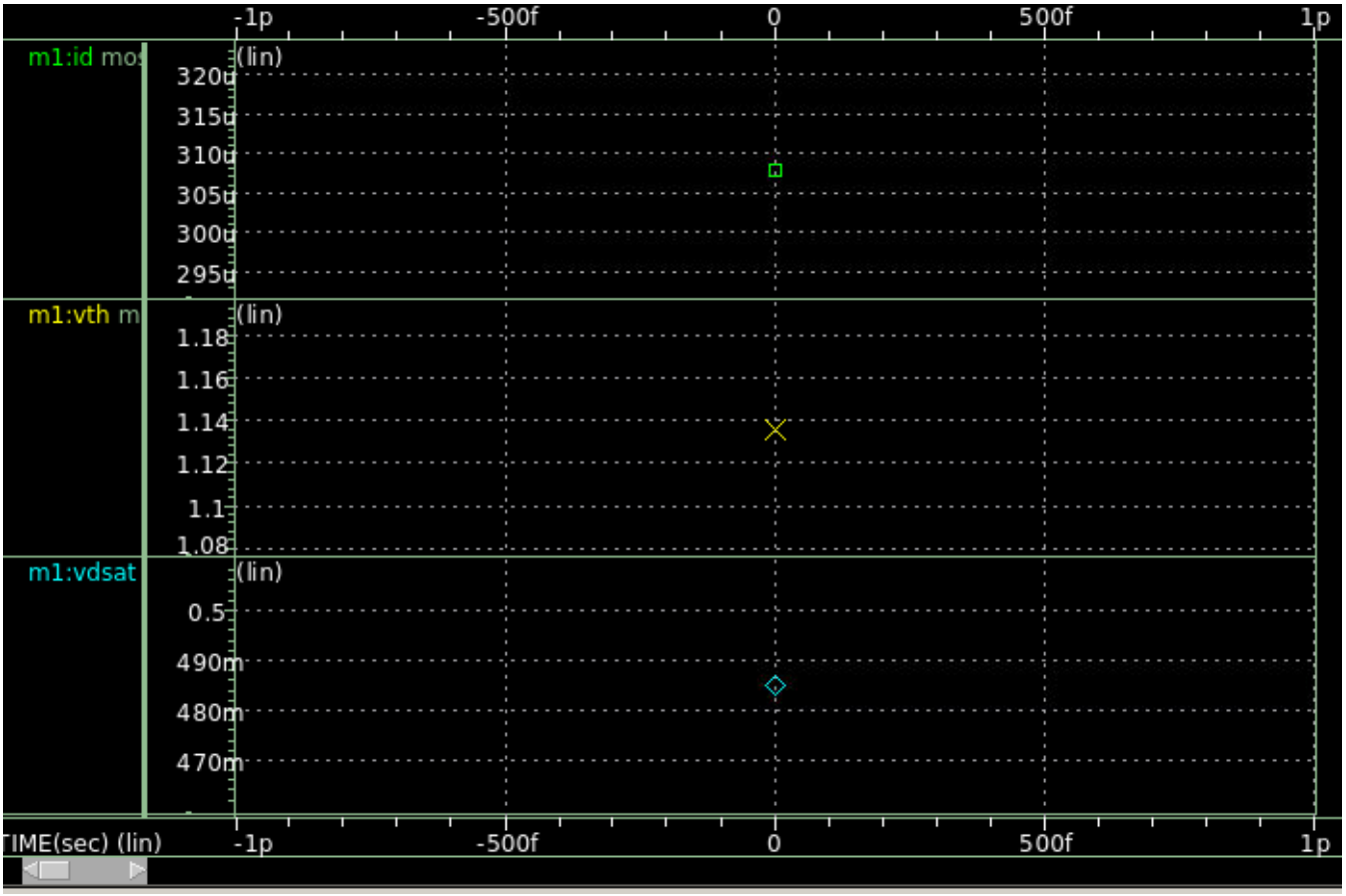
[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_wdf.op1 &

1:v=2.5	m1:vdsat=584m	m1:cbtot=12.5f
2:v=2	m1:beta=116u	m1:cgs=15.9f
m1:region=2	m1:gam eff=1.17	m1:cgd=0.52f
m1:id=30.6u	m1:gm=52.5u	v1:volts=2
m1:ibs=1.93f	m1:gds=398n	v1:current=-1.15n
m1:ibd=-2.51p	m1:gmb=44.1u	v1:power=2.31n
m1:vgs=2	m1:vod=919m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.13f	v2:current=-31.3u
m1:vbs=0	m1:cgtot=18f	v2:power=78.3u
m1:vth=1.08	m1:cstot=15.7f	



[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_wdf.op2 &

1:v=2.5	m1:vdsat=485m	m1:cbtot=12.3f
2:v=2	m1:beta=1.56m	m1:cgs=16.3f
m1:region=2	m1:gam eff=1.17	m1:cgd=0.52f
m1:id=308u	m1:gm=668u	v1:volts=2
m1:ibs=3.08E-17	m1:gds=4.27u	v1:current=-778p
m1:ibd=-2.51p	m1:gmb=350u	v1:power=1.56n
m1:vgs=2	m1:vod=864m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-324u
m1:vbs=0	m1:cgtot=18.2f	v2:power=810u
m1:vth=1.14	m1:cstot=15.7f	




```
* operating analysis of nmos at different temperature
.option search=/data/pdk/pdk_cc/hspice

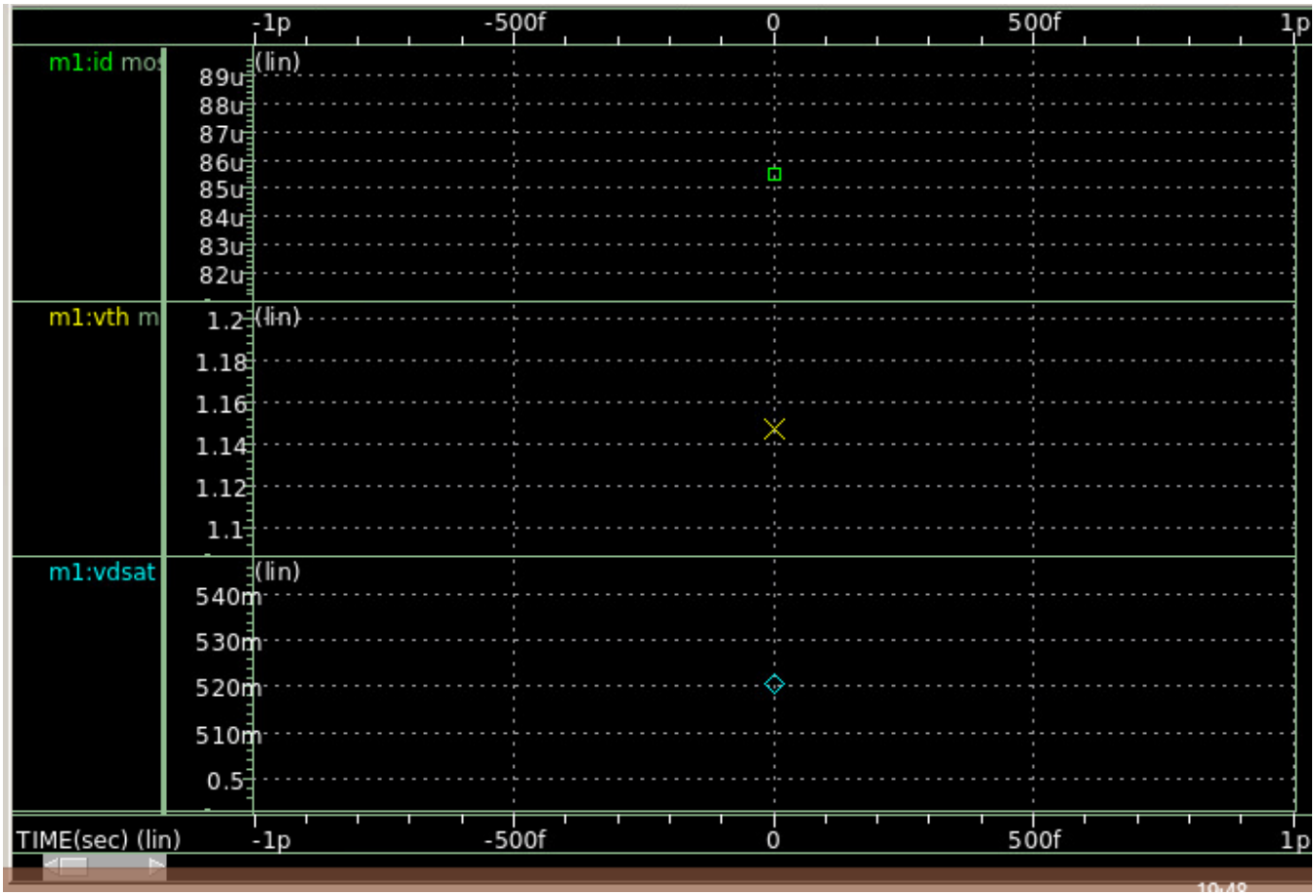
.temp 25 125 -25
.lib reference40_models.lib SS

.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 0 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

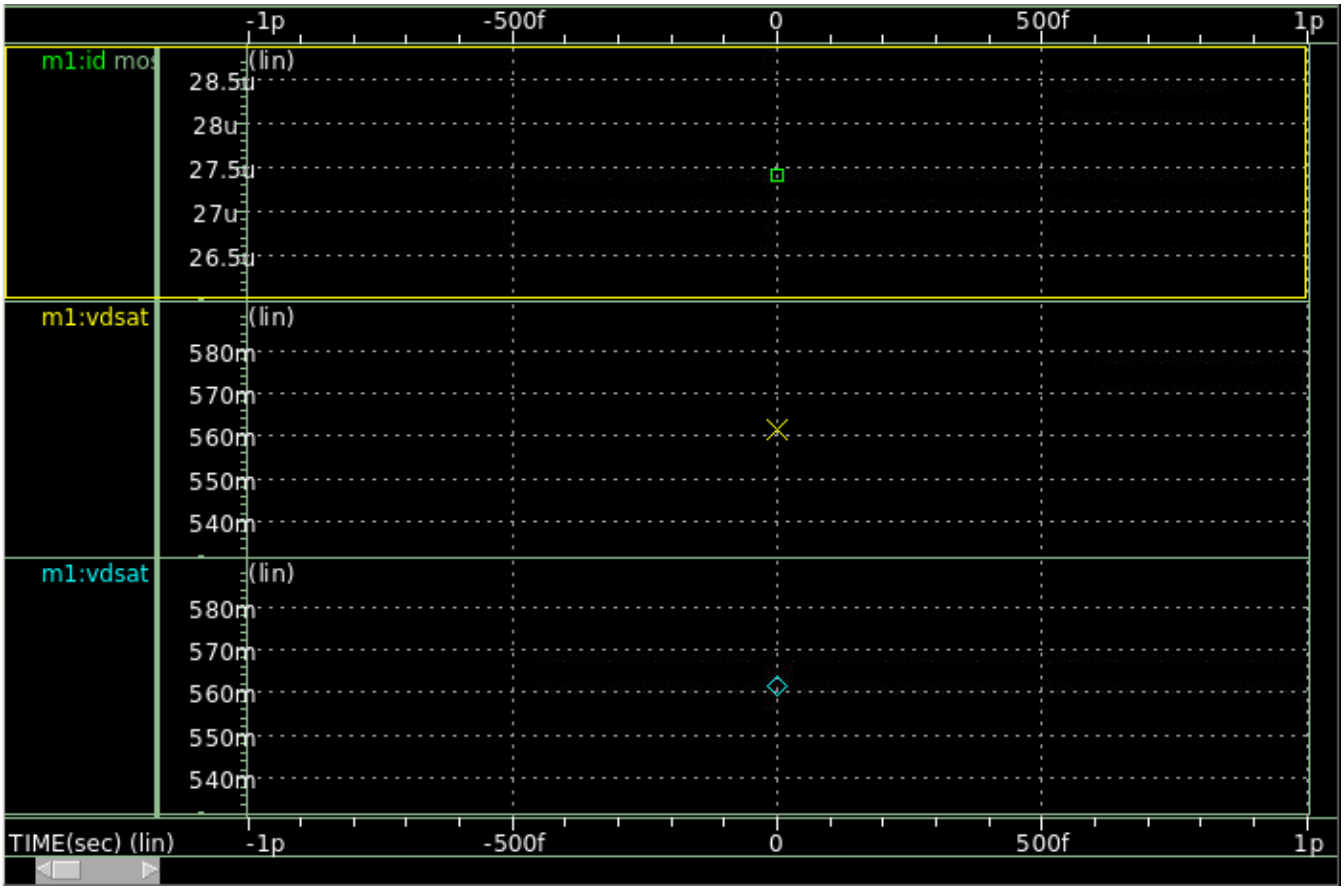
[b23n_sarif_727@prac3 experiment3]$ vi mosfet_temp_ss.sp



1:v=2.5	m1:vdsat=520m	m1:cbtot=12.7f
2:v=2	m1:beta=394u	m1:cgs=16.1f
m1:region=2	m1:gam eff=1.2	m1:cgd=0.52f
m1:id=85.5u	m1:gm=167u	v1:volts=2
m1:ibs=6.48E-18	m1:gds=1.1u	v1:current=-840p
m1:ibd=-2.51p	m1:gmb=128u	v1:power=1.68n
m1:vgs=2	m1:vod=853m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-88.9u
m1:vbs=0	m1:cgtot=18.2f	v2:power=222u
m1:vth=1.15	m1:cstot=15.8f	

[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_ss_wdf.op0 &

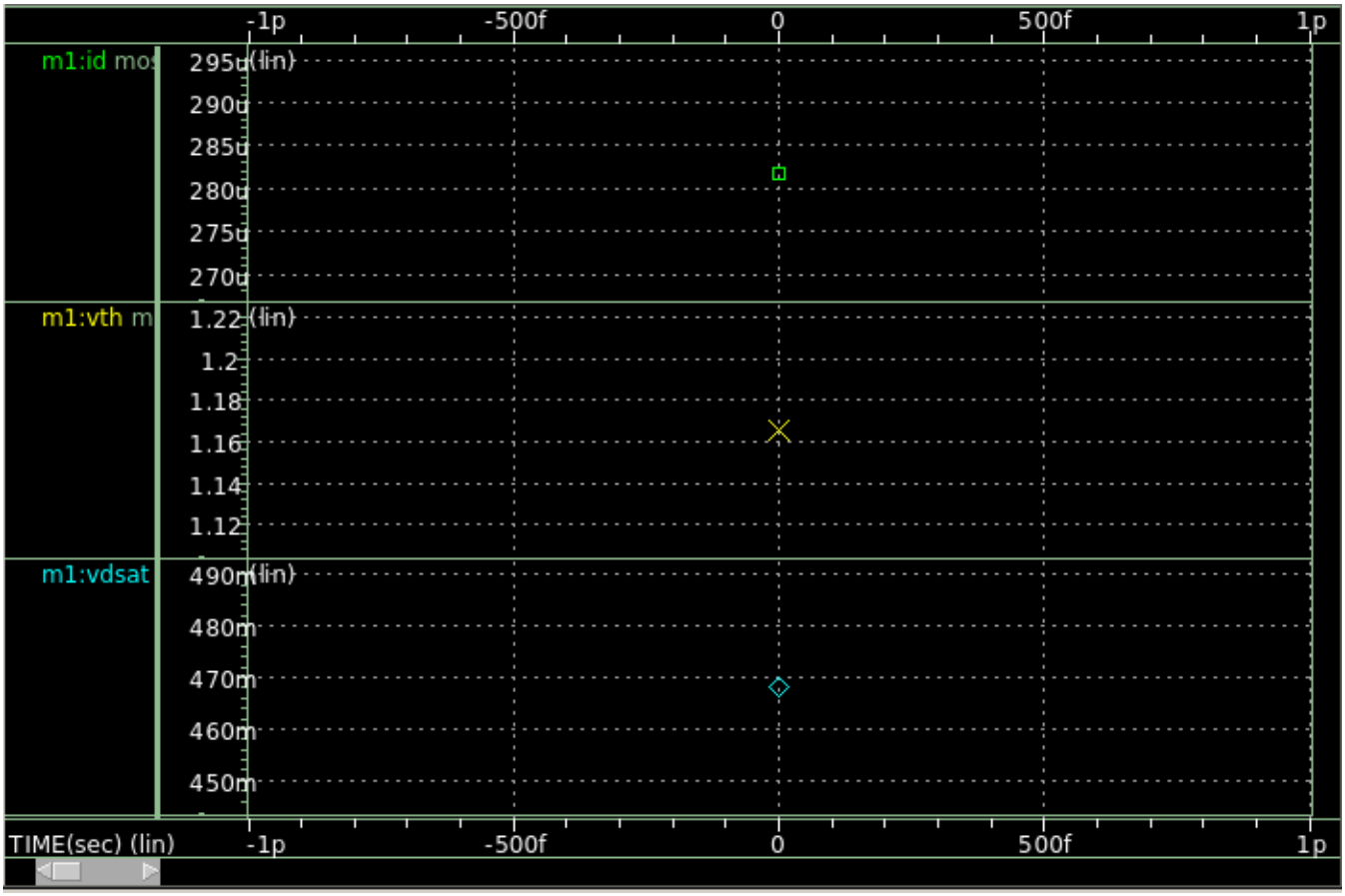
1:v=2.5	m1:vdsat=561m	m1:cbtot=12.8f
2:v=2	m1:beta=112u	m1:cgs=15.9f
m1:region=2	m1:gam eff=1.2	m1:cgd=0.52f
m1:id=27.4u	m1:gm=49.3u	v1:volts=2
m1:ibs=2.09f	m1:gds=343n	v1:current=-1.09n
m1:ibd=-2.51p	m1:gmb=41.8u	v1:power=2.17n
m1:vgs=2	m1:vod=890m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.13f	v2:current=-28.2u
m1:vbs=0	m1:cgtot=18.1f	v2:power=70.5u
m1:vth=1.11	m1:cstot=15.8f	



[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_ss_wdf.op1 &

[b23n_sarif_727@prac3 experiment3]$ wv mosfet_temp_ss_wdf.op2 &

1:v=2.5	m1:vdsat=468m	m1:cbtot=12.6f
2:v=2	m1:beta=1.53m	m1:cgs=16.3f
m1:region=2	m1:gam eff=1.2	m1:cgd=0.52f
m1:id=282u	m1:gm=635u	v1:volts=2
m1:ibs=3.21E-17	m1:gds=3.73u	v1:current=-733p
m1:ibd=-2.51p	m1:gmb=343u	v1:power=1.47n
m1:vgs=2	m1:vod=835m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-299u
m1:vbs=0	m1:cgtot=18.3f	v2:power=747u
m1:vth=1.17	m1:cstot=15.8f	



[b23n_sarif_727@prac3 experiment3]$ vi mosfet_vb_neg.sp

[b23n_sarif_727@prac3 experiment3]$ wv mosfet_vb_neg_wdf.op0

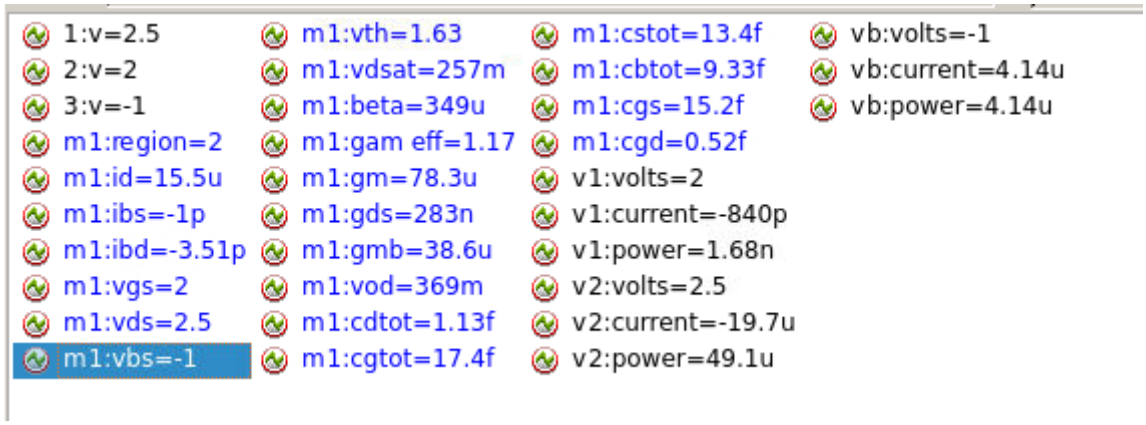
```
* mosfet_operating point analysis
.option search=/data/pdk/pdk_cc/hspice

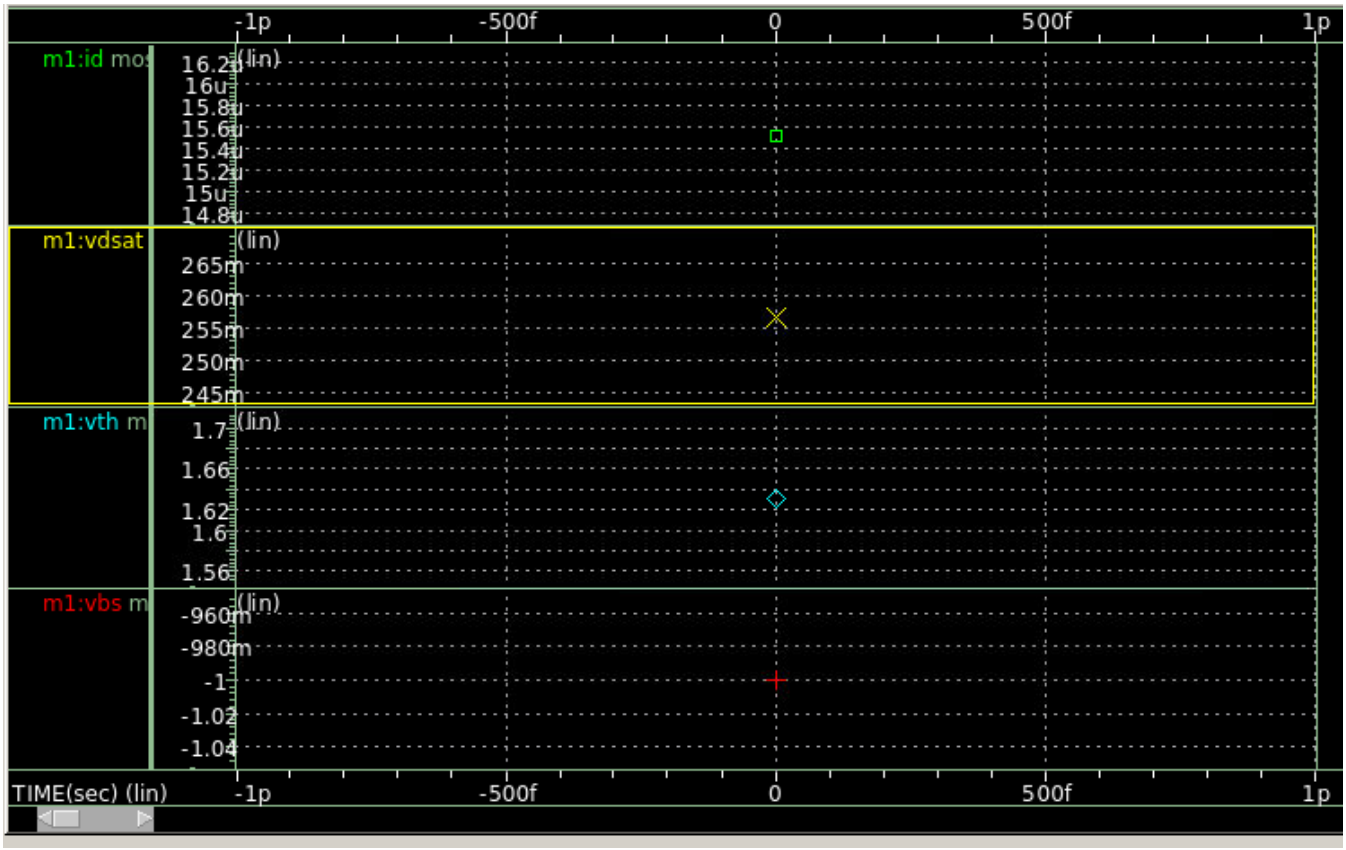

.lib reference40_models.lib TT

.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 3 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5
VB 3 0 -1

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

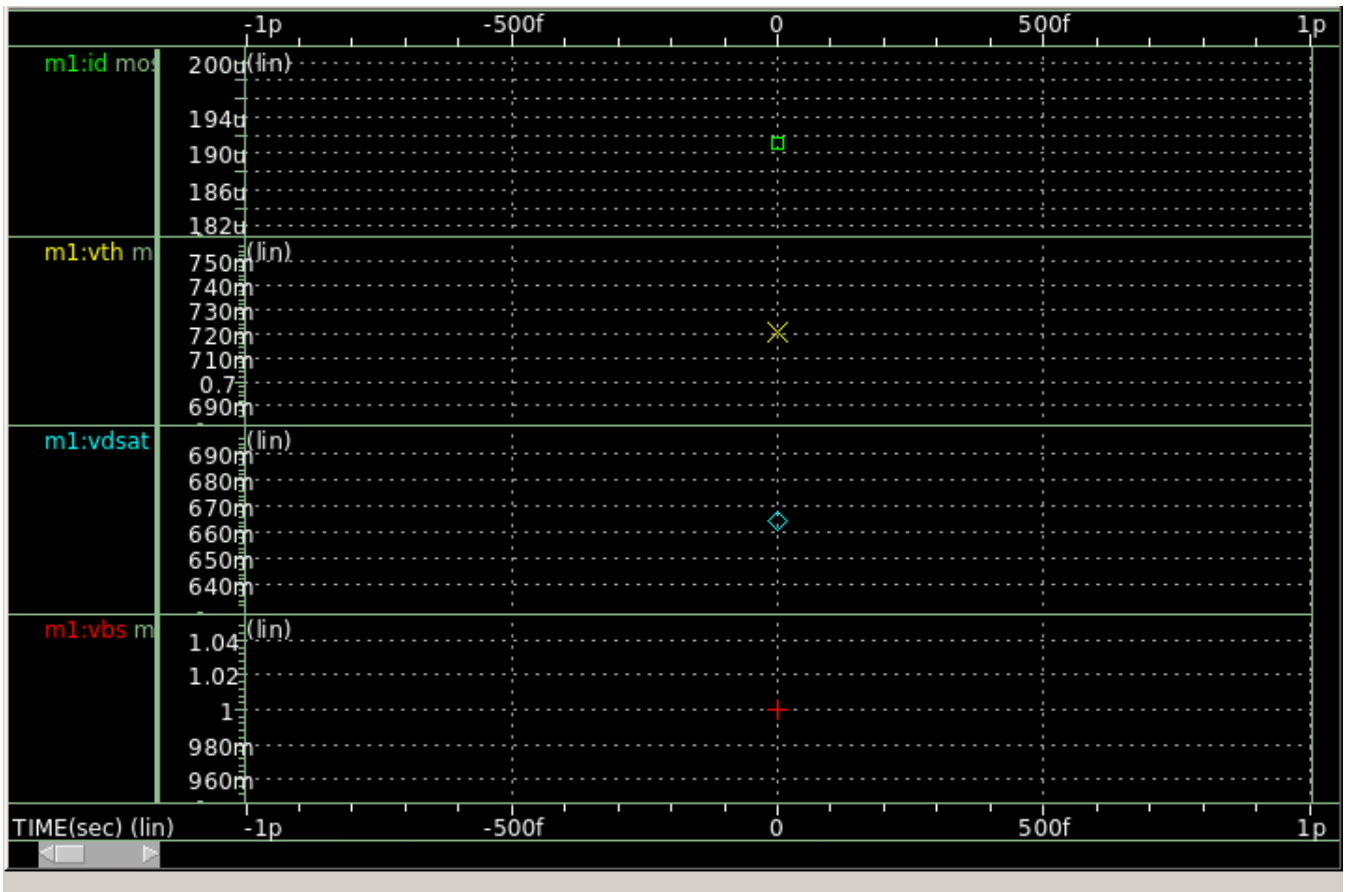




[b23n_sarif_727@prac3 experiment3]$ vi mosfet_vb_pos.sp

[b23n_sarif_727@prac3 experiment3]$ wv mosfet_vb_pos_wdf.op0 &

1:v=2.5	m1:vth=721m	m1:cstot=13.3f	vb:volts=1
2:v=2	m1:vdsat=664m	m1:cbtot=6.19f	vb:current=-19m
3:v=1	m1:beta=460u	m1:cgs=18.6f	vb:power=19m
m1:region=2	m1:gam eff=1.17	m1:cgd=0.52f	
m1:id=191u	m1:gm=227u	v1:volts=2	
m1:ibs=19m	m1:gds=2.32u	v1:current=-175p	
m1:ibd=-1.53p	m1:gmb=68.6u	v1:power=351p	
m1:vgs=2	m1:vod=1.28	v2:volts=2.5	
m1:vds=2.5	m1:cdtot=1.26f	v2:current=-193u	
m1:vbs=1	m1:cgtot=18.8f	v2:power=483u	



```
* mosfetoperating point analysis
.option search=/data/pdk/pdk_cc/hspice

.lib reference40_models.lib TT

.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 3 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5
VB 3 0 1

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

[b23n_sarif_727@prac3 experiment3]$ vi mpsfet_temp_ff.sp

[b23n_sarif_727@prac3 experiment3]$ wv mpsfet_temp_ff_wdf.op0 &

```
* operating analysis of nmos at different temperature
.option search=/data/pdk/pdk_cc/hspice

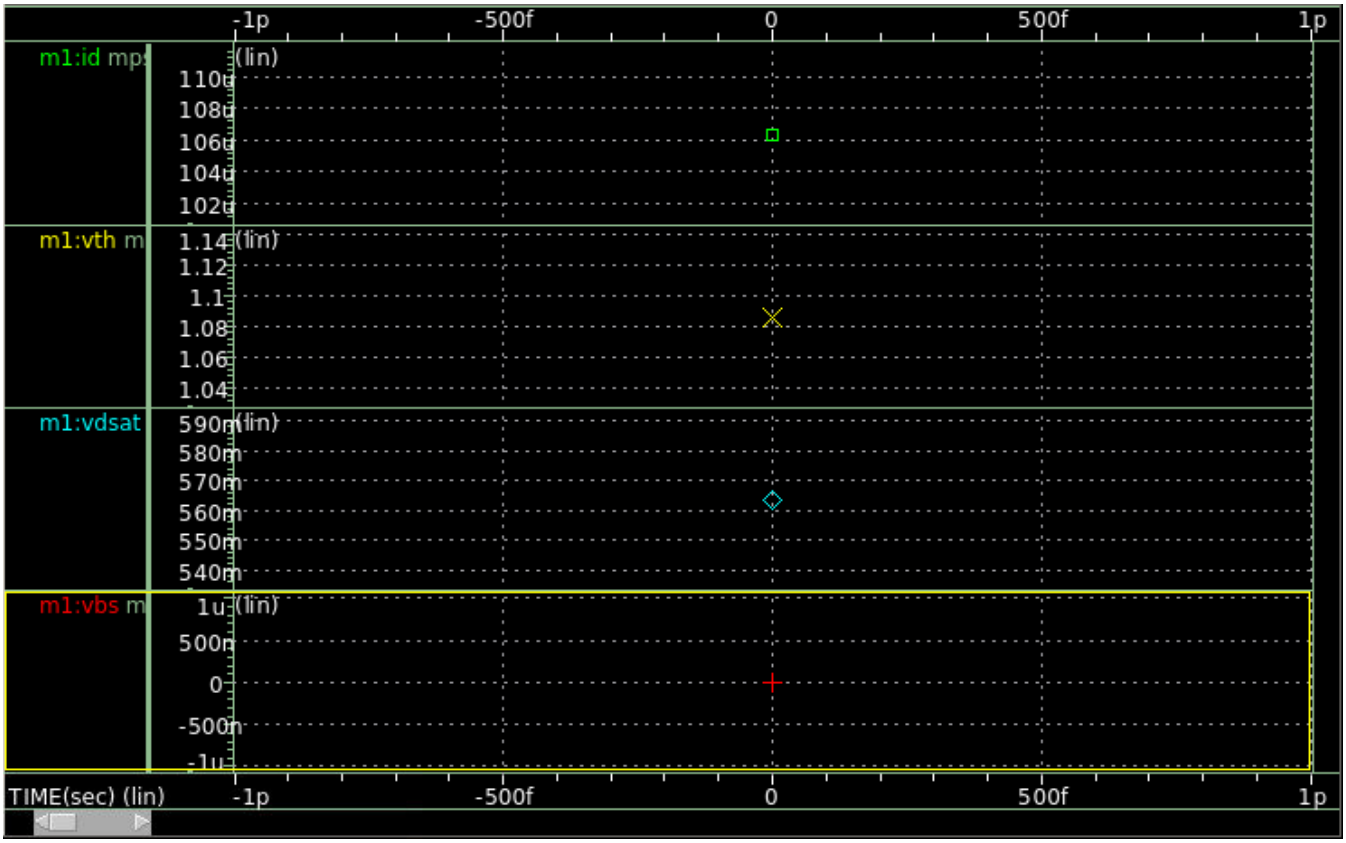

.temp 25 125 -25
.lib reference40_models.lib FF

.option WDF=1
*.model  N25  nmos  level = 54
M1 1 2 0 0 N25 W=2u L=1u
V1 2 0 2
V2 1 0 2.5

.OP
.probe V(*) I(*)
.print V(*) I(*)

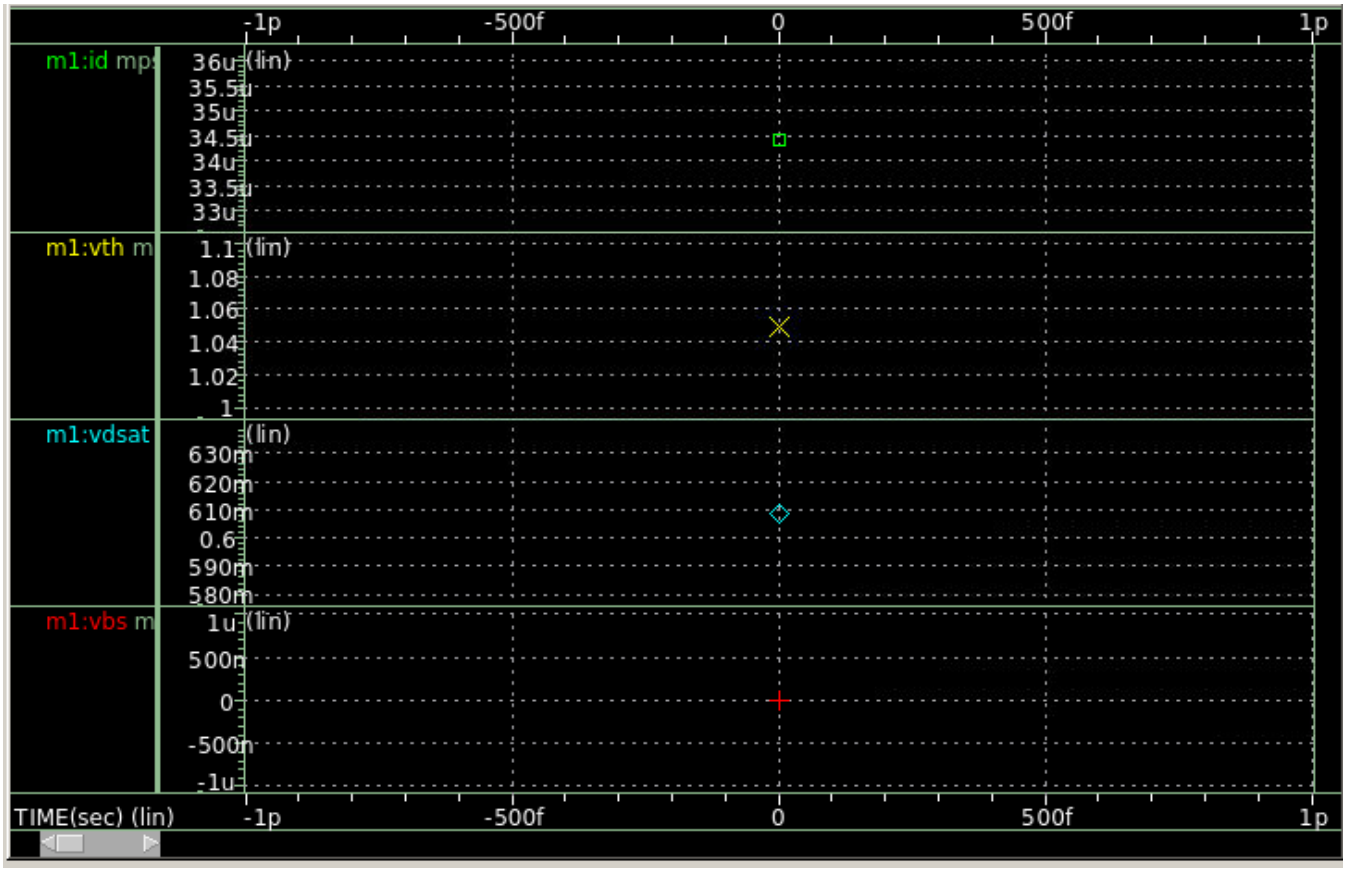
.end
```

1:v=2.5	m1:vdsat=563m	m1:cbtot=12.1f
2:v=2	m1:beta=422u	m1:cgs=16.1f
m1:region=2	m1:gam eff=1.14	m1:cgd=0.52f
m1:id=106u	m1:gm=190u	v1:volts=2
m1:ibs=5.67E-18	m1:gds=1.5u	v1:current=-960p
m1:ibd=-2.51p	m1:gmb=141u	v1:power=1.92n
m1:vgs=2	m1:vod=914m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-109u
m1:vbs=0	m1:cgtot=18.1f	v2:power=273u
m1:vth=1.09	m1:cstot=15.6f	



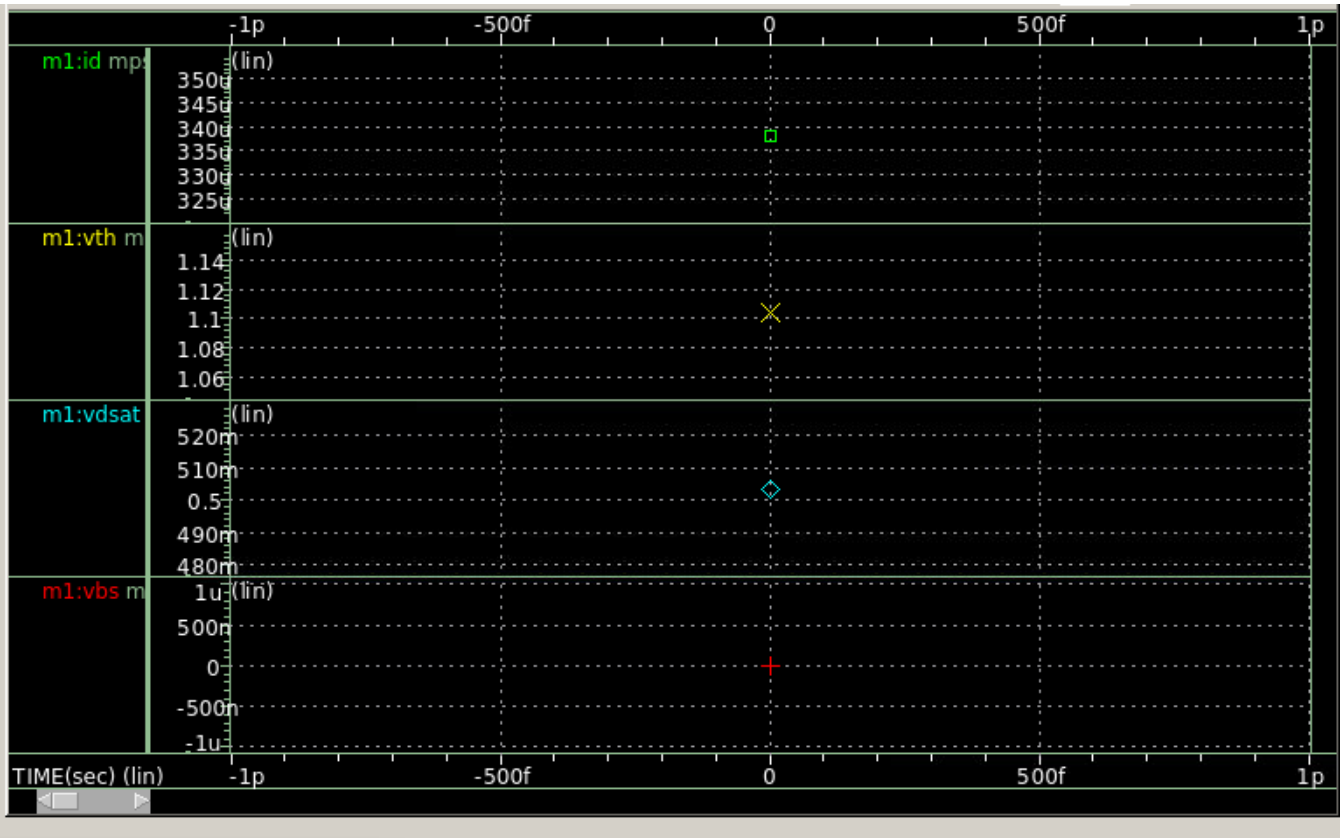
[b23n_sarif_727@prac3 experiment3]$ wv mpsfet_temp_ff_wdf.op1 &

1:v=2.5	m1:vdsat=609m	m1:cbtot=12.2f
2:v=2	m1:beta=121u	m1:cgs=15.9f
m1:region=2	m1:gam eff=1.14	m1:cgd=0.52f
m1:id=34.4u	m1:gm=56.2u	v1:volts=2
m1:ibs=1.75f	m1:gds=469n	v1:current=-1.25n
m1:ibd=-2.51p	m1:gmb=46.8u	v1:power=2.49n
m1:vgs=2	m1:vod=951m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.13f	v2:current=-35.1u
m1:vbs=0	m1:cgtot=18f	v2:power=87.7u
m1:vth=1.05	m1:cstot=15.6f	



[b23n_sarif_727@prac3 experiment3]$ wv mpsfet_temp_ff_wdf.op2 &

1:v=2.5	m1:vdsat=503m	m1:cbtot=12.1f
2:v=2	m1:beta=1.6m	m1:cgs=16.3f
m1:region=2	m1:gam eff=1.14	m1:cgd=0.52f
m1:id=338u	m1:gm=704u	v1:volts=2
m1:ibs=2.93E-17	m1:gds=4.96u	v1:current=-838p
m1:ibd=-2.51p	m1:gmb=357u	v1:power=1.68n
m1:vgs=2	m1:vod=896m	v2:volts=2.5
m1:vds=2.5	m1:cdtot=1.18f	v2:current=-354u
m1:vbs=0	m1:cgtot=18.2f	v2:power=884u
m1:vth=1.1	m1:cstot=15.6f	



[b23n_sarif_727@prac3 experiment3]$ vi operating_point_analysis.sp

[b23n_sarif_727@prac3 experiment3]$ wv operating_point_analysis_wdf.op0 &

[b23n_sarif_727@prac3 experiment3]$ wv operating_point_analysis_wdf.tr0 &

```
* opearating point  analysis of RC circuit means steady syate analysis

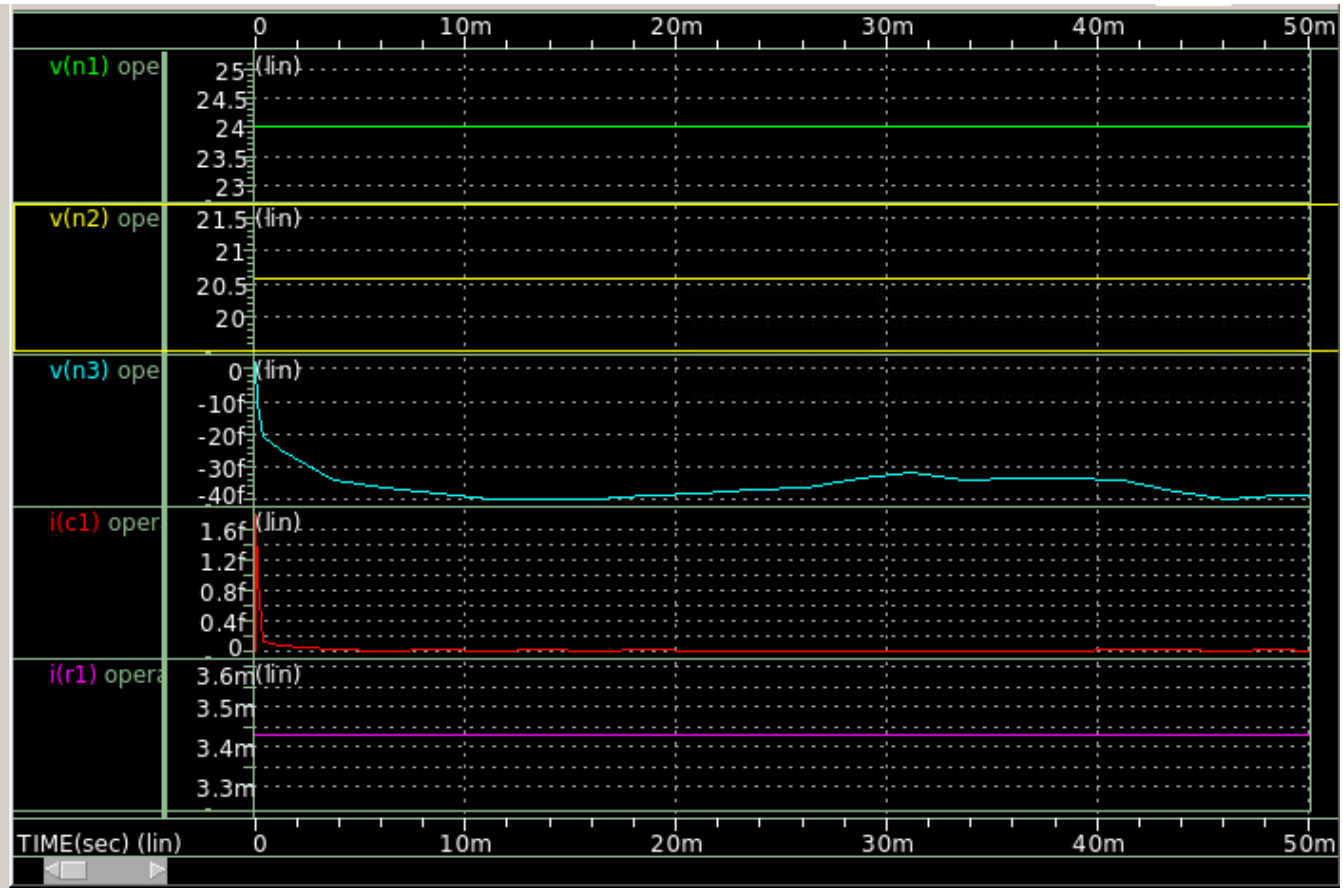
.OPTION WDF=1

R1 N1 N2 1K
R2 N2 0 6K
C1 N2 N3 10u
R3 N3 0 3K
V1 N1 0 24

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

c1:c=10u
c1:volts=20.6
c1:current=0
c1:power=0
n1:v=24
n2:v=20.6
n3:v=0
r1:res=1k
r1:v=3.43
r1:current=3.43m
r1:power=11.8m
r2:res=6k
r2:v=20.6
r2:current=3.43m
r2:power=70.5m
r3:res=3k
r3:v=0
r3:current=0
r3:power=0
v1:volts=24
v1:current=-3.43m
v1:power=82.3m



[b23n_sarif_727@prac3 experiment3]$ vi p_mosfet.sp

```
* mosfet operating point analysis in pmos
.option search=/data/pdk/pdk_cc/hspice

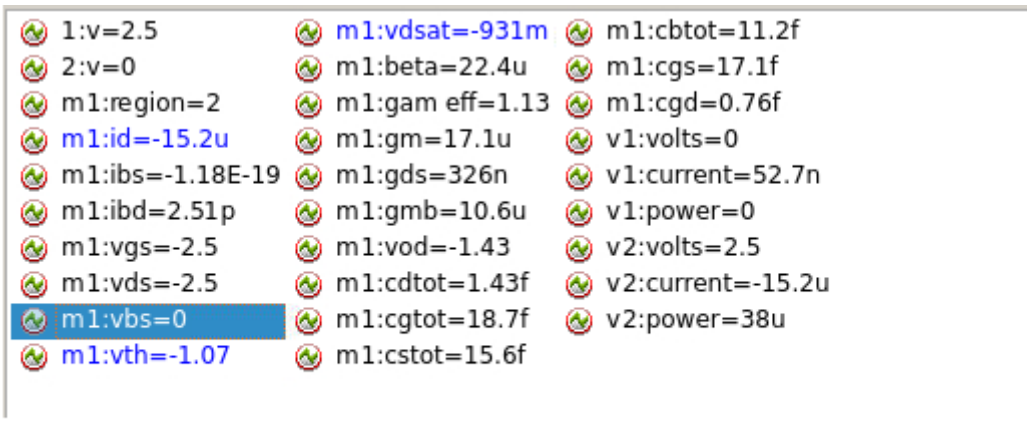
.lib reference40_models.lib TT

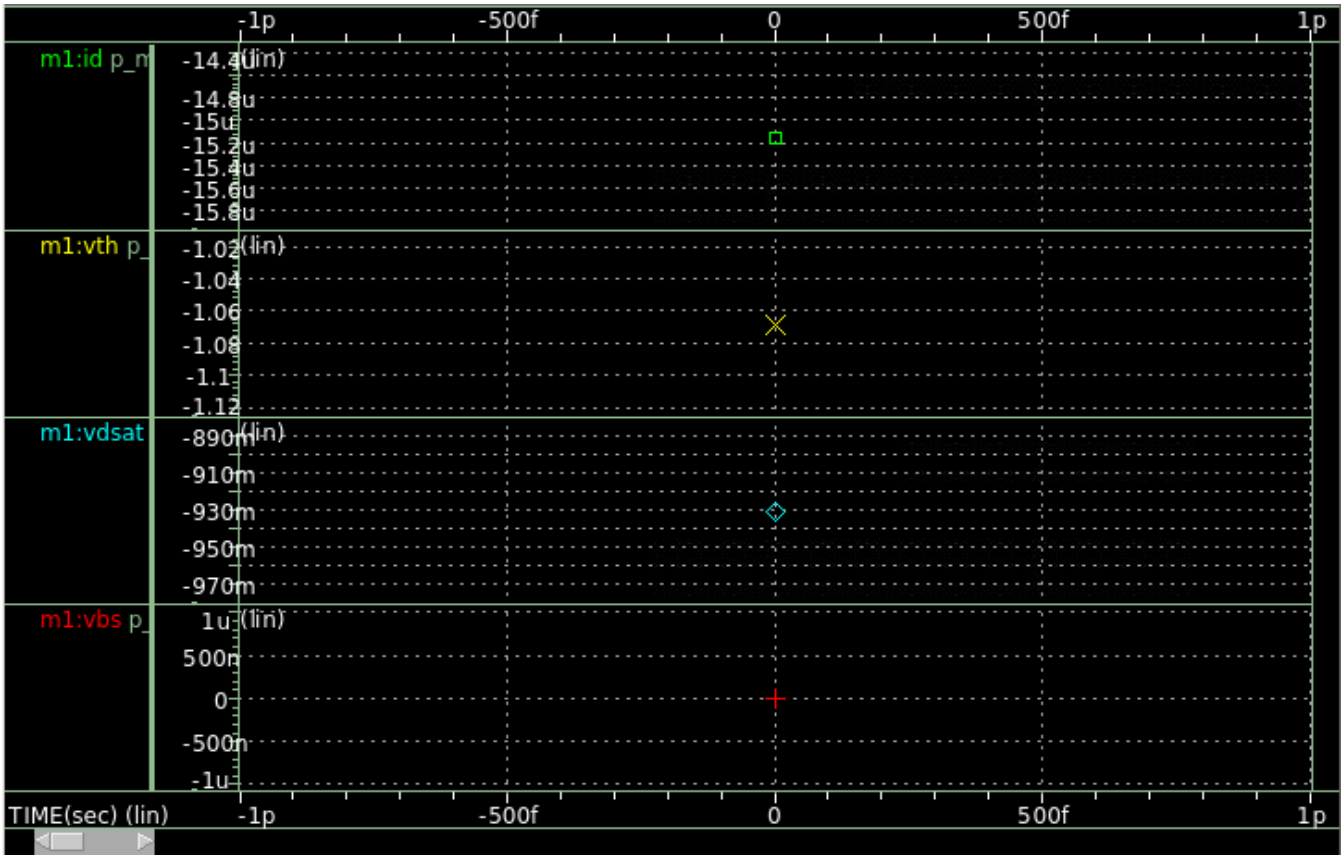
.option WDF=1
*.model  P25  pmos  level = 54
M1 0 2 1 1 P25 W=2u L=1u
V1 2 0 0
V2 1 0 2.5

.OP
.probe V(*) I(*)
.print V(*) I(*)

.end
```

[b23n_sarif_727@prac3 experiment3]$ wv p_mosfet_wdf.op0 &





[b23n_sarif_727@prac3 experiment3]$ vi p_mosfet_1.sp

[b23n_sarif_727@prac3 experiment3]$ wv p_mosfet_1_wdf.sw0 &

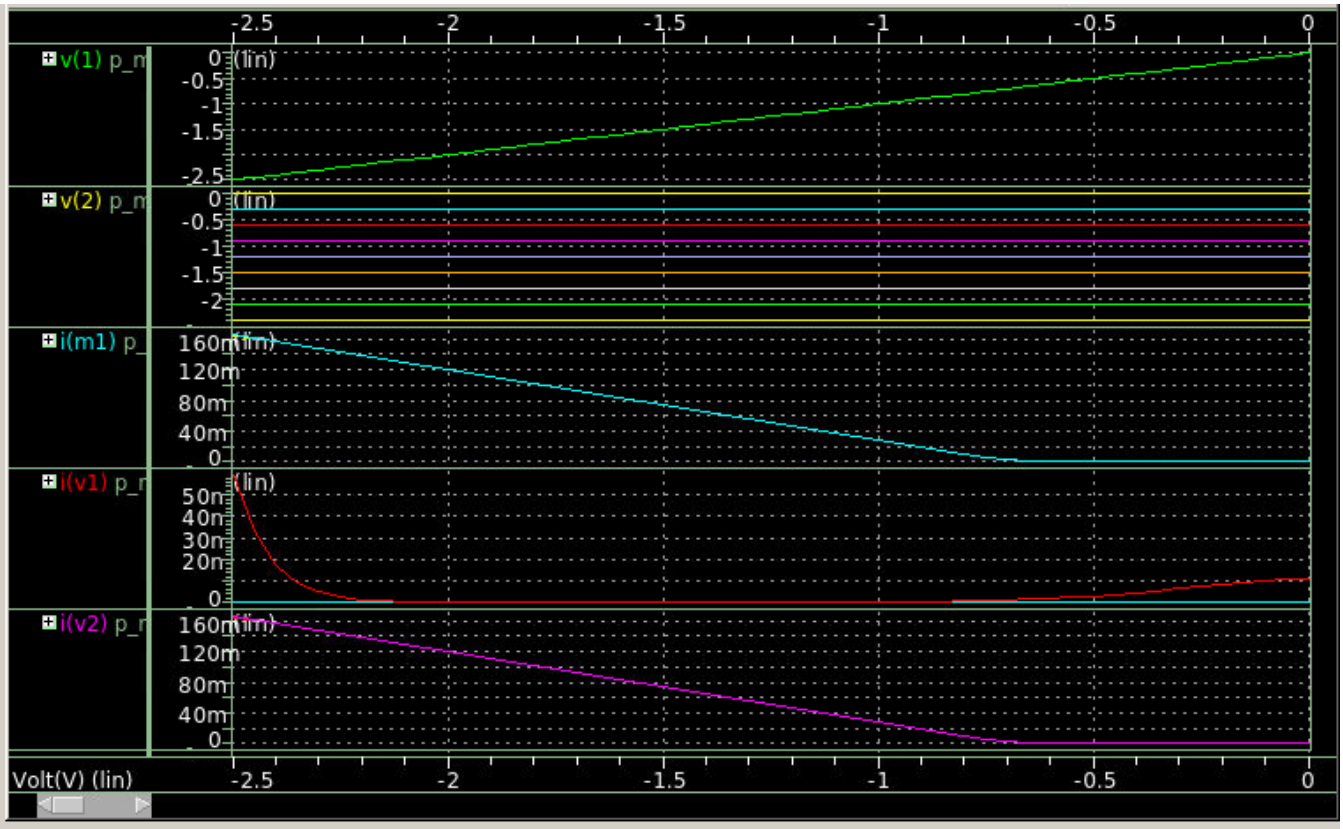
```
* mosfet operating point analysis in pmos
.option search=/data/pdk/pdk_cc/hspice

.lib reference40_models.lib TT

.option WDF=1
*.model  P25  pmos  level = 54
M1 0 2 1 1 P25 W=2u L=1u
V1 2 0 0
V2 1 0 2.5

.dc V2 0 -2.5 0.05 V1 0 -2.5 0.3
.probe V(*) I(*)
.print dc V(*) I(*)

.end
```



[b23n_sarif_727@prac3 experiment3]$ vi R_mosfet.sp

[b23n_sarif_727@prac3 experiment3]$ wv R_mosfet_wdf.op0 &

[b23n_sarif_727@prac3 experiment3]$ wv R_mosfet_wdf.sw0 &

```
* mosfet operating point analysis
.option search=/data/pdk/pdk_cc/hspice

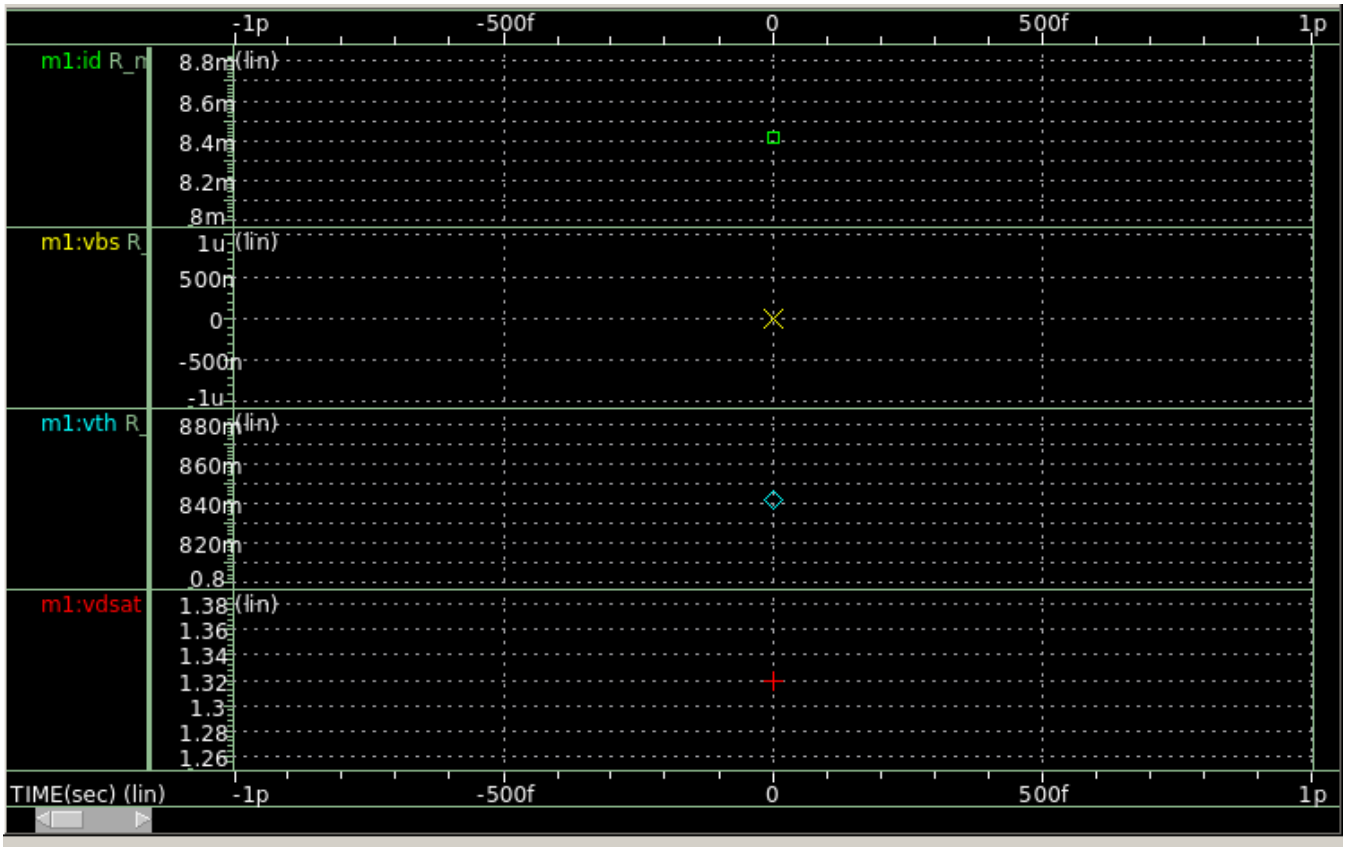

.lib reference40_models.lib FF

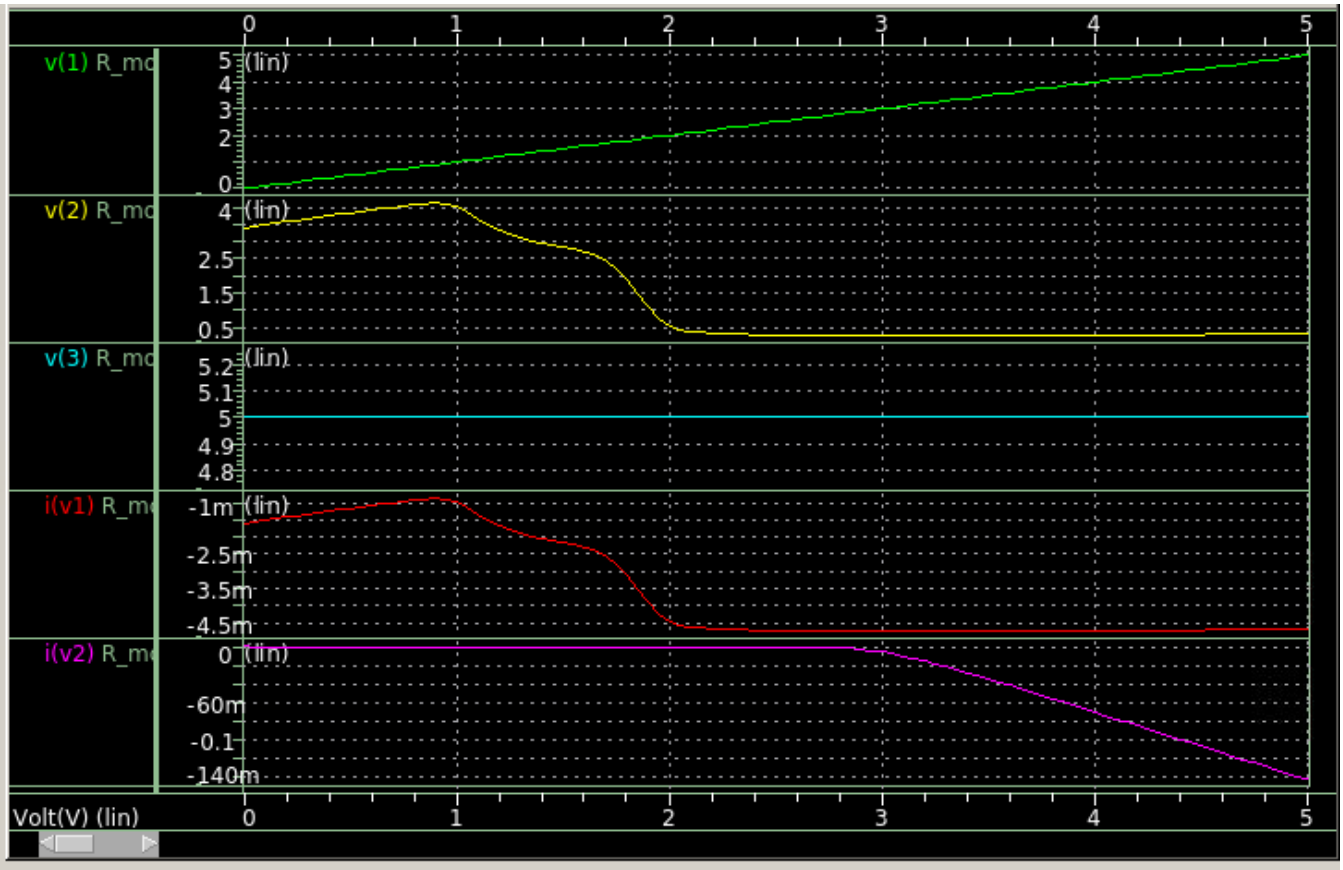
.option WDF=1
*.model  N25  nmos  level = 54
M1 2 1 0 0 N25 W=180u L=2u
R 3 2 1K
V1 3 0 5
V2 1 0 5

.dc V2 0 5 0.05
.OP
.probe V(2)
.print V(2)

.end
```

1:v=5	m1:vth=842m	m1:cstot=1.64p	v1:power=23.5m
2:v=295m	m1:vdsat=1.32	m1:cbtot=1.36p	v2:volts=5
3:v=5	m1:beta=14.4m	m1:cgs=2.21p	v2:current=-144m
m1:region=1	m1:gam eff=1.14	m1:cgd=1.78p	v2:power=719m
m1:id=8.41m	m1:gm=1.62m	r:res=1k	
m1:ibs=289p	m1:gds=24.9m	r:v=4.7	
m1:ibd=-48.6f	m1:gmb=2.22m	r:current=4.7m	
m1:vgs=5	m1:vod=4.16	r:power=22.1m	
m1:vds=295m	m1:cdtot=2.53p	v1:volts=5	
m1:vbs=0	m1:cgtot=3.8p	v1:current=-4.7m	





Experiment 4

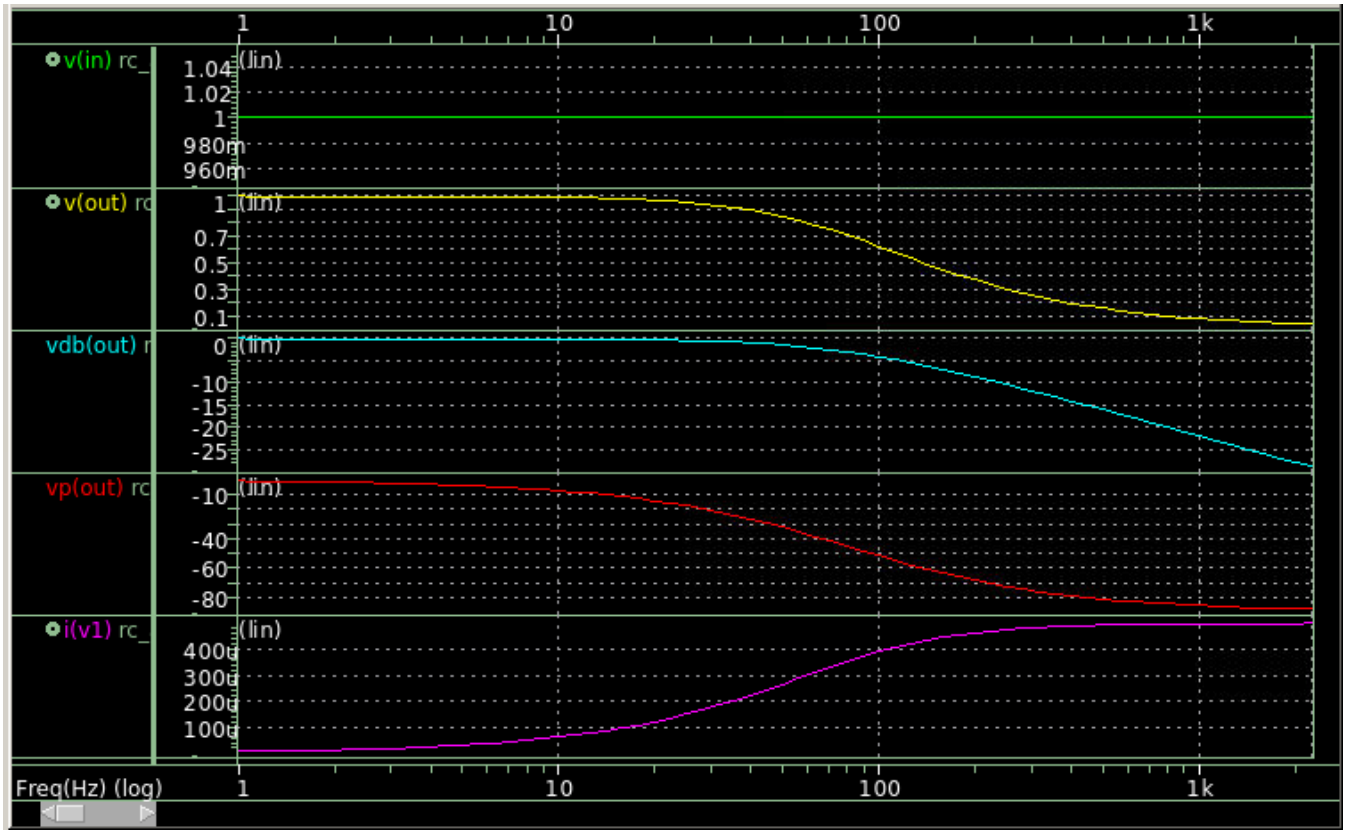
```
* rc circuit ac response

.option WDF=1
R1 IN OUT 2K
C1 OUT 0 1U

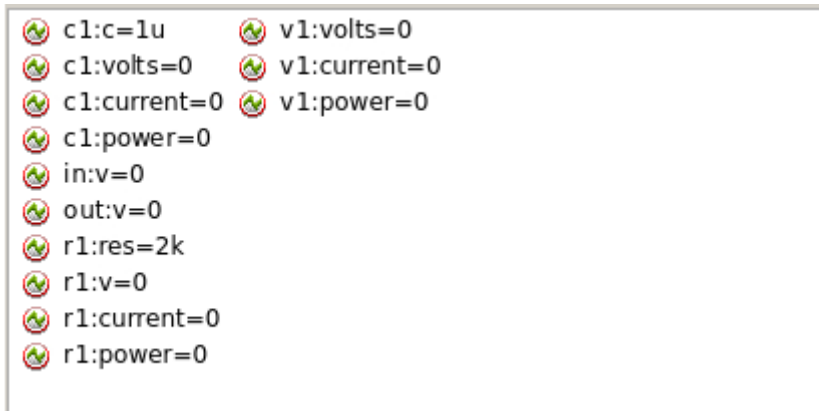
V1 IN 0 AC=1

.PROBE I(*) V(*)
.plot AC VDB(OUT) VP(OUT)
.ac dec 20 1 2k
.end
~
```

[b23n_sarif_727@prac3 experiment4]$ vi rc_ac_response.sp



[b23n_sarif_727@prac3 experiment4]$ wv rc_ac_response_wdf.ac0 &



[b23n_sarif_727@prac3 experiment4]$ wv rc_ac_response_wdf.op0 &

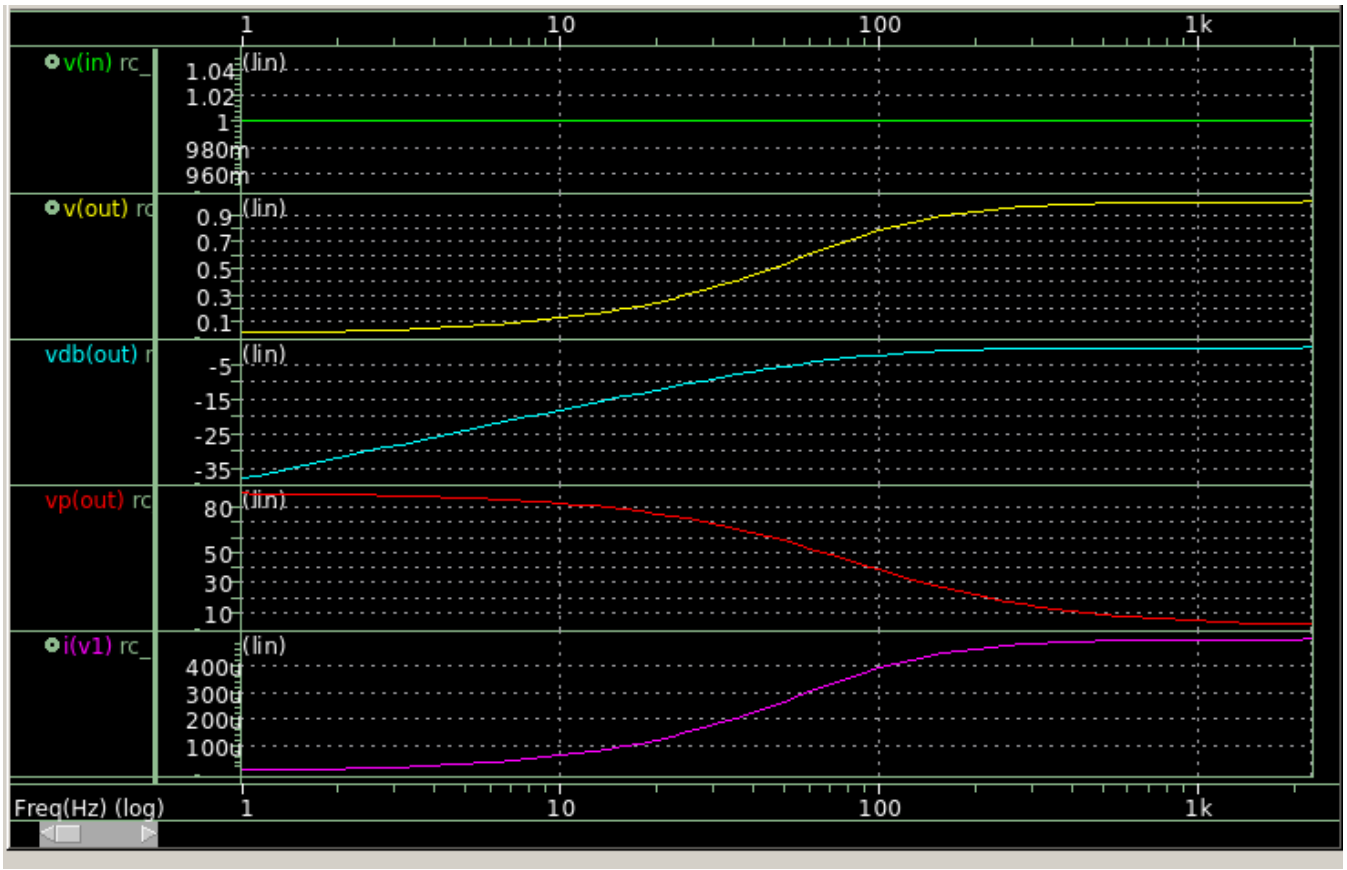
```
* rc circuit ac response for high pass filter

.option WDF=1
R1 OUT 0 2K
C1 IN OUT 1u

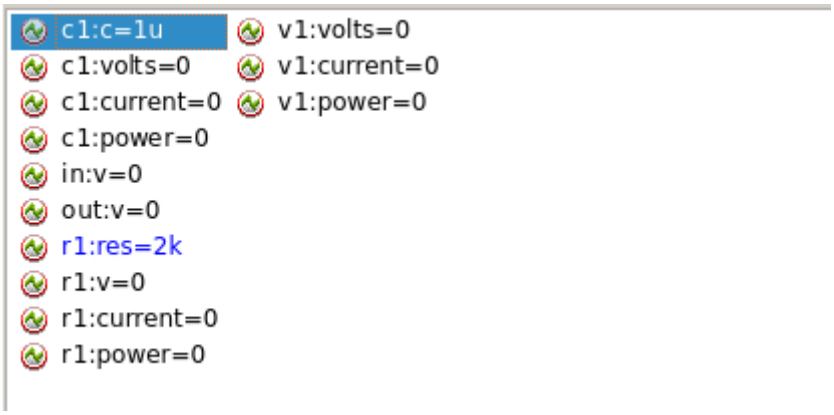
V1 IN 0 AC=1

.PROBE I(*) V(*)
.plot AC VDB(OUT) VP(OUT)
.ac dec 20 1 2k
.end
```

[b23n_sarif_727@prac3 experiment4]$ vi rc_high_acresponse.sp



[b23n_sarif_727@prac3 experiment4]$ wv rc_high_acresponse_wdf.ac0 &



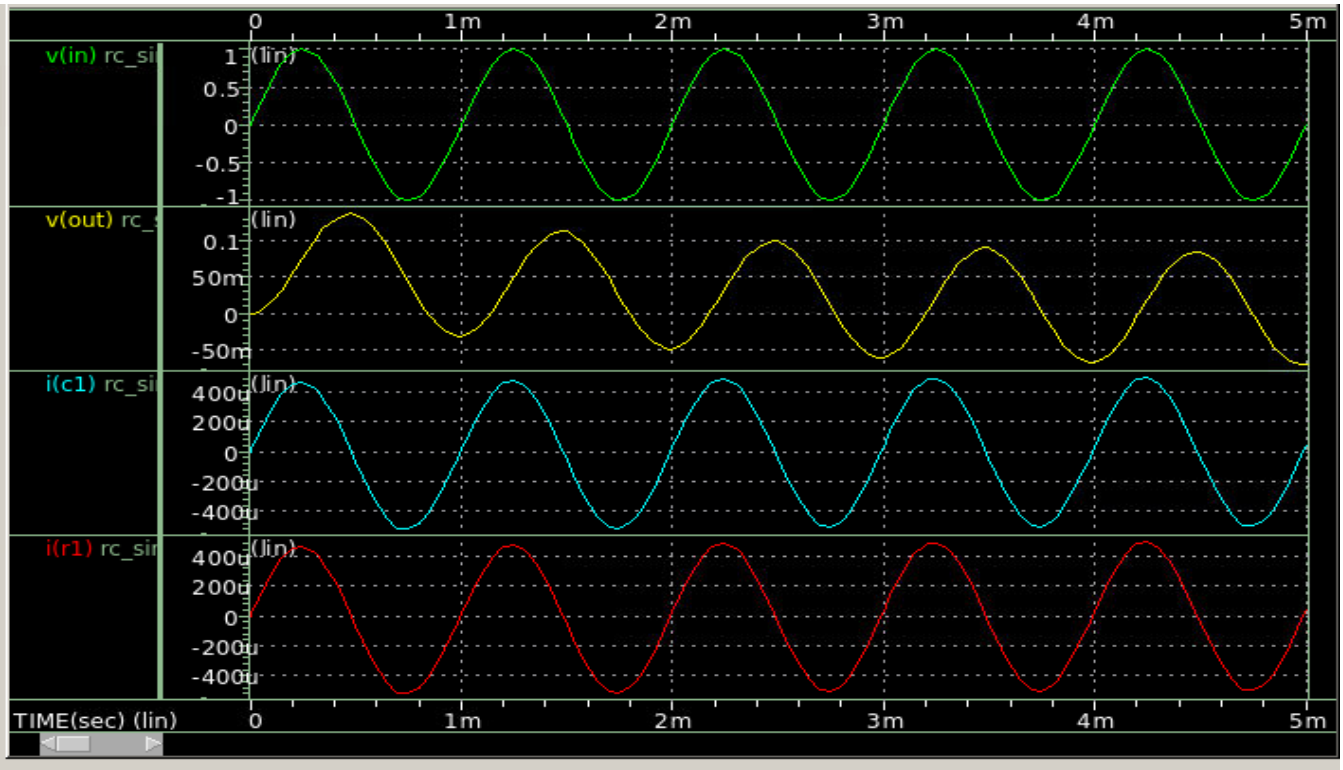
[b23n_sarif_727@prac3 experiment4]$ wv rc_high_acresponse_wdf.op0 &

```
* rc circuit sinusoidal signal response

.option WDF=1
R1 IN OUT 2k
C1 OUT 0 1u

V1 IN 0 SIN(0 1 1k 0 0 0)
.TRAN .01M 5M
.PROBE V(*) I(*)
.END
```

[b23n_sarif_727@prac3 experiment4]$ vi rc_sinwave1k.sp



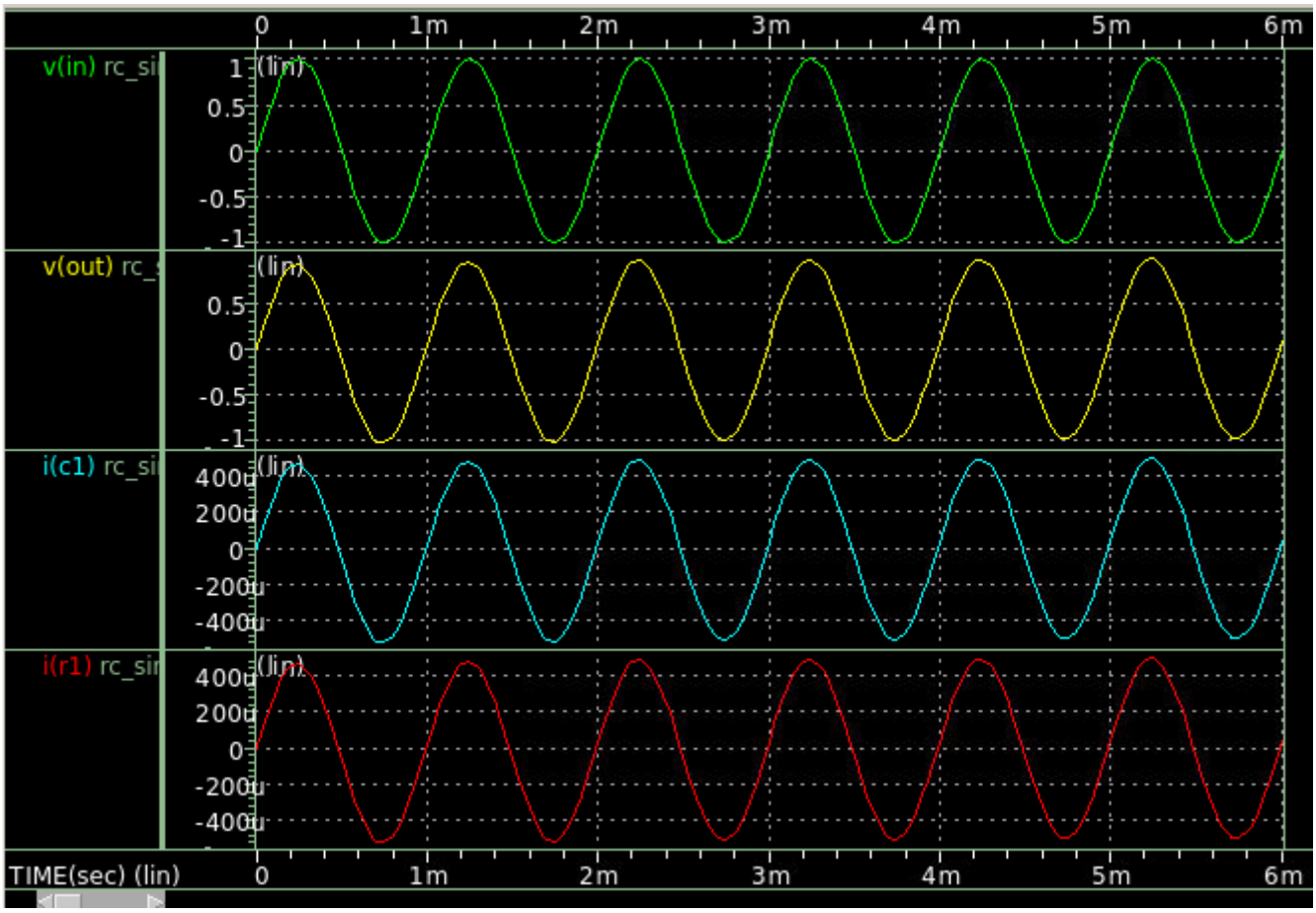
[b23n_sarif_727@prac3 experiment4]$ wv rc_sinwave1k_wdf.tr0 &

```
* rc circuit sinusoidal signal response of high filter

.option WDF=1
R1 OUT 0 2k
C1 IN OUT 1u

V1 IN 0 SIN(0 1 1k 0 0 0)
.TRAN .01m 6m
.PROBE V(*) I(*)
.END
```

[b23n_sarif_727@prac3 experiment4]$ vi rc_sinwave_highpass1k.sp



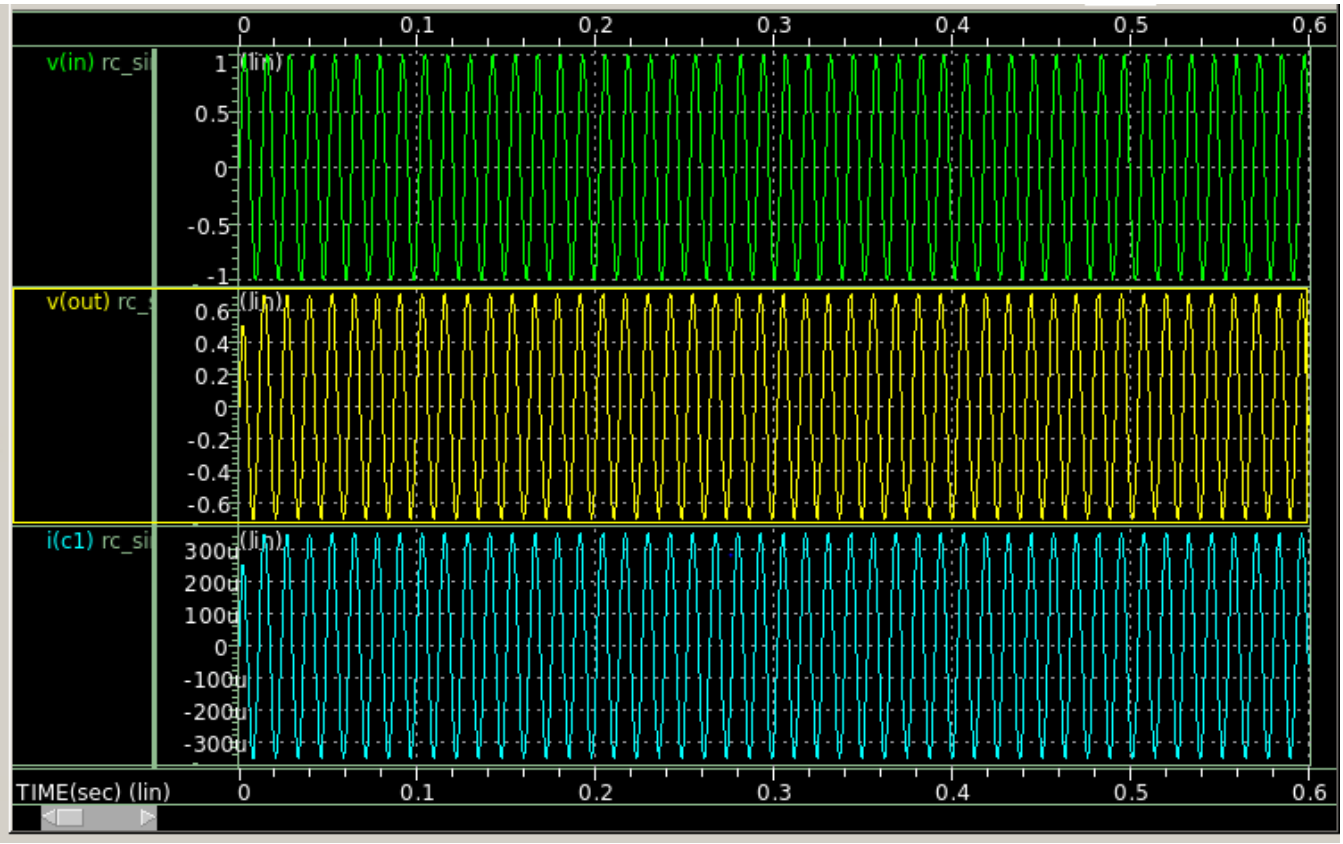
[b23n_sarif_727@prac3 experiment4]$ wv rc_sinwave_highpass1k_wdf.tr0 &

```
*highpass filter response

.option WDF=1

C1 IN OUT 1u
R1 OUT 0 2K
*V1 IN 0 SIN(0 1 1 0 0 0)
V1 IN 0 SIN(0 1 79 0 0 0)
.TRAN .01m 600m
.PROBE V(*) I(*)
.END
```

[b23n_sarif_727@prac3 experiment4]$ vi rc_sinwave_highpass.sp



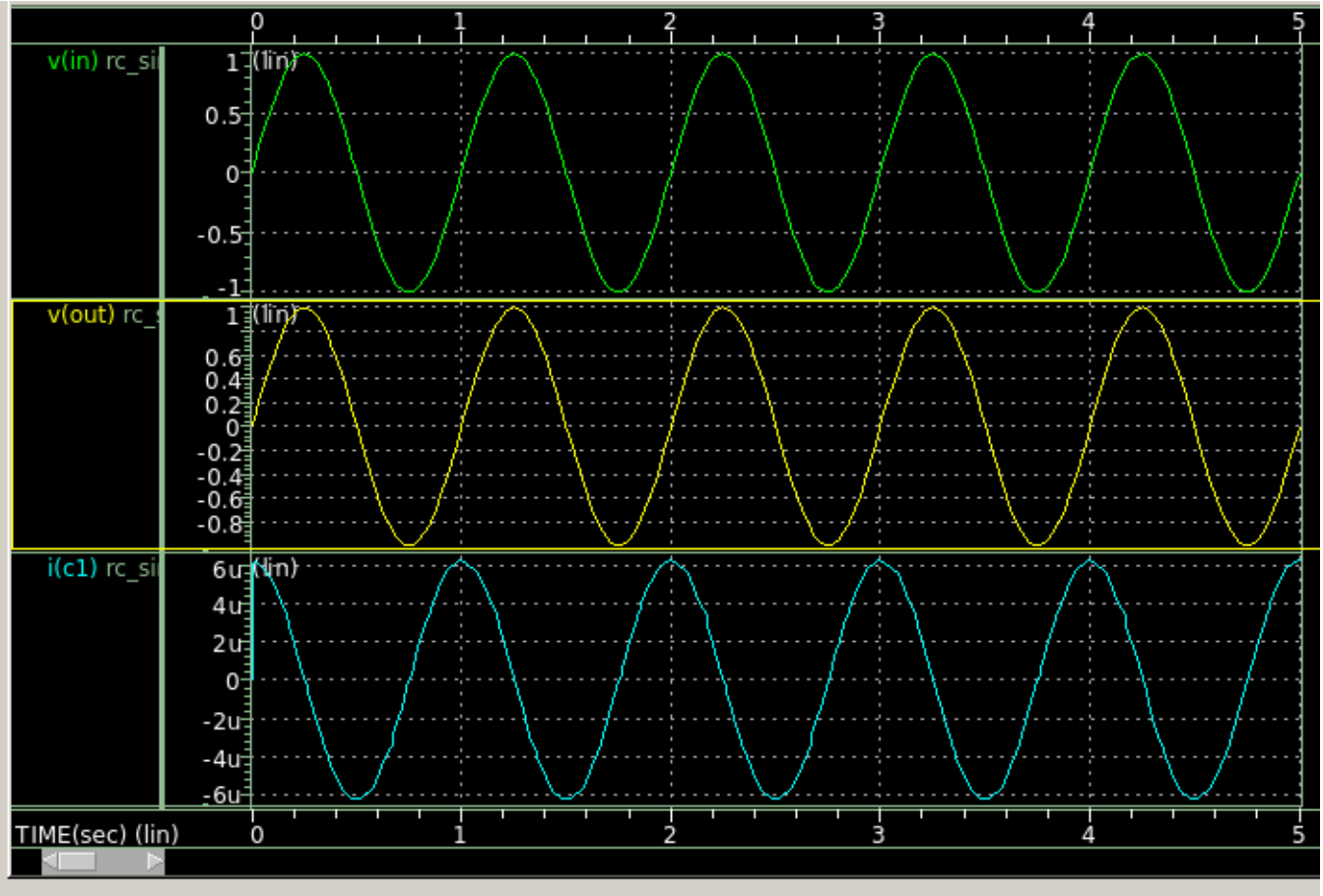
[b23n_sarif_727@prac3 experiment4]$ wv rc_sinwave_highpass_wdf.tr0 &

[b23n_sarif_727@prac3 experiment4]$ vi rc_sinwave.sp

[b23n_sarif_727@prac3 experiment4]$ wv rc_sinwave_wdf.tr0 &

```
* rc circuit sinusoidal signal response

.option WDF=1
R1 IN OUT 2K
C1 OUT 0 1u
*V1 IN 0 SIN(0 1 1 0 0 0)
v1 IN 0 SIN(0 1 79 0 0 0)
.TRAN .01m 60m
.PROBE I(*) V(*)
.END
```



Experiment5

[b23n_sarif_727@prac3 experiment5]$ vi capactor_varying_tphl_tplh.sp

[b23n_sarif_727@prac3 experiment5]$ vi capactor_varying_tphl_tplh.mt0

[b23n_sarif_727@prac3 experiment5]$ wv capactor_varying_tphl_tplh_wdf.tr0 &

```
RC step response BY varying capactor -case_measure tplh and tphl and

.OPTION POST WDF=1

R1 IN OUT 1k
C1 OUT 0 8u
V1 IN 0 pulse(0 5 0 0 0 4m 8m)

.probe V(*) I(*)
.print V(*) I(*)

.tran .05m 32m

.MEAS TRAN vmax MAX V(OUT) FROM=0 TO=32m
.MEAS TRAN vmin MIN V(OUT) FROM=0 TO=32m

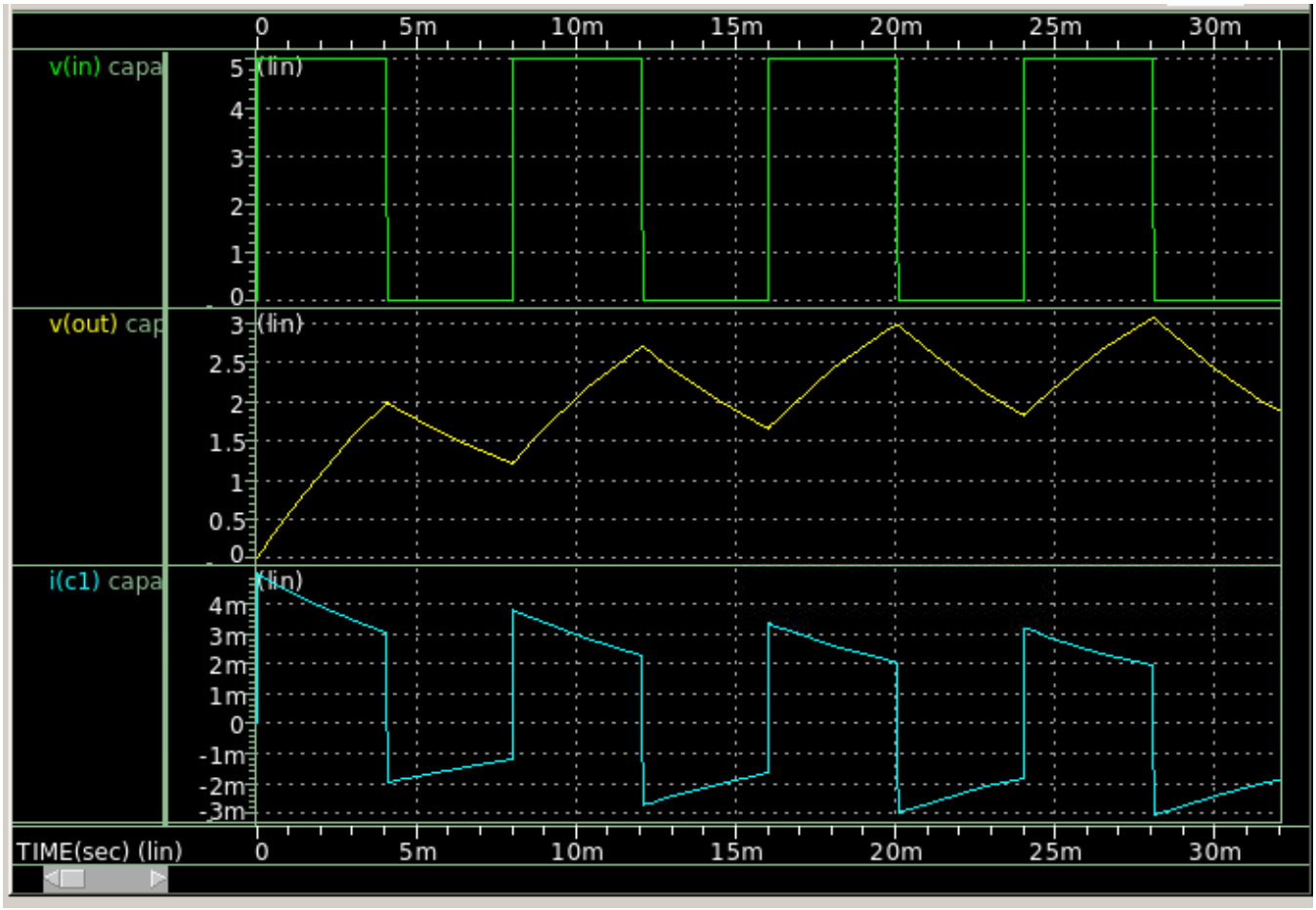
.MEAS TRAN Trise TRIG V(OUT) VAL='vmin+0.1*(vmax-vmin)' RISE=1
+ TARG V(OUT) VAL='vmin+0.9*(vmax-vmin)' RISE=1

.MEAS TRAN Tfall TRIG V(OUT) VAL='vmin+0.9*(vmax-vmin)' FALL=1
+ TARG V(OUT) VAL='vmin+0.1*(vmax-vmin)' FALL=1

.MEASURE TRAN TPLH TRIG V(IN) VAL=2.5 RISE=2 TARG V(OUT) VAL=2.5 RISE=2
.MEASURE TRAN TPHL TRIG V(IN) VAL=2.5 FALL=2 TARG V(OUT) VAL=2.5 FALL=2

.end
```

```
$DATA1 SOURCE='PrimeSim HSPICE' VERSION='V-2023.12-SP2 linux64' PARAM_COUNT=0
.TITLE 'rc step response by varying capactor -case_measure tplh and tphl and'
 vmax            vmin            trise           tfall
 tplh            tphl            temper          alter#
    3.0825           0.0           1.876e-02          failed
  1.033e-02        9.441e-03         25.0000        1
~
```



[b23n_sarif_727@prac3 experiment5]$ vi practic_subckt.sp

[b23n_sarif_727@prac3 experiment5]$ vi practic_subckt.pa0

[b23n_sarif_727@prac3 experiment5]$ wv practic_subckt_wdf.tr0 &

```
*define global node for this subckt

.global gnd!
****************
*********define subckt*****
.subckt rc A B
.print V(A) V(B)
r1 A B 1k
c1 B gnd! 1u
.ends rc

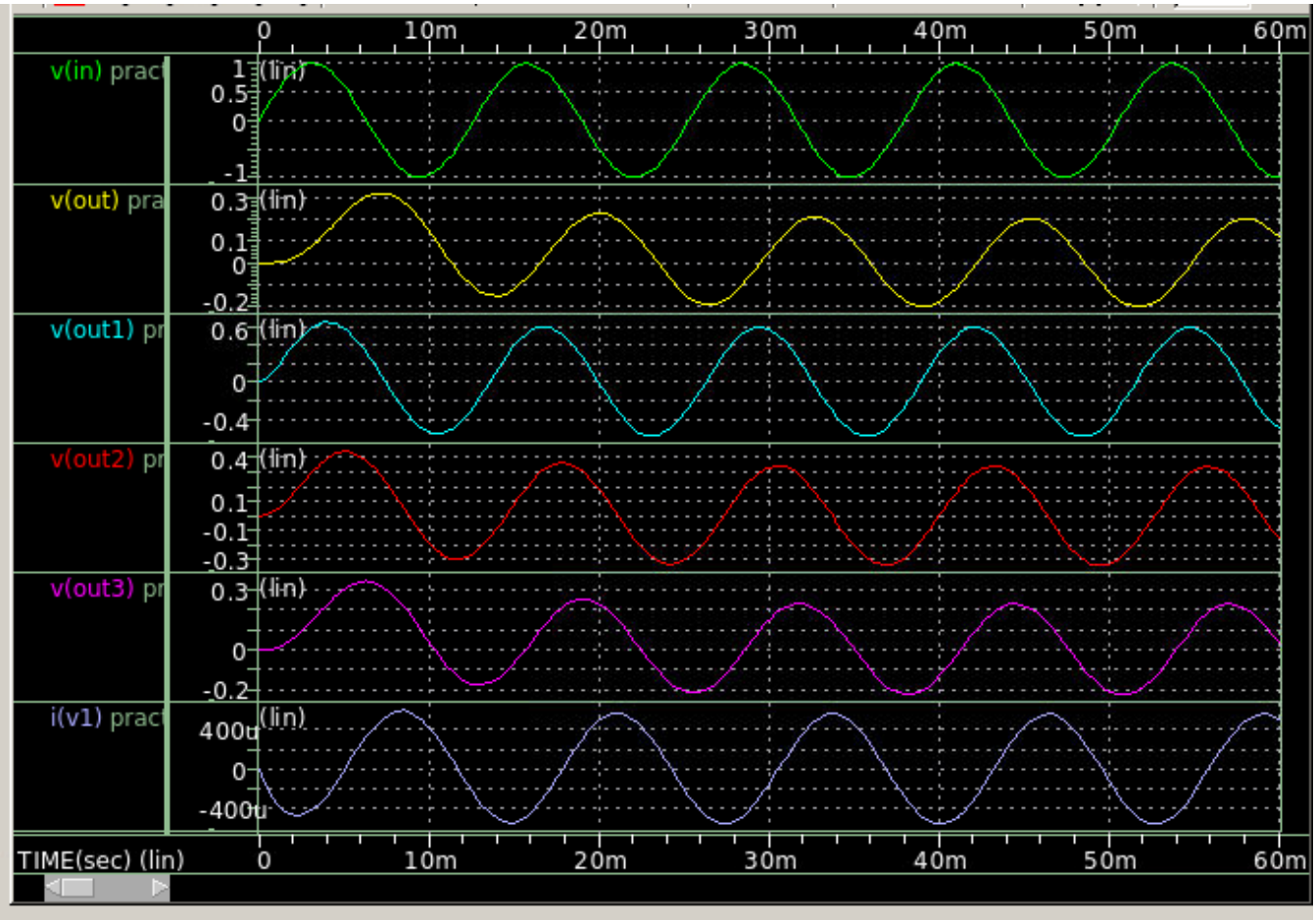
**********************
.option RUNLVL=3
.option WDF=1

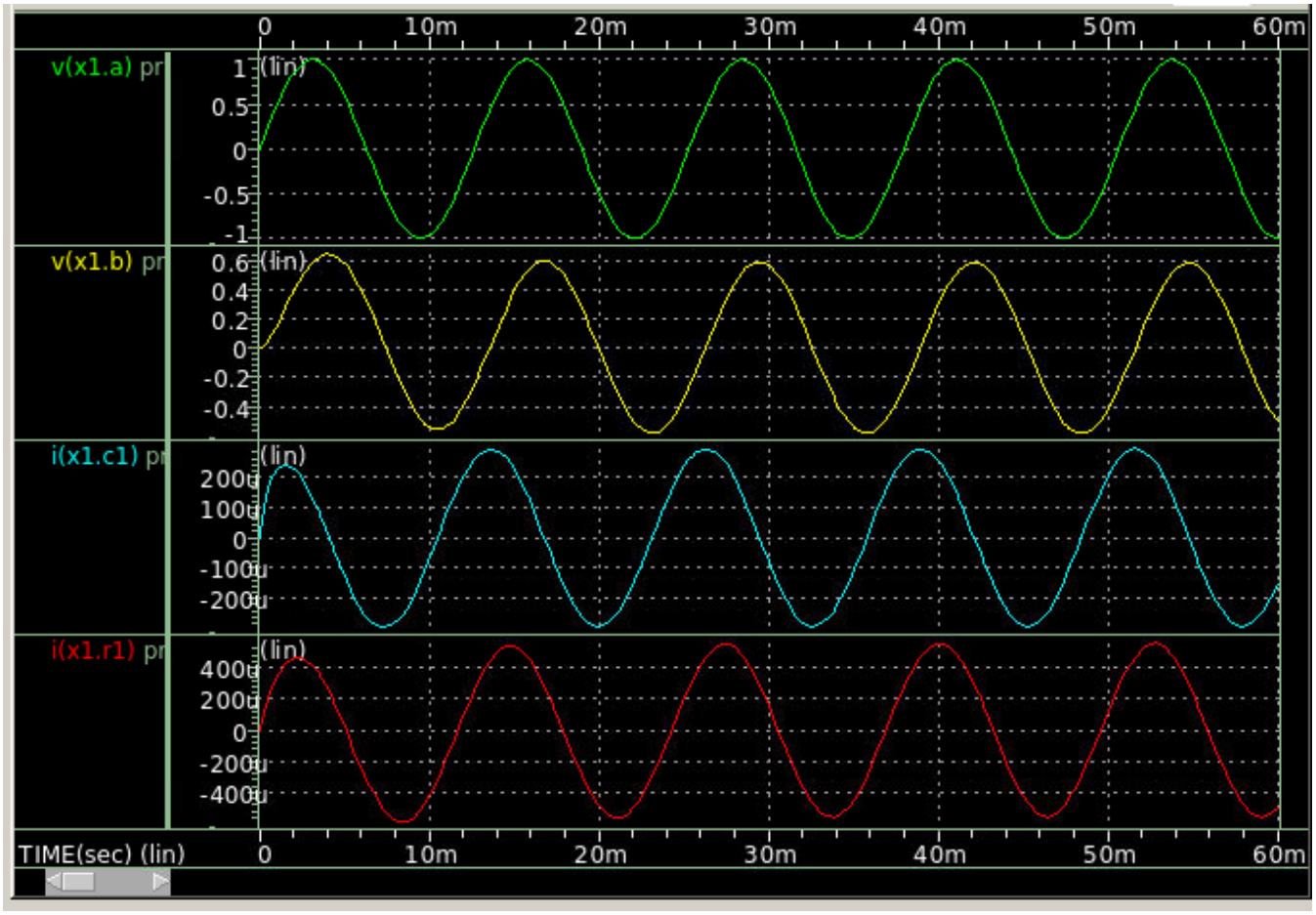
*******************
.temp 25
**********************
X1 IN OUT1 rc
X2 OUT1 OUT2 rc
X3 OUT2 OUT3 rc
X4 OUT3 OUT rc
*************input stimuli*******
V1 IN 0 SIN(0 1 79 0 0 0)

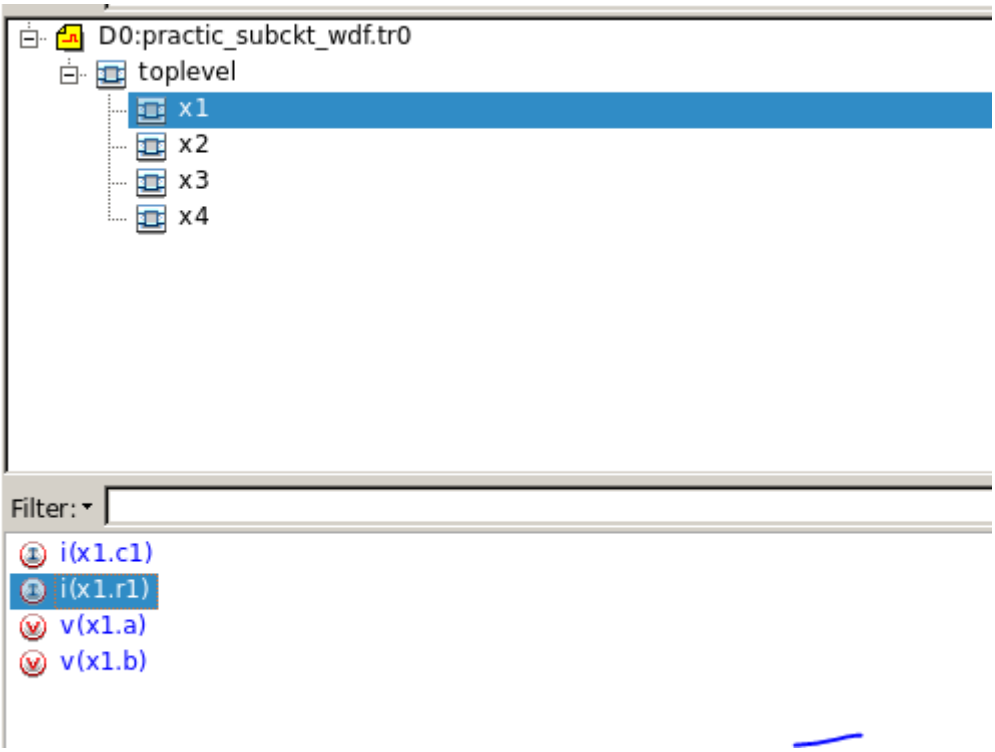
**************analysis option*******
.tran .01m 60m
.option probe=1
.print(X1.A)

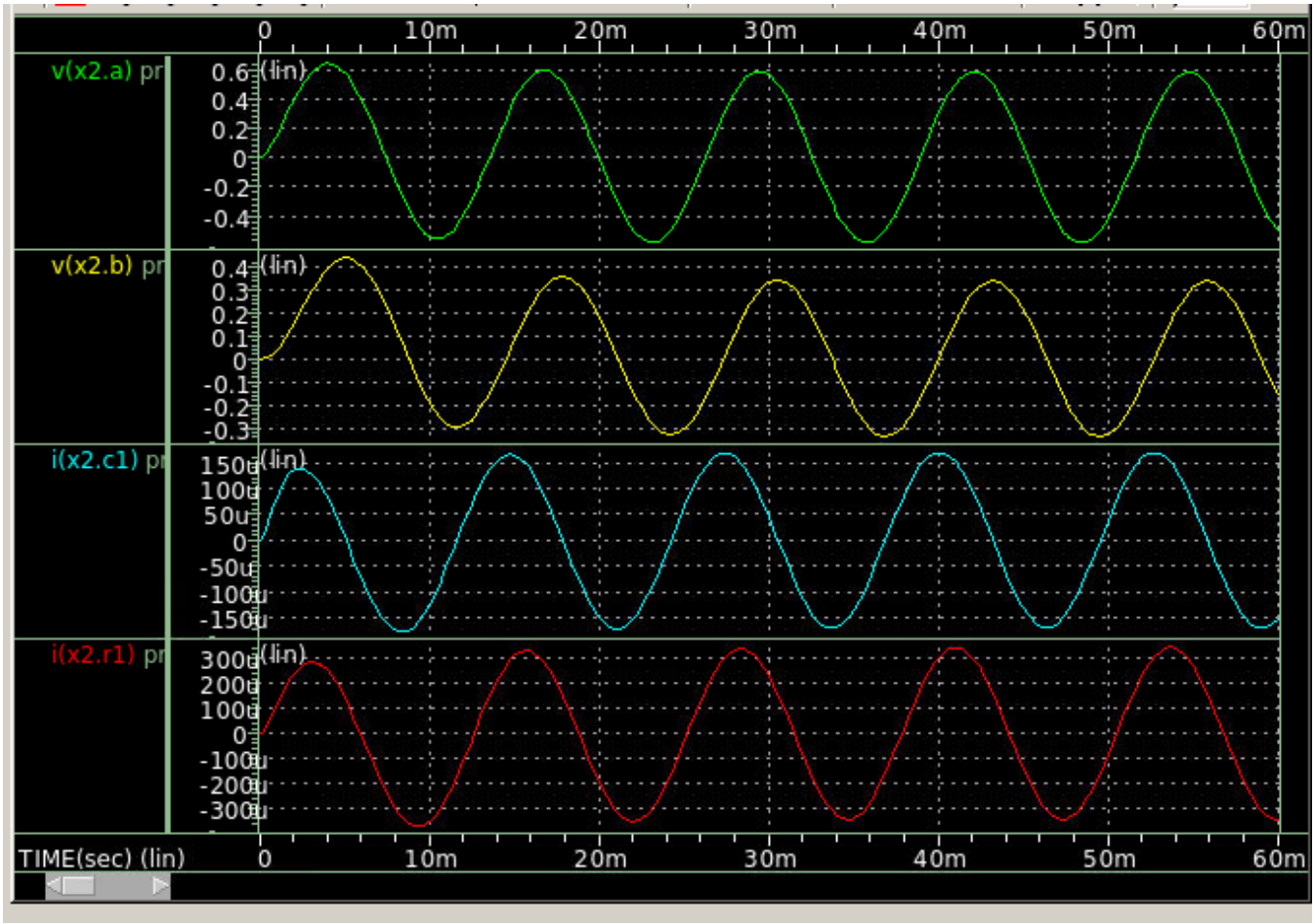
.probe tran V(*) I(*)
.end
```

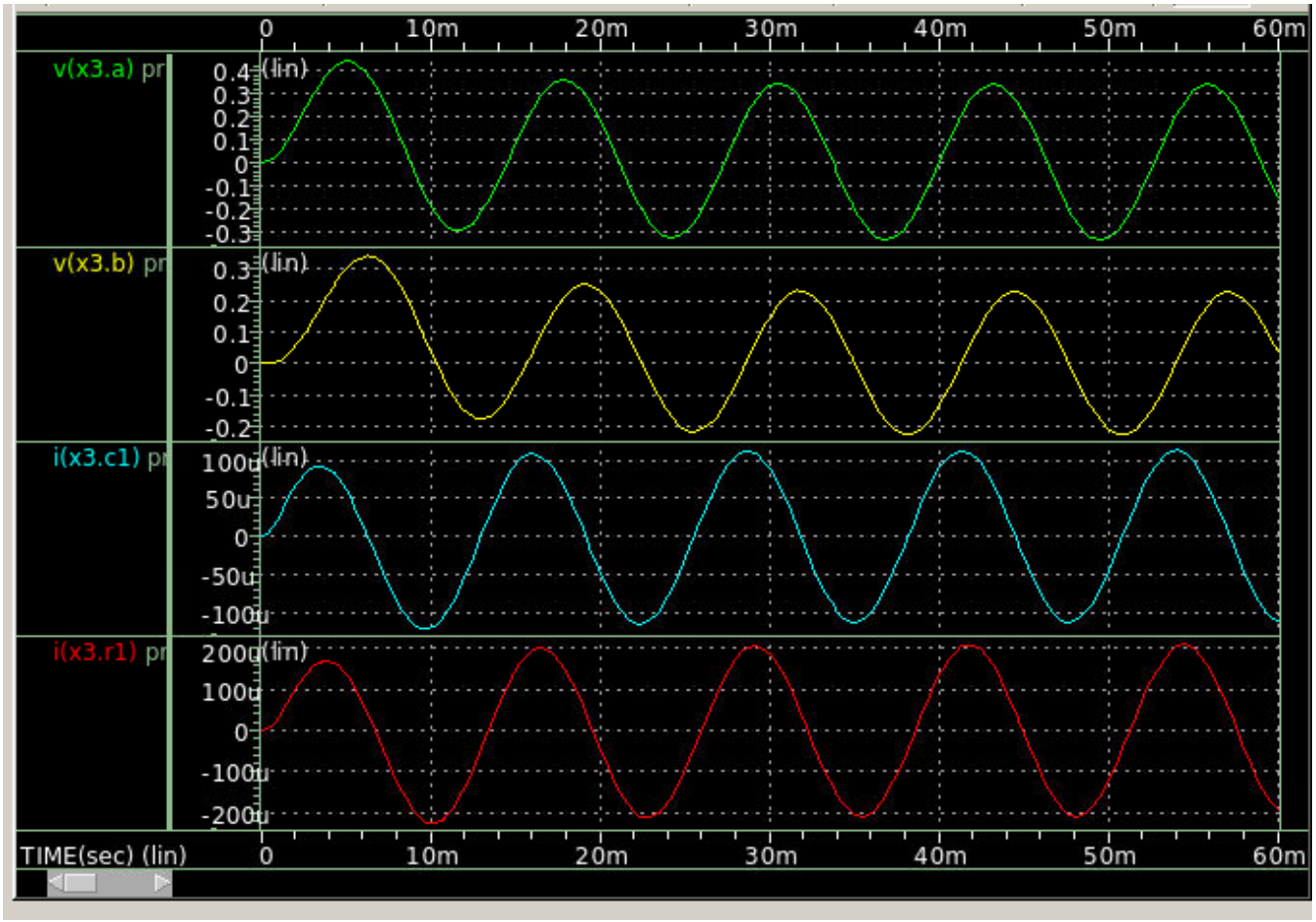
```
      1 x1.
      2 x2.
      3 x3.
      4 x4.
~
```

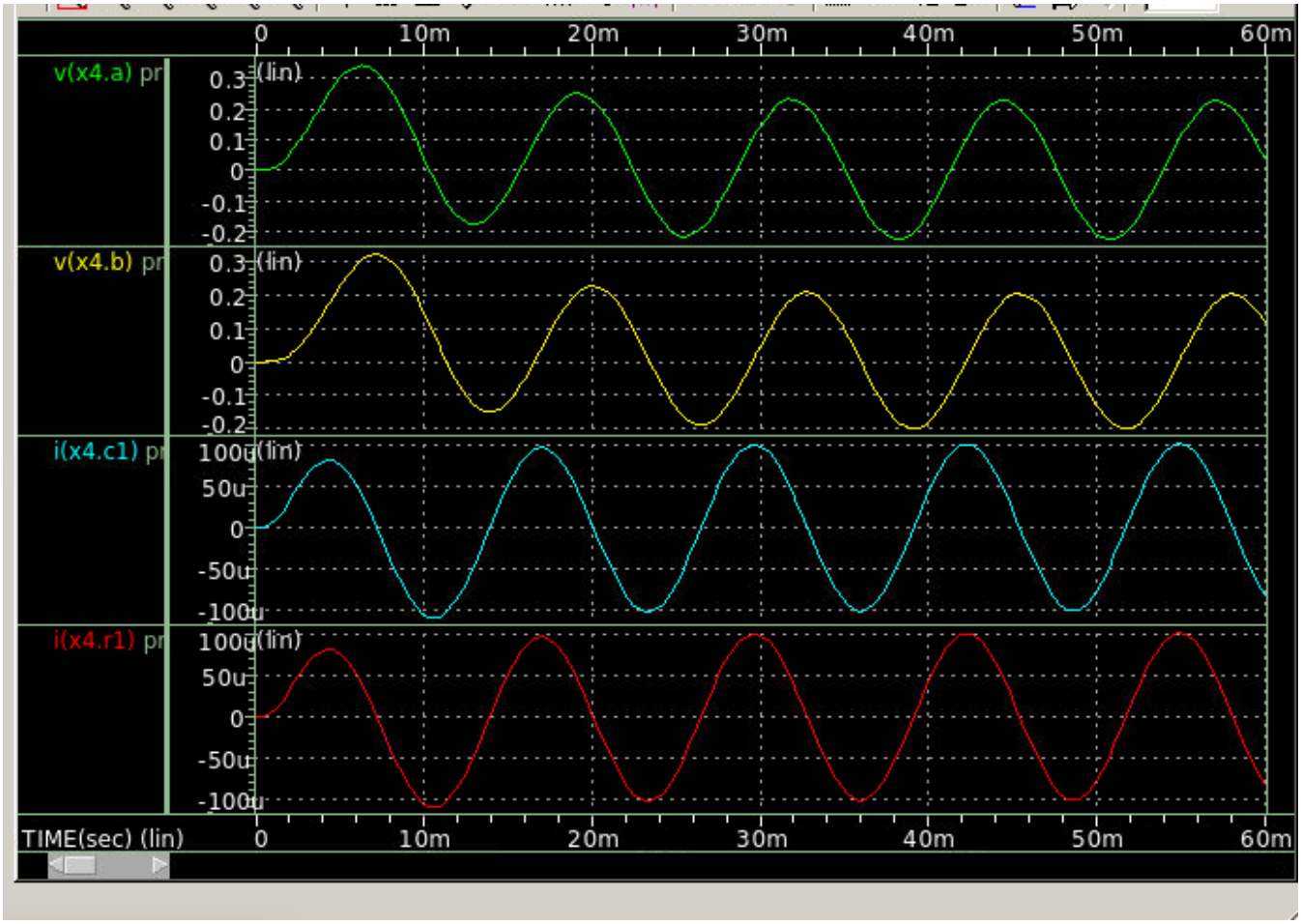

0
10m
20m
30m
40m
50m
60m
v(in) prac
1
0.5
0
-1
(lin)
v(out) pra
0.3
0.1
0
-0.2
(lin)
v(out1) pr
0.6
0
-0.4
(lin)
v(out2) pr
0.4
0.1
-0.1
-0.3
(lin)
v(out3) pr
0.3
0
-0.2
(lin)
i(v1) prac
400u
0
-400u
(lin)
TIME(sec) (lin)


0
10m
20m
30m
40m
50m
60m
v(x1.a) pr
1
0.5
0
-0.5
-1
(lin)
v(x1.b) pr
0.6
0.4
0.2
0
-0.2
-0.4
(lin)
i(x1.c1) pr
200u
100u
0
-100u
-200u
(lin)
i(x1.r1) pr
400u
200u
0
-200u
-400u
(lin)
TIME(sec) (lin)
0
10m
20m
30m
40m
50m
60m


D0:practic_subckt_wdf.tr0
toplevel
x1
x2
x3
x4
Filter:
i(x1.c1)
i(x1.r1)
v(x1.a)
v(x1.b)


0
10m
20m
30m
40m
50m
60m
v(x2.a) pr
(lin)
0.6
0.4
0.2
0
-0.2
-0.4
v(x2.b) pr
(lin)
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
i(x2.c1) pr
(lin)
150u
100u
50u
0
-50u
-100u
-150u
i(x2.r1) pr
(lin)
300u
200u
100u
0
-100u
-200u
-300u
TIME(sec) (lin)


0
10m
20m
30m
40m
50m
60m
v(x3.a) pr
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
(lin)
v(x3.b) pr
0.3
0.2
0.1
0
-0.1
-0.2
(lin)
i(x3.c1) pr
100u
50u
0
-50u
-100u
(lin)
i(x3.r1) pr
200u
100u
0
-100u
-200u
(lin)
TIME(sec) (lin)


0
10m
20m
30m
40m
50m
60m
v(x4.a) pr
(lin)
0.3
0.2
0.1
0
-0.1
-0.2
v(x4.b) pr
(lin)
0.3
0.2
0.1
0
-0.1
-0.2
i(x4.c1) pr
100u
(lin)
50u
0
-50u
-100u
i(x4.r1) pr
100u
(lin)
50u
0
-50u
-100u
TIME(sec) (lin)
0
10m
20m
30m
40m
50m
60m

```
*define global node for this subckt

.global gnd!
****************
*********define subckt*****
.subckt rc A B
.print V(A) V(B)
r1 A B 1k
c1 B gnd! 1u
.ends rc

**********************
.option RUNLVL=3
.option WDF=1

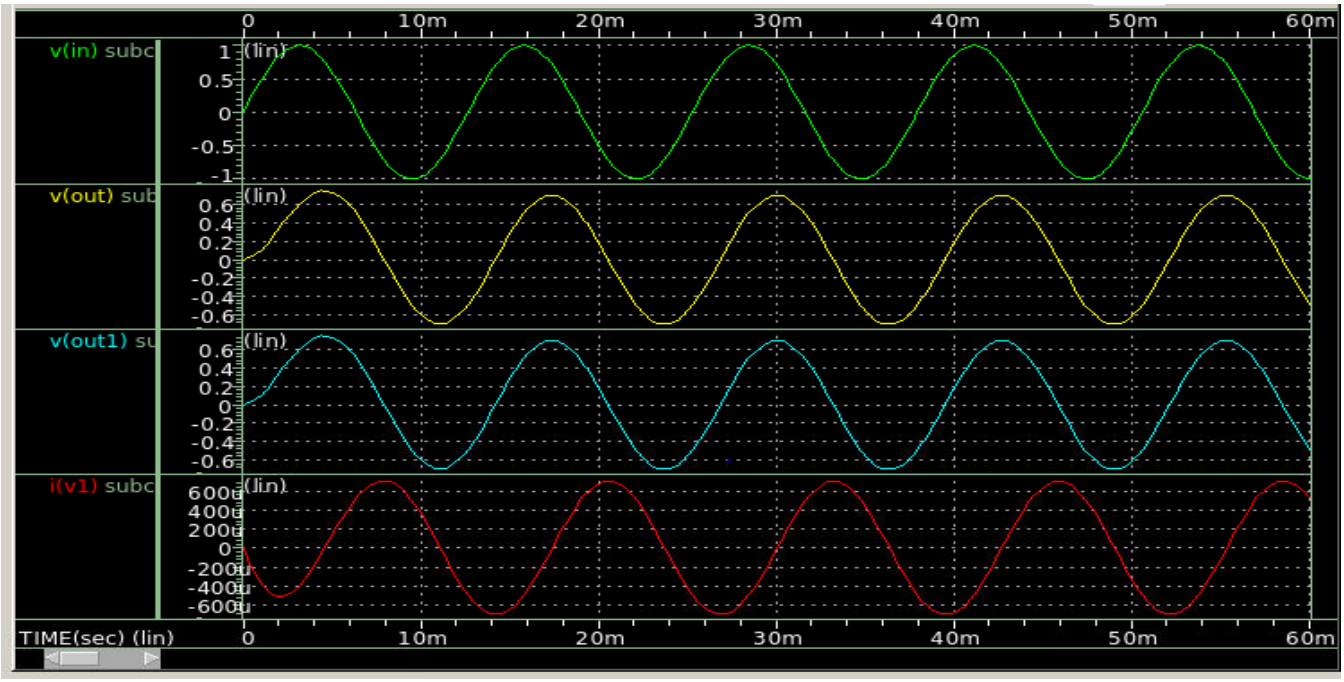
******************
.temp 25
**********************
X1 IN OUT rc
X2 OUT1 OUT rc

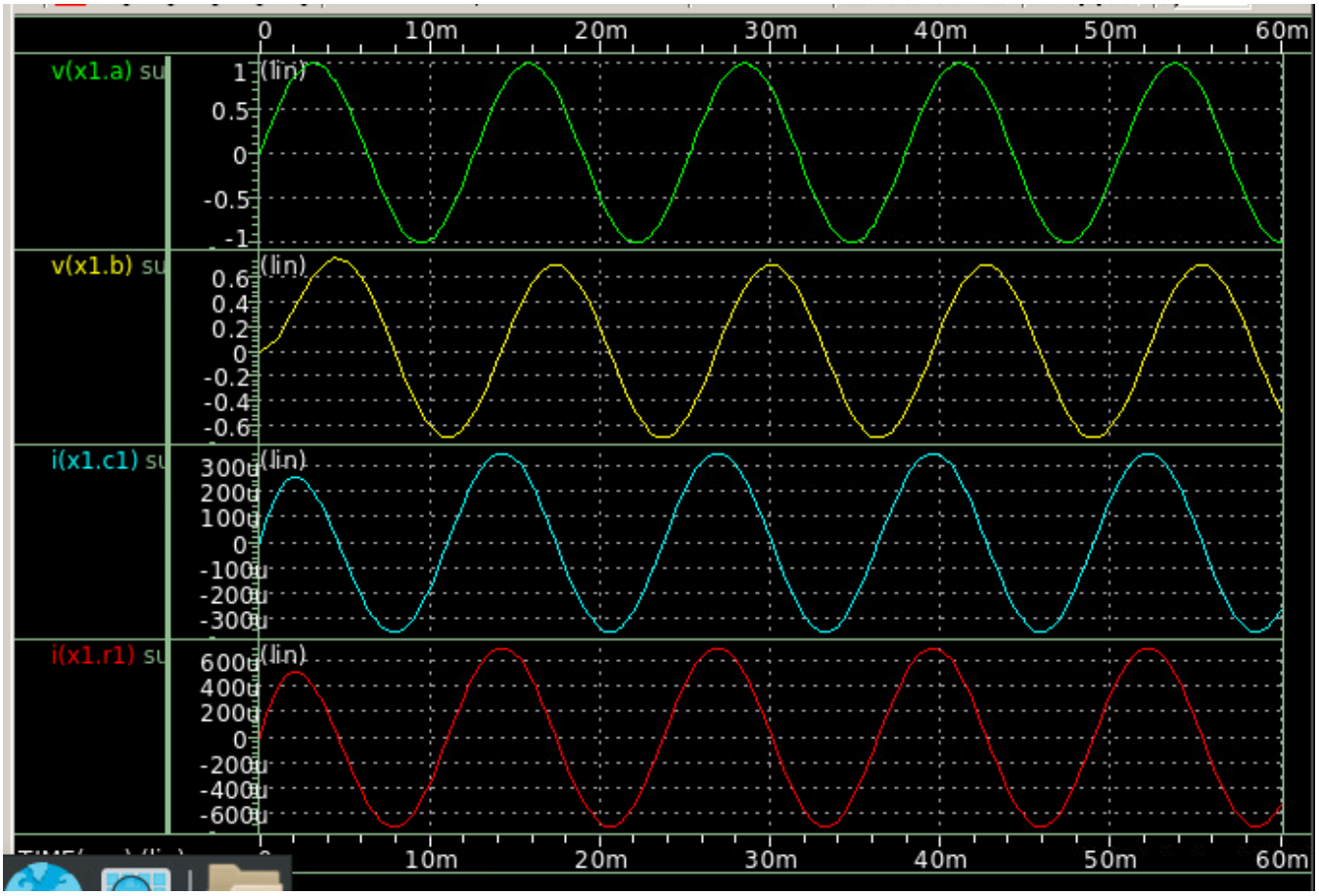
*************input stimuli*******
V1 IN 0 SIN(0 1 79 0 0 0)

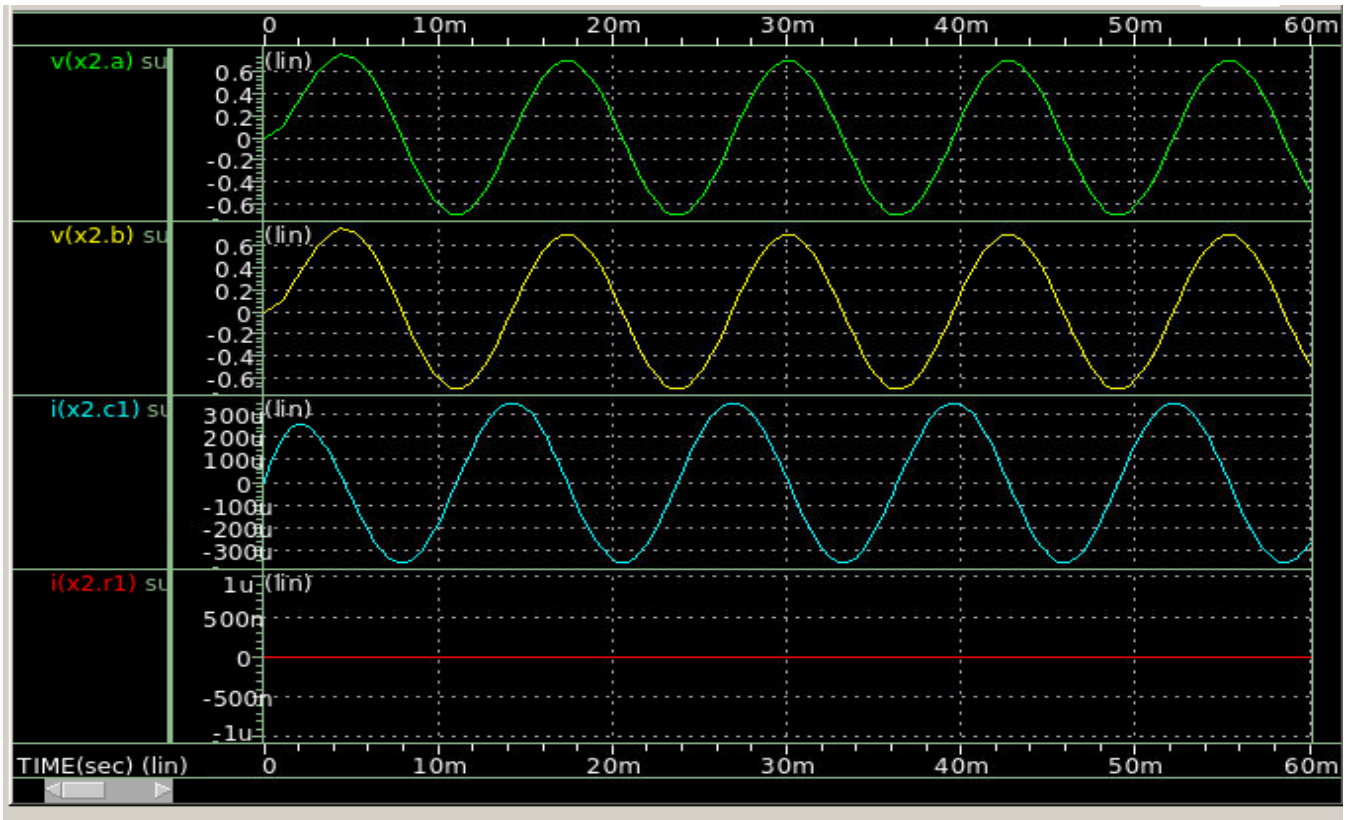
**************analysis option*******
.tran .01m 60m
.option probe=1
.print(X1.A)
.probe tran V(*) I(*)
.end
```

[b23n_sarif_727@prac3 experiment5]$ vi subckt.sp

[b23n_sarif_727@prac3 experiment5]$ wv subckt_wdf.tr0 &




0
10m
20m
30m
40m
50m
60m
v(x1.a) su
1
0.5
0
-0.5
-1
(lin)
v(x1.b) su
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
(lin)
i(x1.c1) su
300u
200u
100u
0
-100u
-200u
-300u
(lin)
i(x1.r1) su
600u
400u
200u
0
-200u
-400u
-600u
(lin)


0
10m
20m
30m
40m
50m
60m
v(x2.a) su
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
(lin)
v(x2.b) su
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
(lin)
i(x2.c1) su
300u
200u
100u
0
-100u
-200u
-300u
(lin)
i(x2.r1) su
1u
500n
0
-500n
-1u
(lin)
TIME(sec) (lin)

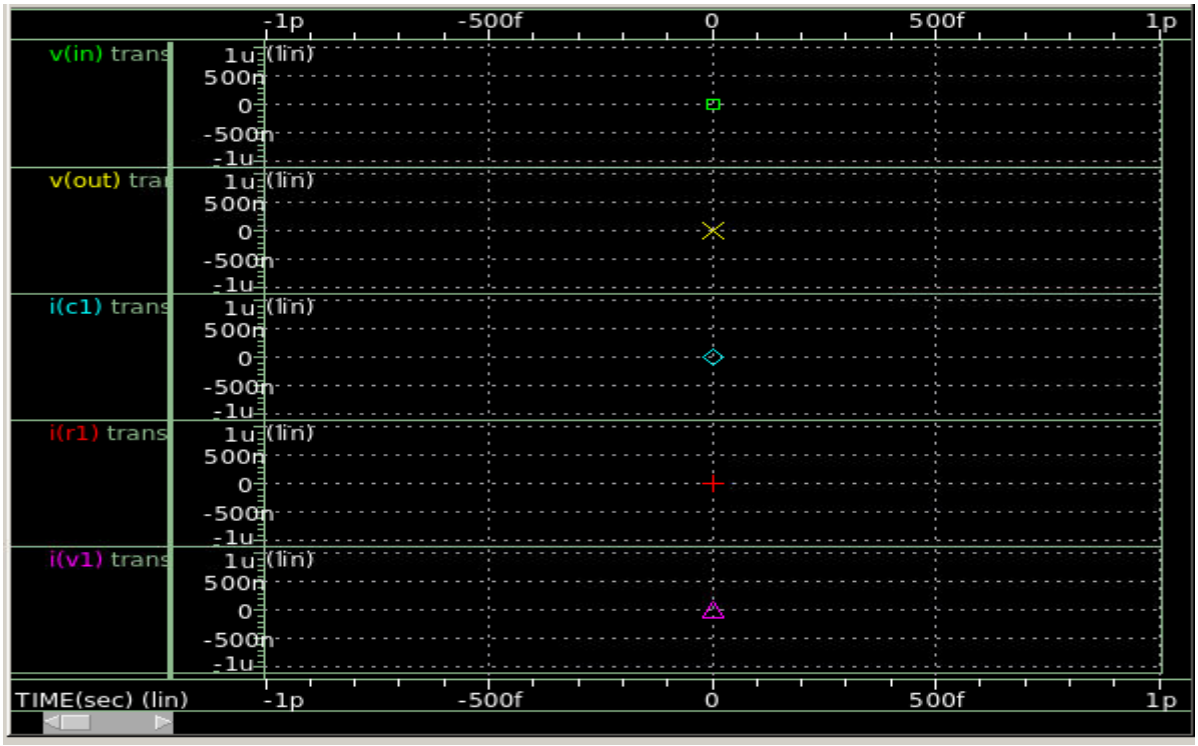
```
*transient analysis of rc circuit

.option WDF=1
.temp 25

R1 IN OUT 1K
C1 OUT 0 1U
V1 IN 0 PULSE(0 5 0 0.0001N 0.0001N 5M 10M)
.PROBE I(*) V(*)
.TRAN 0.1M 40M
.END
~
```

[b23n_sarif_727@prac3 experiment5]$ vi transient_rc_analysis.sp

[b23n_sarif_727@prac3 experiment5]$ wv transient_rc_analysis_wdf.tr0 &



Experiment6

[b23n_sarif_727@prac3 experiment6]$ vi netlist1.sp

[b23n_sarif_727@prac3 experiment6]$ vi netlist1.mt0

[b23n_sarif_727@prac3 experiment6]$ wv netlist1_wdf.tr0 &

```
* netlist1 characteristics without load capcatnace

.option search='/data/pdk/pdk_cc/hspice'

.lib 'reference40_models.lib' FF
.option WDF=1

M1 OUT IN VDD VDD P25 L=100n W=.12u
M2 OUT IN 0 0 N25 L=100n W=.1u
VDD VDD 0 2.5

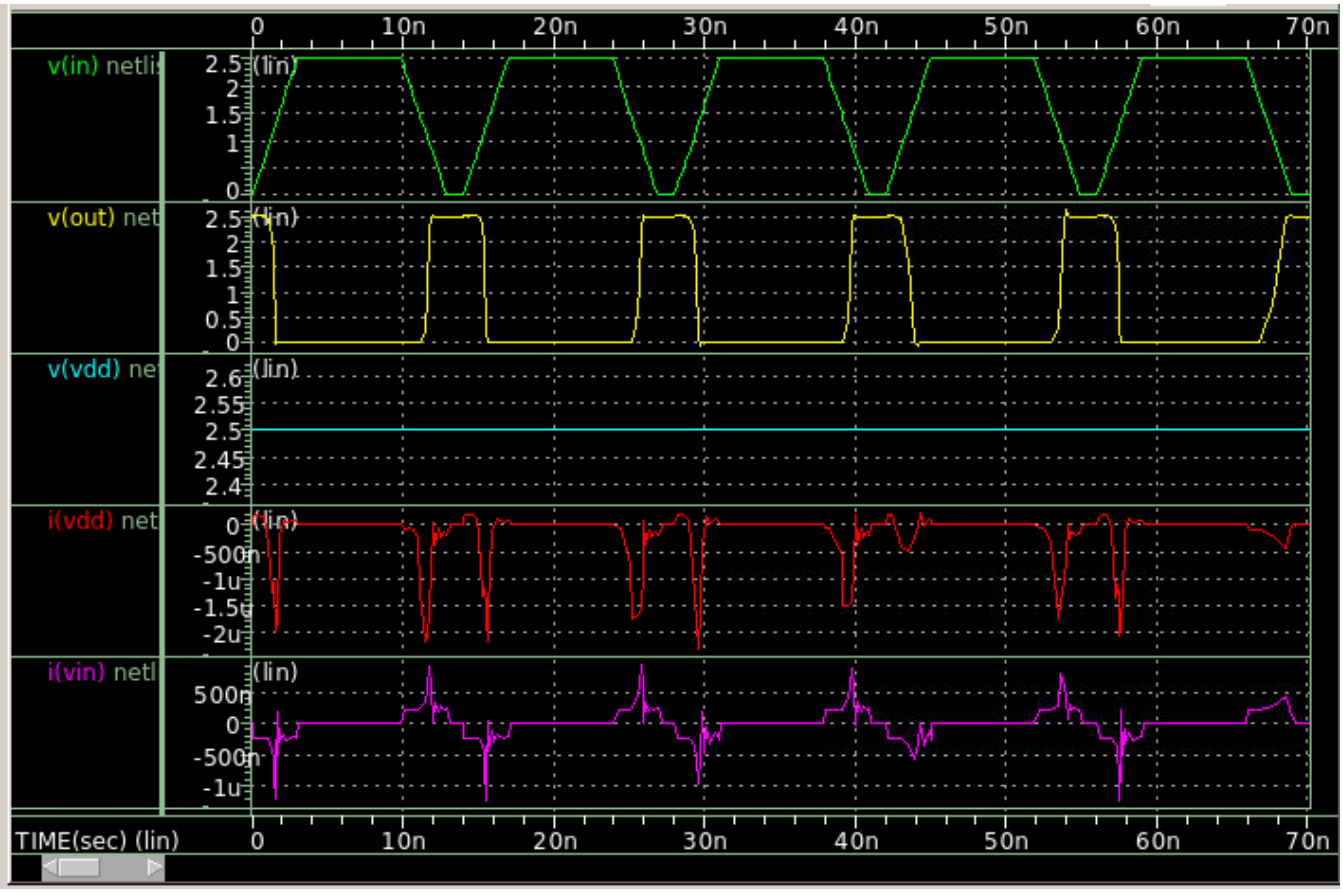
VIN IN 0 PULSE(0 2.5 0n 3n 3n 6.9n 14n)

.TRAN 100p 70n

.MEASURE TRAN TPHL TRIG V(IN) VAL=1.25 RISE=3 TARG V(OUT) VAL=1.25 FALL=3
.MEASURE TRAN TPLH TRIG V(IN) VAL=1.25 FALL=3 TARG V(OUT) VAL=1.25 RISE=3

.PRINT tran V(IN) V(OUT)
.PROBE tran V(IN) V(OUT)
.END
```

```
$DATA1 SOURCE='PrimeSim HSPICE' VERSION='V-2023.12-SP2 linux64' PARAM_COUNT=0
.TITLE '* netlist1 characteristics without load capcatnace'
 tphl             tplh             temper           alter#
 -4.565e-11        2.837e-10         25.0000         1
~
```



[b23n_sarif_727@prac3 experiment6]$ vi netlist2.sp

[b23n_sarif_727@prac3 experiment6]$ vi netlist2.mt0

[b23n_sarif_727@prac3 experiment6]$ wv netlist2_wdf.tr0 &

```
** netlist by using load capactor obtain the result
.option search='/data/pdk/pdk_cc/hspice'

.lib 'reference40_models.lib' FF
.option WDF=1

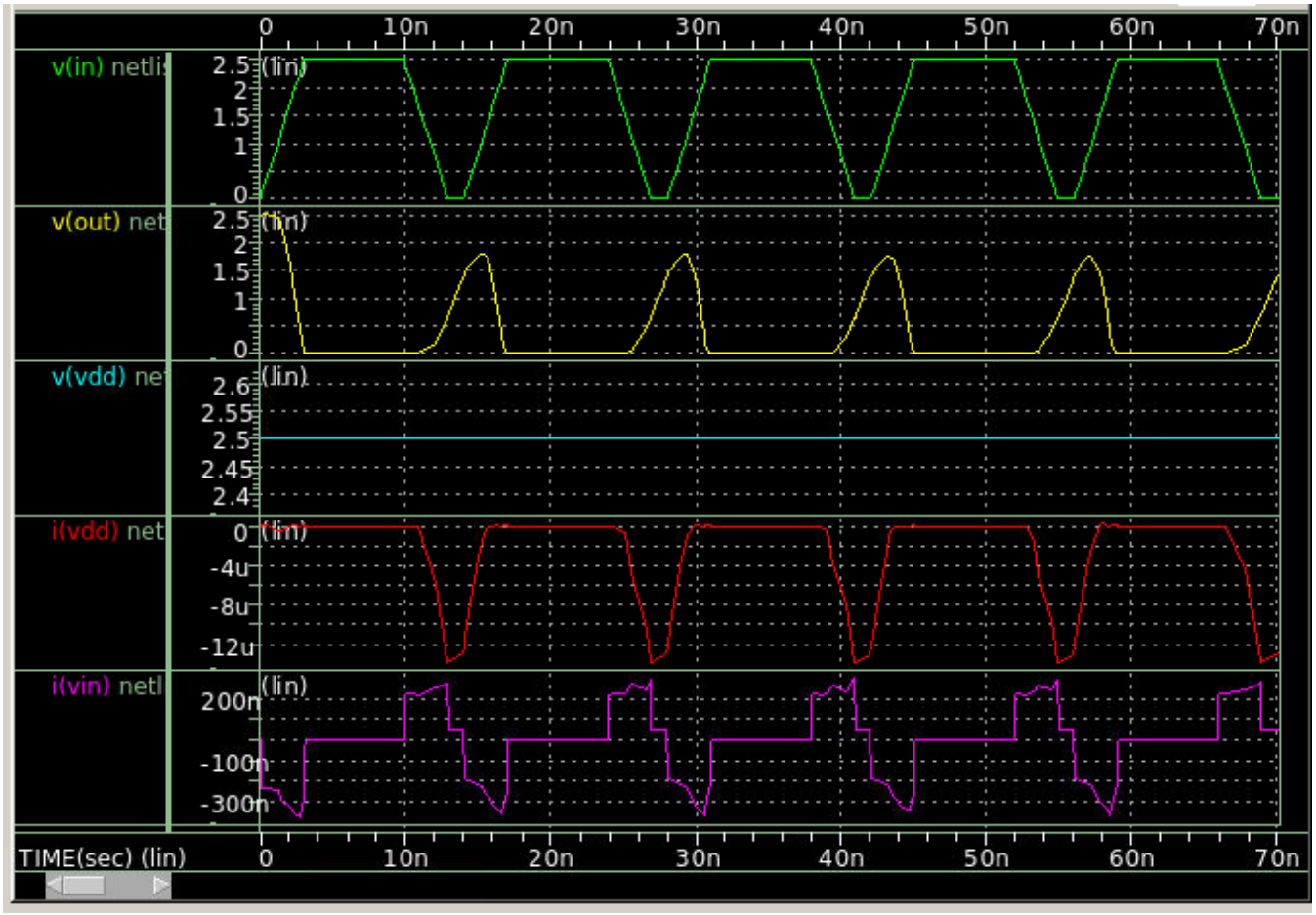
M1 OUT IN VDD VDD P25 L=100n W=.12u
M2 OUT IN 0 0 N25 L=100n W=.1u
VDD VDD 0 2.5
CL OUT 0 20f

VIN IN 0 PULSE(0 2.5 0n 3n 3n 6.9n 14n)
.TRAN 100p 70n

.MEASURE TRAN TPHL TRIG V(IN) VAL=1.25 RISE=3 TARG V(OUT) VAL=1.25 FALL=3
.MEASURE TRAN TPLH TRIG V(IN) VAL=1.25 FALL=3 TARG V(OUT) VAL=1.25 RISE=3

.PRINT tran V(IN) V(OUT)
.PROBE tran V(IN) V(OUT)
.END
```

```
$DATA1 SOURCE='PrimeSim HSPICE' VERSION='V-2023.12-SP2 linux64' PARAM_COUNT=0
.TITLE '** netlist by using load capactor obtain the result'
 tphl             tplh             temper           alter#
  5.604e-10        2.434e-09          25.0000        1
~
```



[b23n_sarif_727@prac3 experiment6]$ vi process_corner.sp

[b23n_sarif_727@prac3 experiment6]$ wv process_corner_wdf.sw0 &

[b23n_sarif_727@prac3 experiment6]$ wv process_corner_wdf.sw1 &

[b23n_sarif_727@prac3 experiment6]$ wv process_corner_wdf.sw2 &

```
* analyse the process corner analysis- SS FF TT

.option search=/data/pdk/pdk_cc/hspice

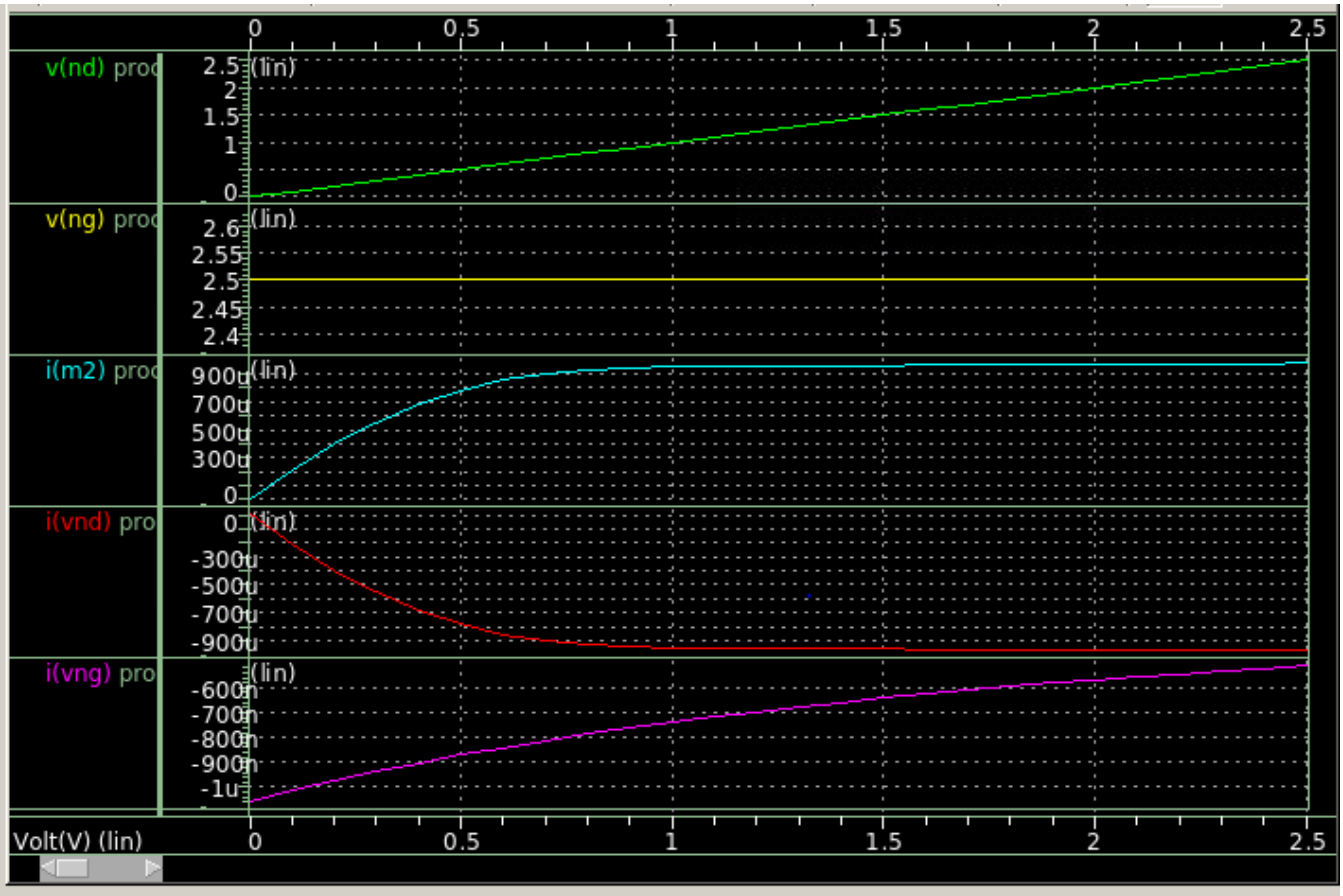
.lib 'reference40_models.lib' TT

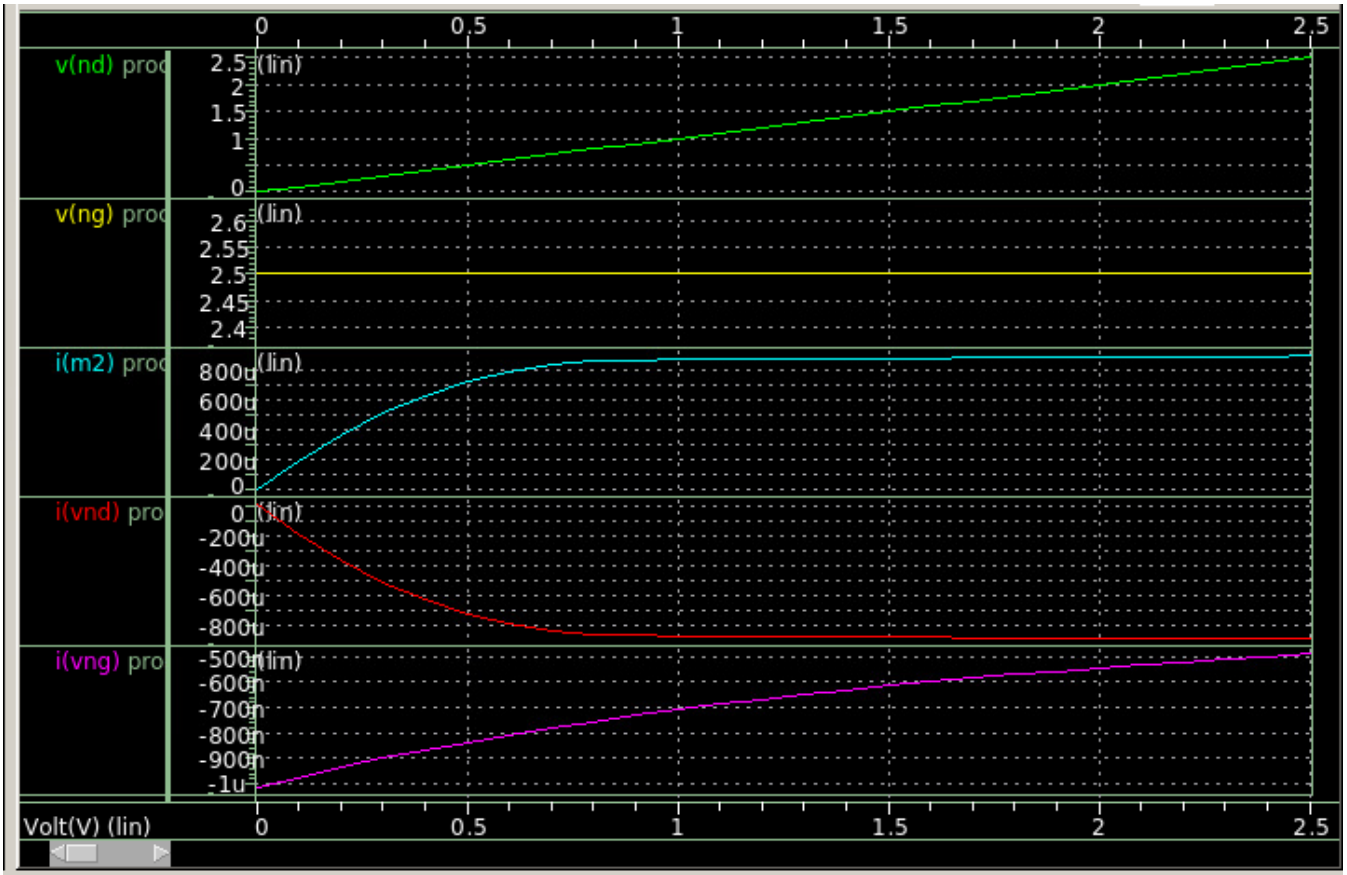
.option WDF=1
*.model  N25  nmos  level = 54
M2 ND NG 0 0 N25 W=10u L=1u
VND ND 0 2.5
VNG NG 0 2.5

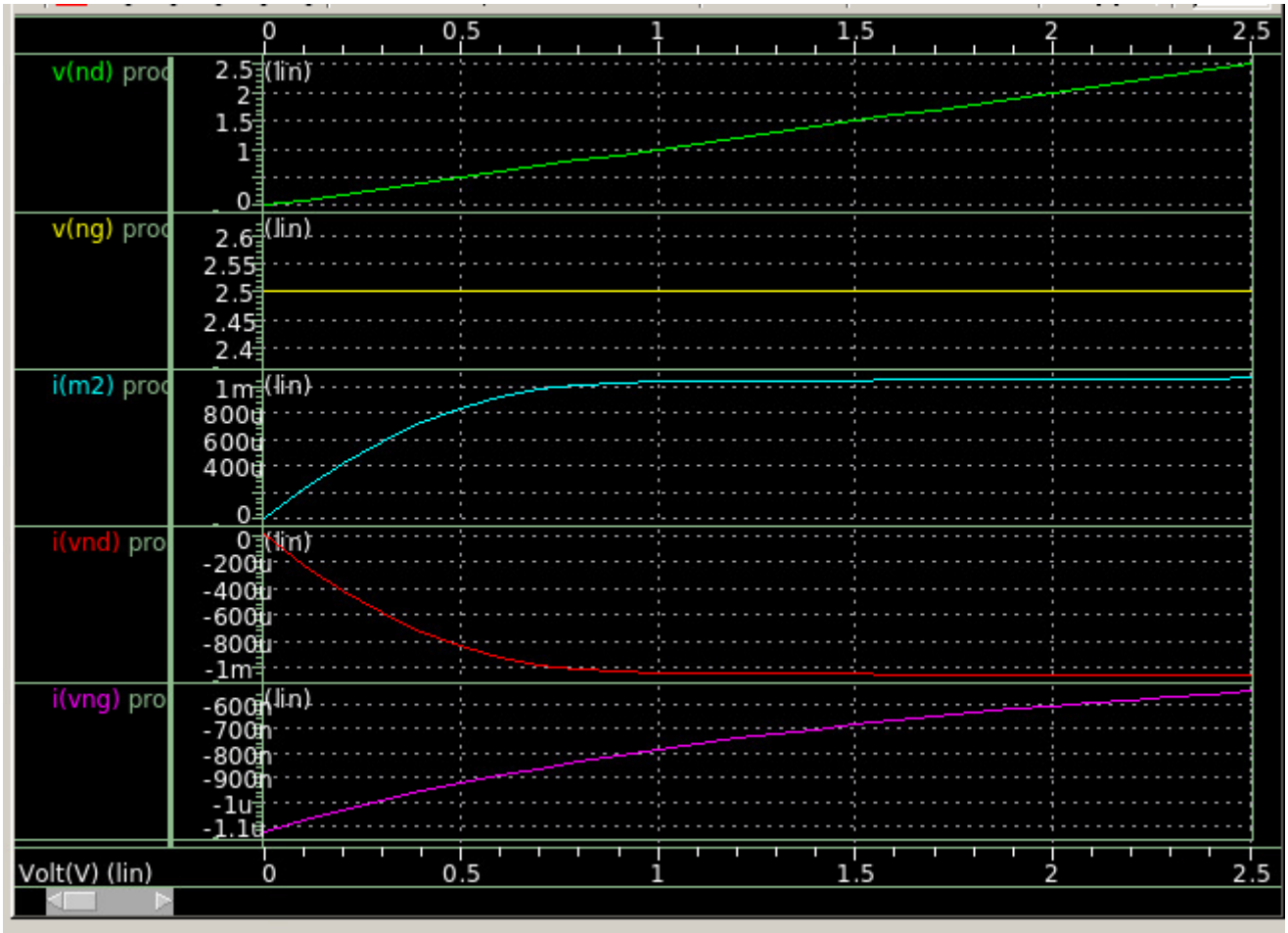
.DC VND 0 2.5 0.1

.probe DC I(M2)
.print DC I(M2)
.alter
.lib 'reference40_models.lib' SS
.alter
.lib 'reference40_models.lib' FF

.end
```




v(nd) proc
v(ng) proc
i(m2) proc
i(vnd) pro
i(vng) pro
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) proc
v(ng) proc
i(m2) proc
i(vnd) pro
i(vng) pro
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5

```
* pvt analysis of nmos N25

.option search=/data/pdk/pdk_cc/hspice

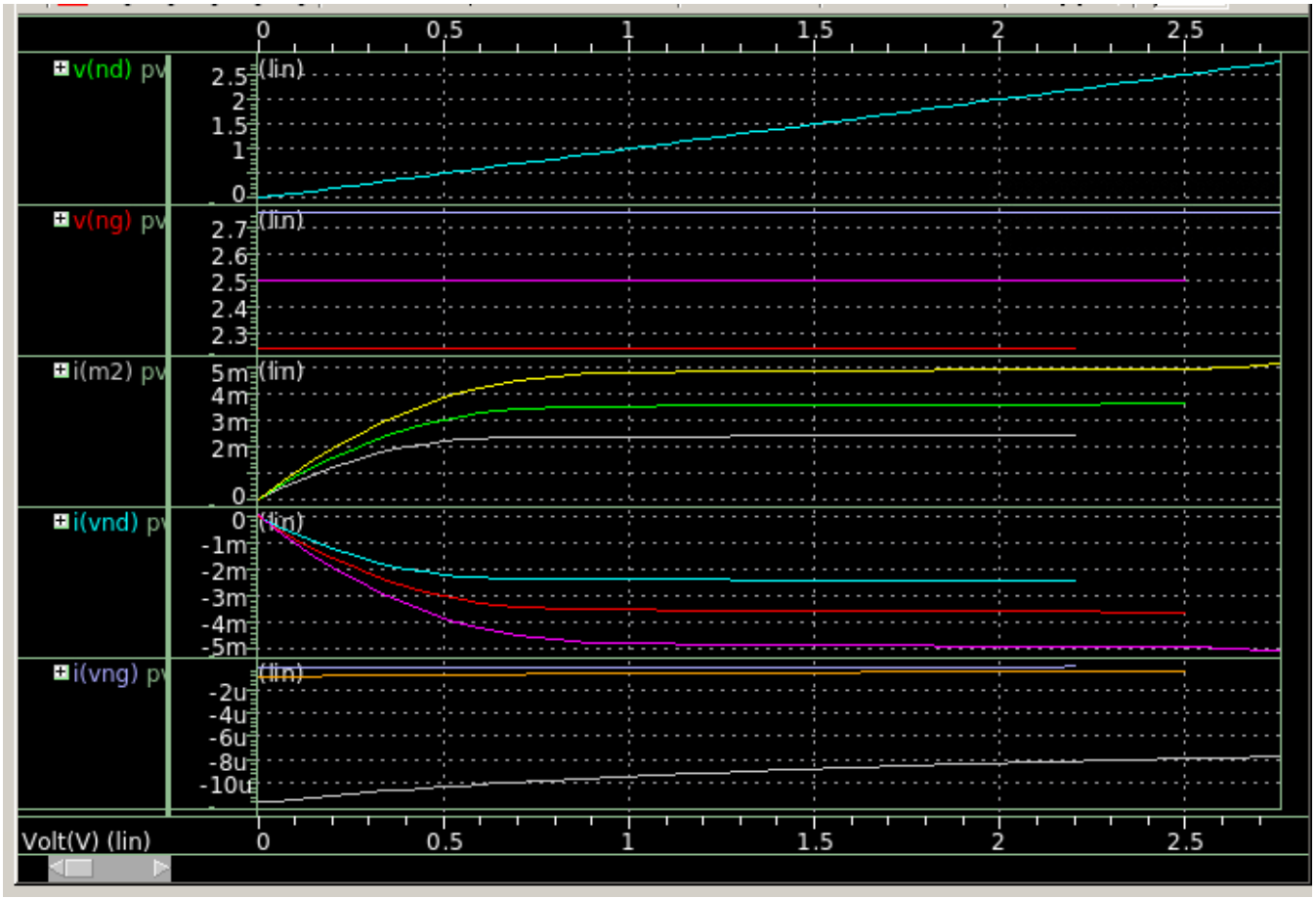
.lib 'reference40_models.lib' TT
.temp -25 25 125
.param VDD=2.5V

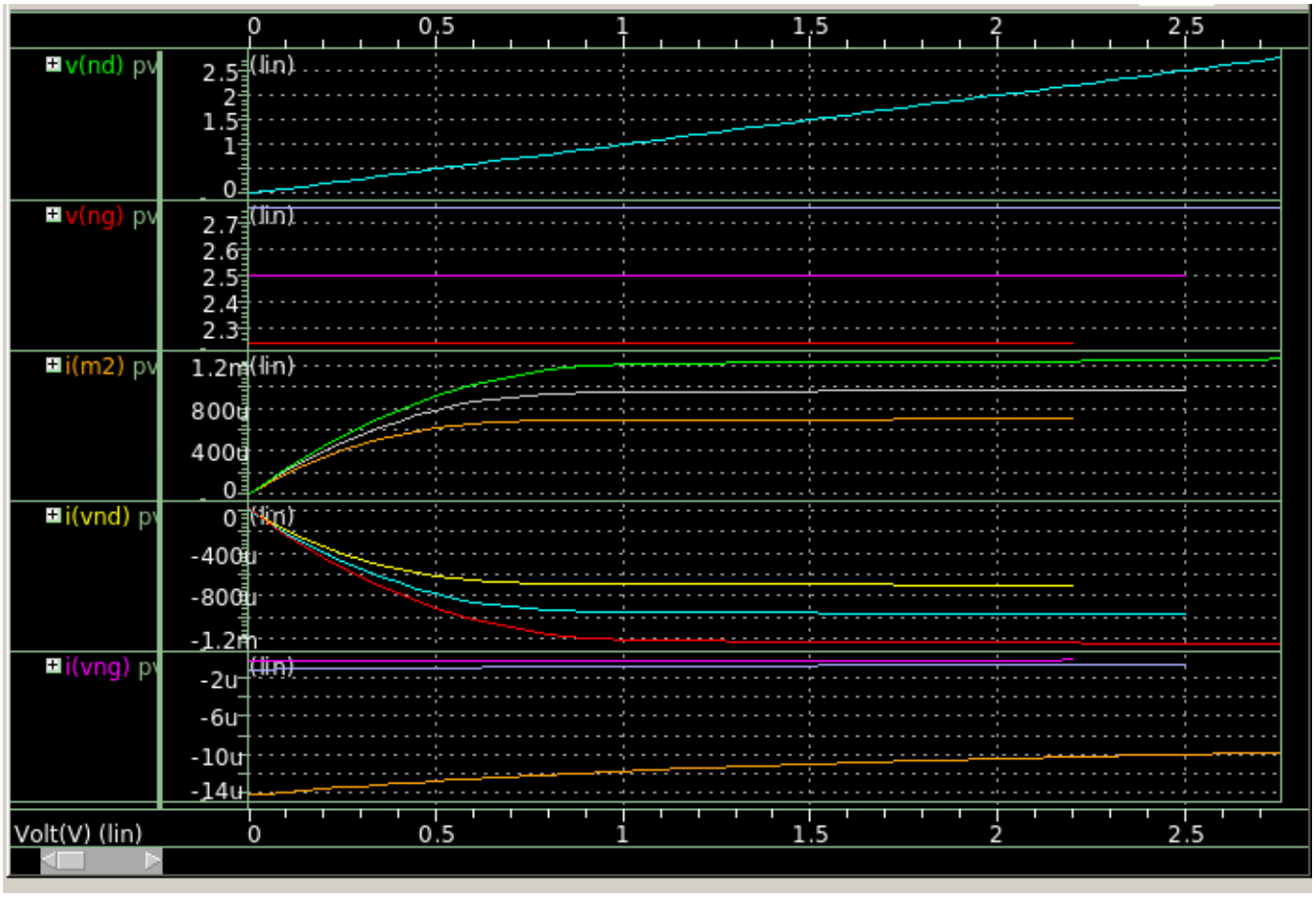
.option WDF=1
*.model  N25  nmos  level = 54
M2 ND NG 0 0 N25 W=10u L=1u
VND ND 0 0
VNG NG 0 VDD

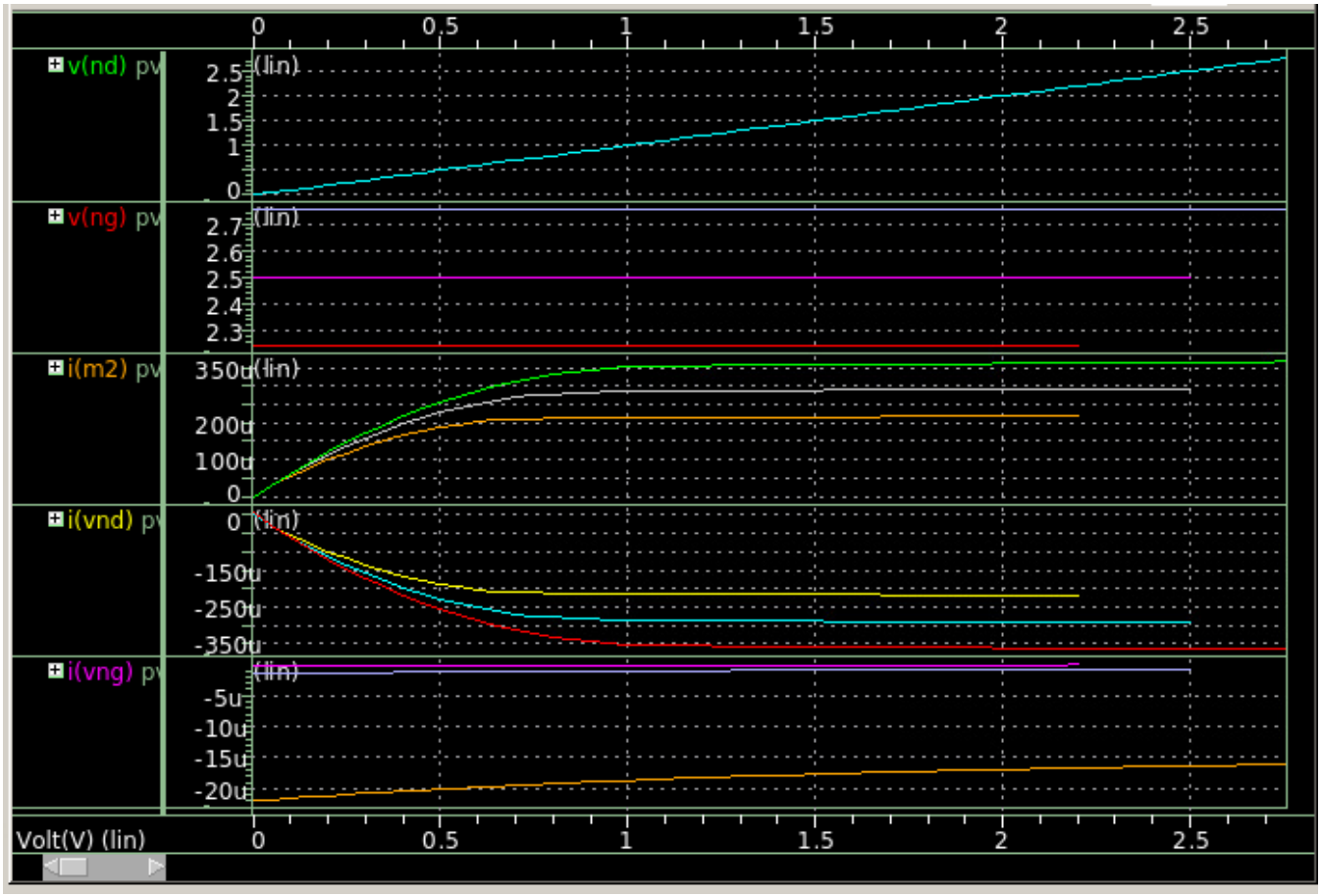
.DC VND 0 VDD 0.05 sweep DATA=supply_values
.DATA supply_values
VDD
2.24
2.5
2.75

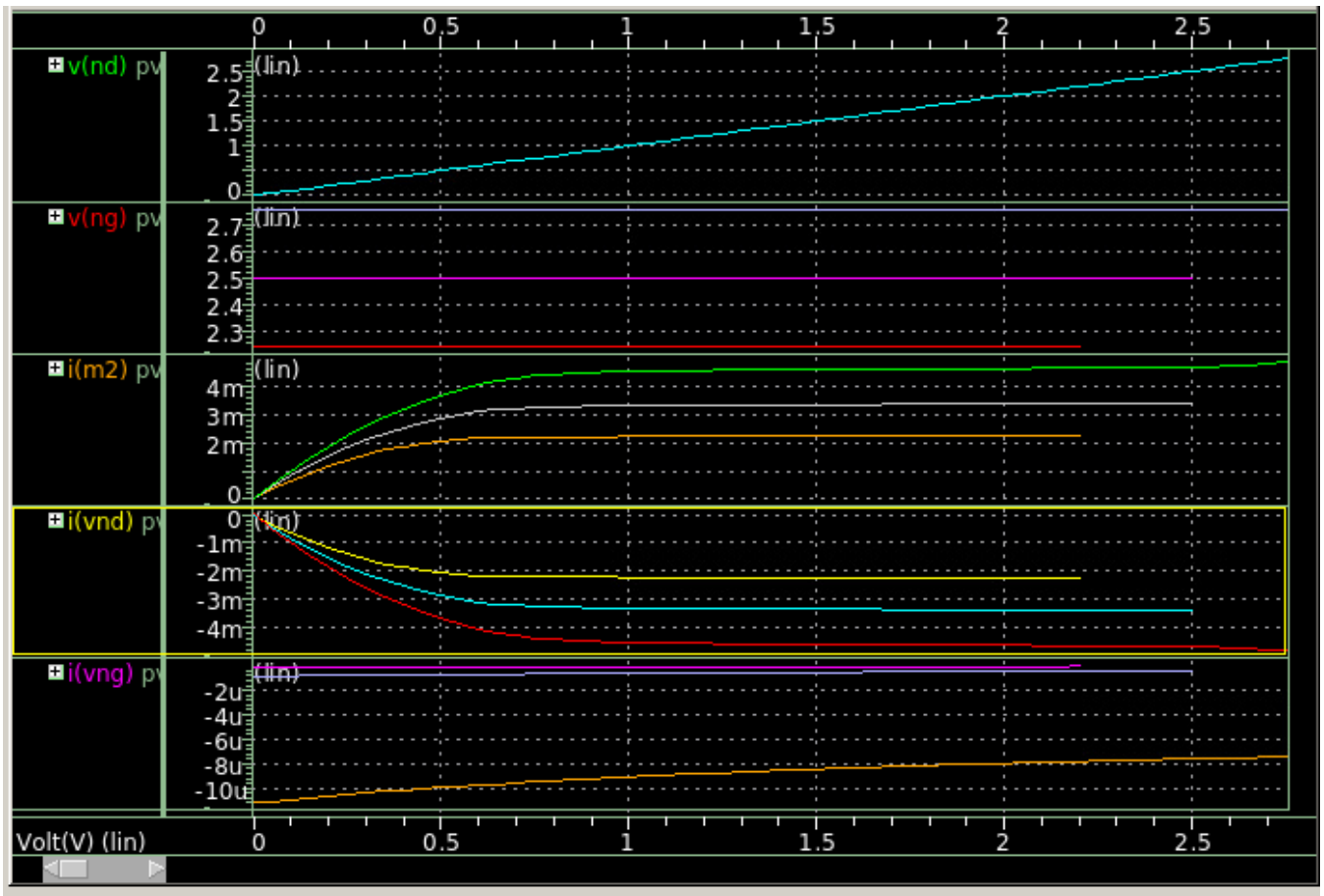
.probe DC I(M2)
.print DC I(M2)
.ENDDATA

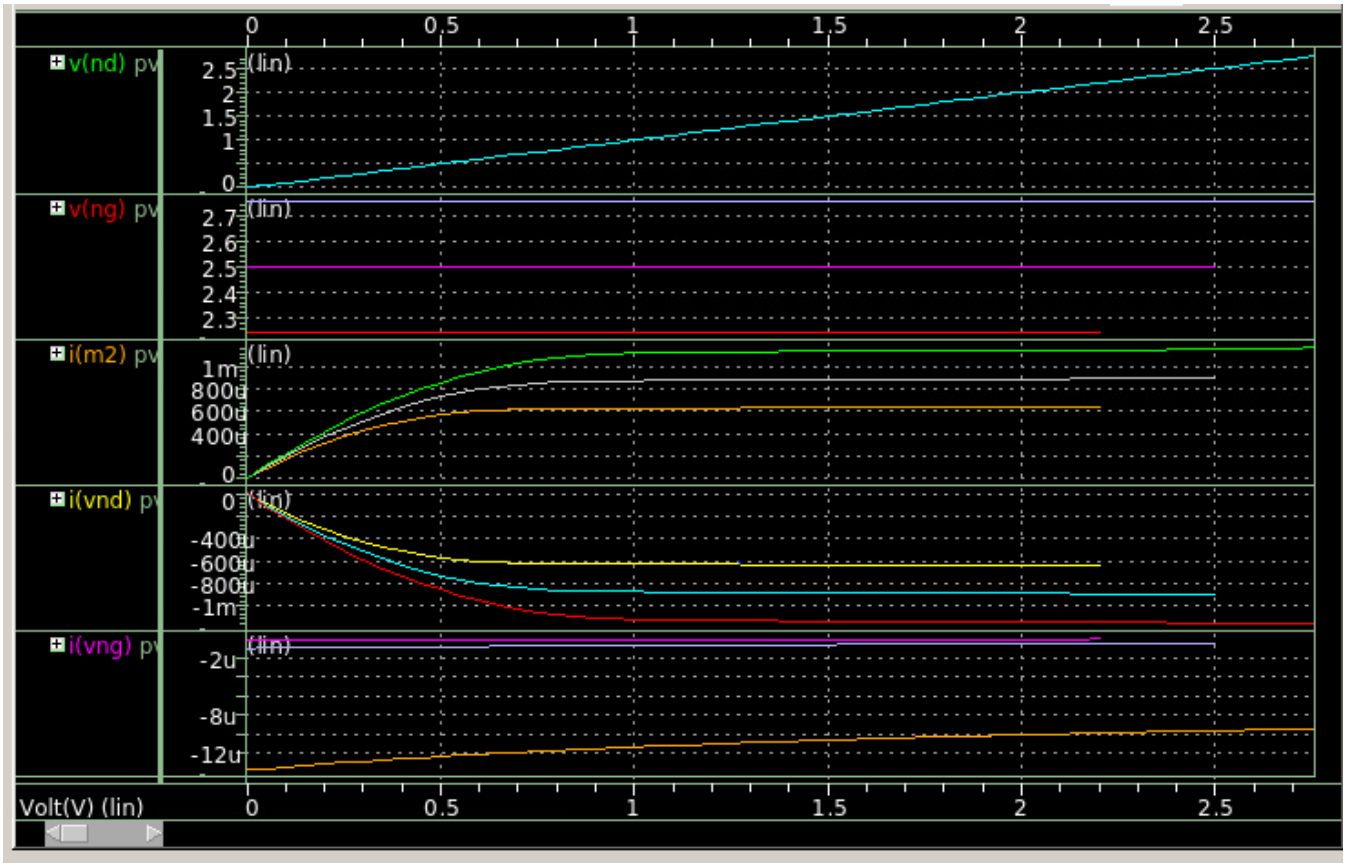
.alter
.lib 'reference40_models.lib' SS
.alter
.lib 'reference40_models.lib' FF
.end
~
```

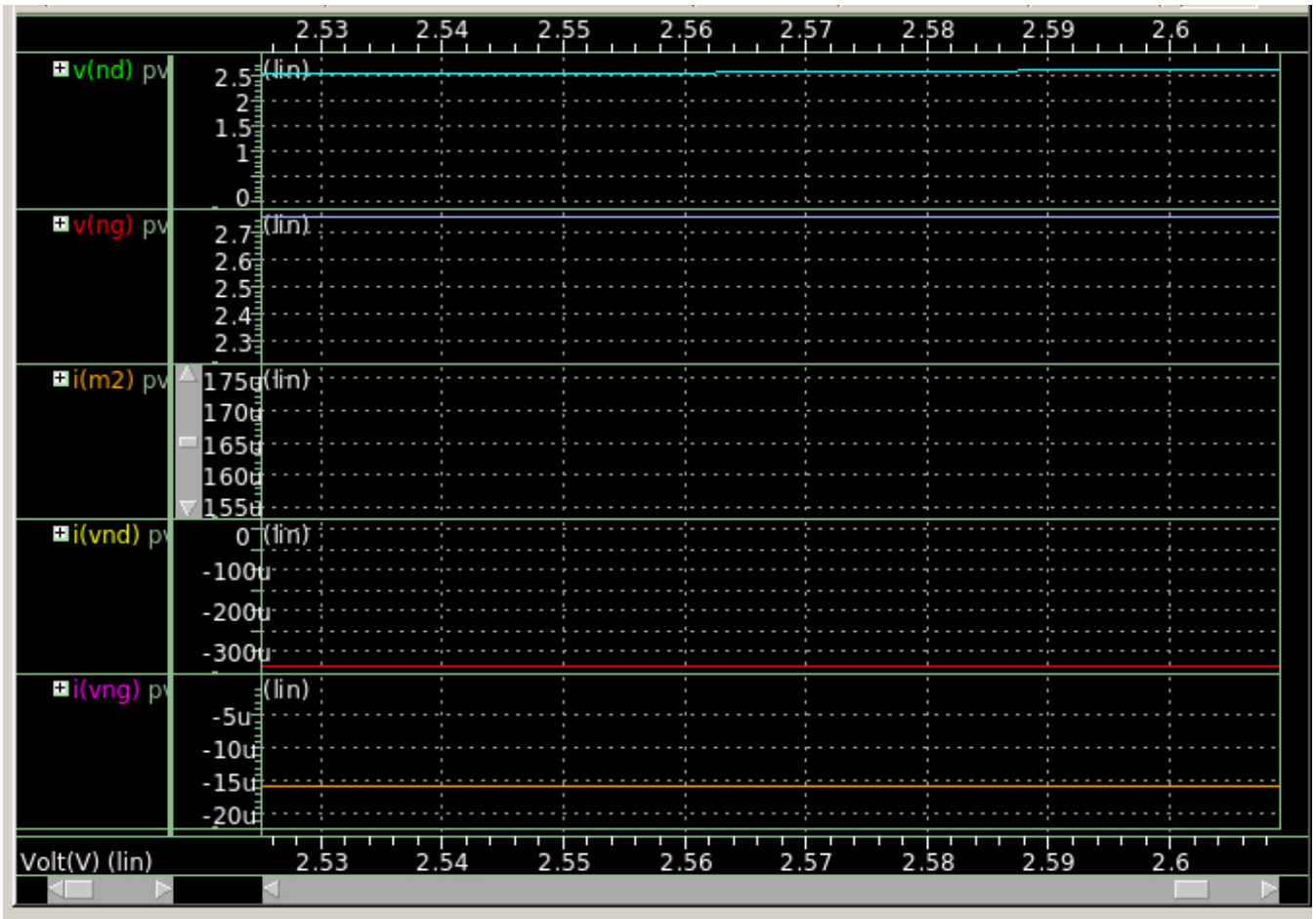

v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5

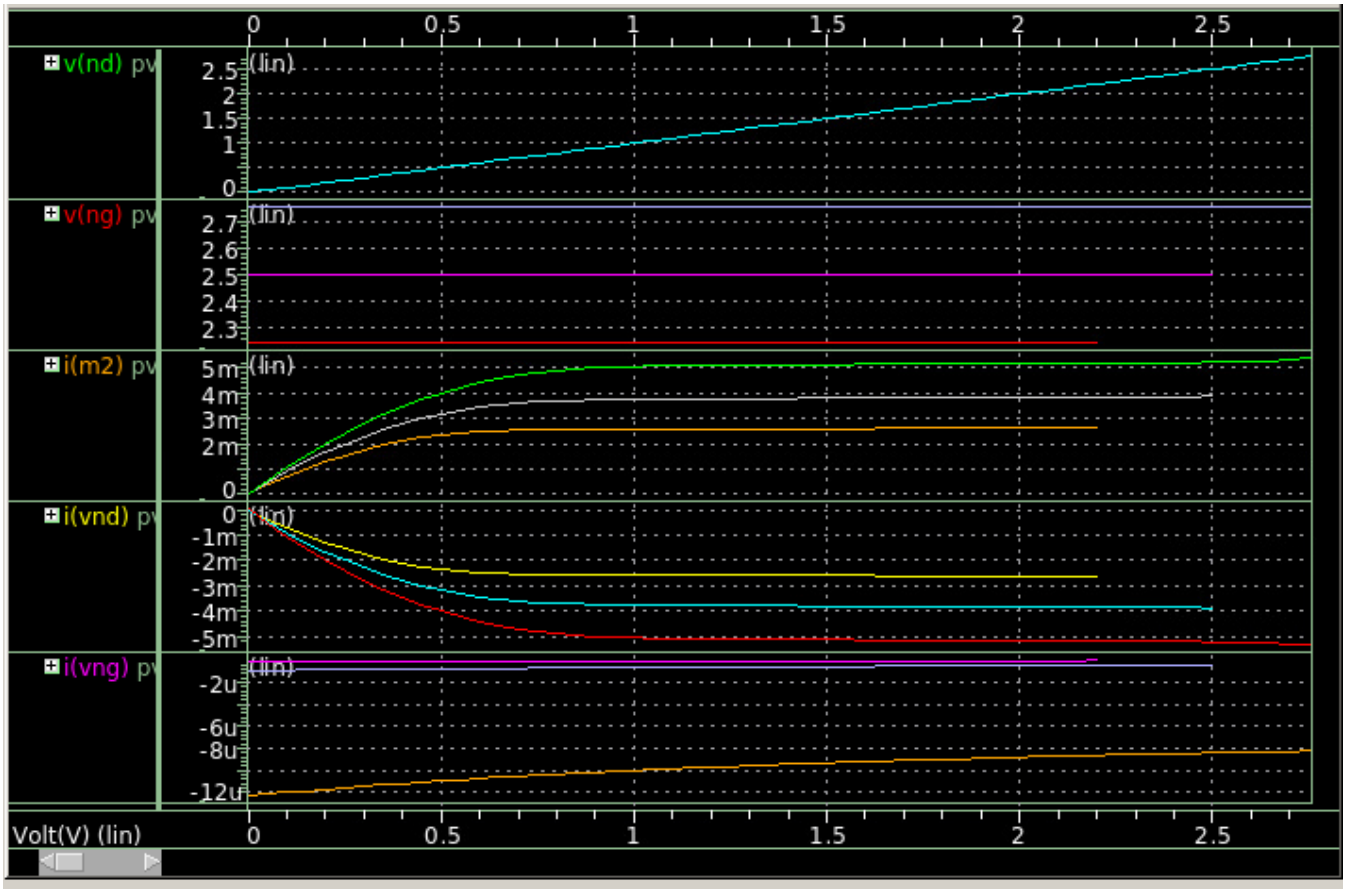

v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5

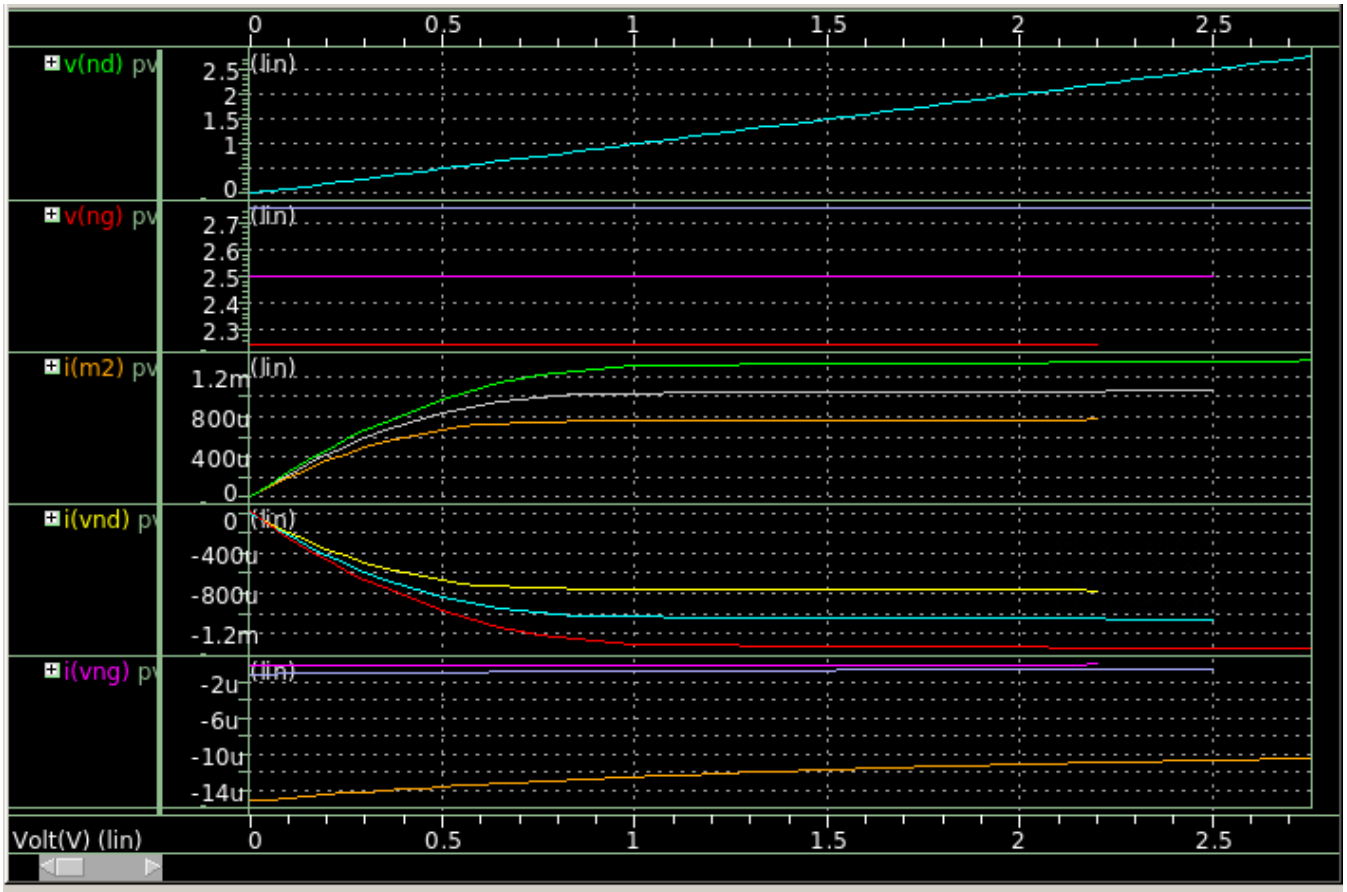

v(nd) pv
(lin)
2.5
2
1.5
1
0
v(ng) pv
(lin)
2.7
2.6
2.5
2.4
2.3
i(m2) pv
(lin)
350u
200u
100u
0
i(vnd) pv
(lin)
0
-150u
-250u
-350u
i(vng) pv
(lin)
-5u
-10u
-15u
-20u
Volt(V) (lin)
0
0.5
1
1.5
2
2.5

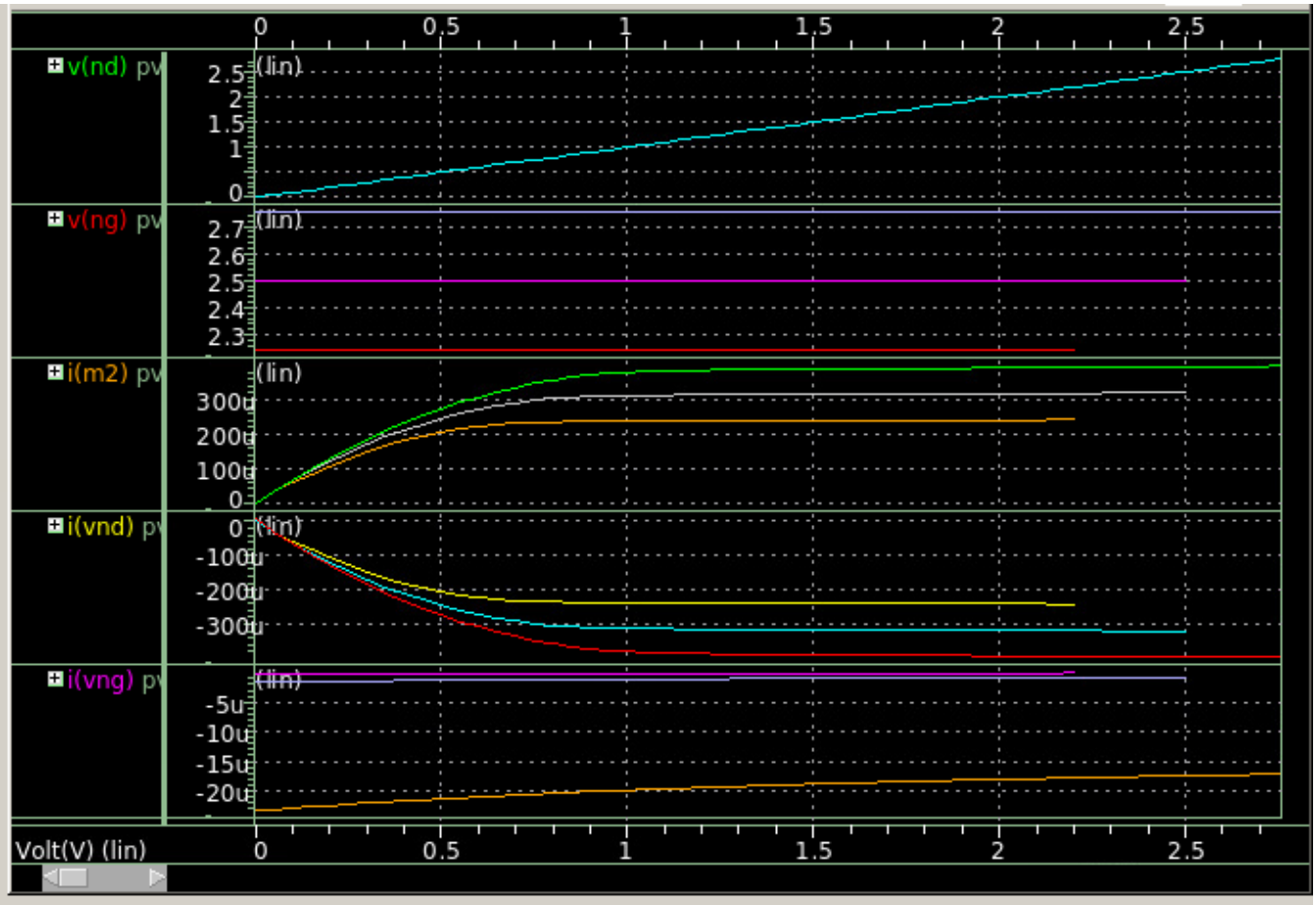

v(nd) pv
(lin)
2.5
2
1.5
1
0
v(ng) pv
(lin)
2.7
2.6
2.5
2.4
2.3
i(m2) pv
(lin)
4m
3m
2m
0
i(vnd) pv
(lin)
0
-1m
-2m
-3m
-4m
i(vng) pv
(lin)
-2u
-4u
-6u
-8u
-10u
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
0
0,5
1
1,5
2
2,5


v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
2.53
2.54
2.55
2.56
2.57
2.58
2.59
2.6


v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
0
0,5
1
1,5
2
2,5


v(nd) pv
v(ng) pv
i(m2) pv
i(vnd) pv
i(vng) pv
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5



[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw0 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw1 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw2 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw3 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw4 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw5 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw6 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw7 &

[b23n_sarif_727@prac3 experiment6]$ wv pvt_corner_nmos_wdf.sw8 &

[b23n_sarif_727@prac3 experiment6]$ vi temperature_corner.sp

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw0 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw1 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw2 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw3 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw4 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw5 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw6 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw7 &

[b23n_sarif_727@prac3 experiment6]$ wv temperature_corner_wdf.sw8 &

```
* analyse the process corner analysis- SS FF TT

.option search=/data/pdk/pdk_cc/hspice

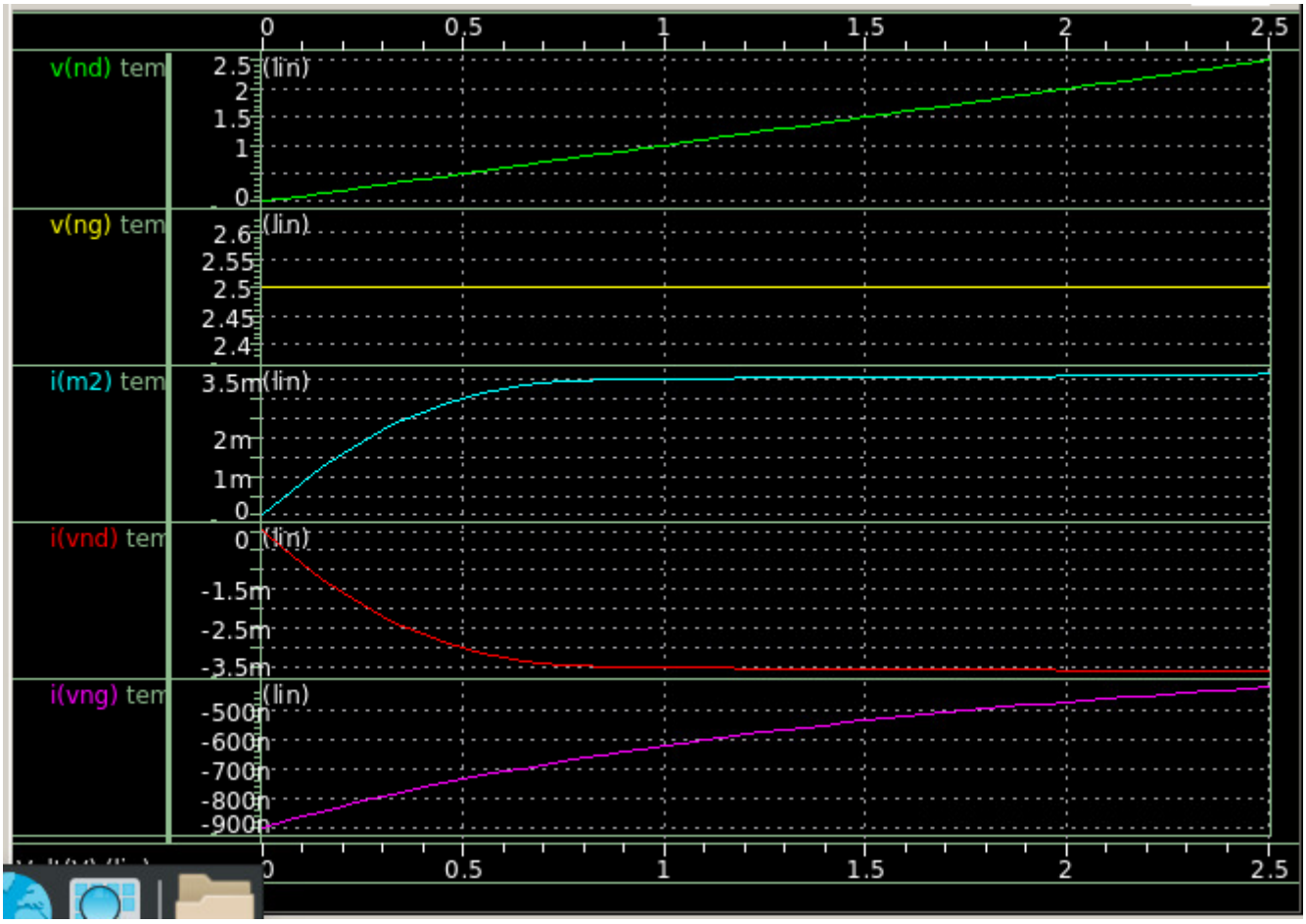
.lib 'reference40_models.lib' TT
.param VDD=2.5V

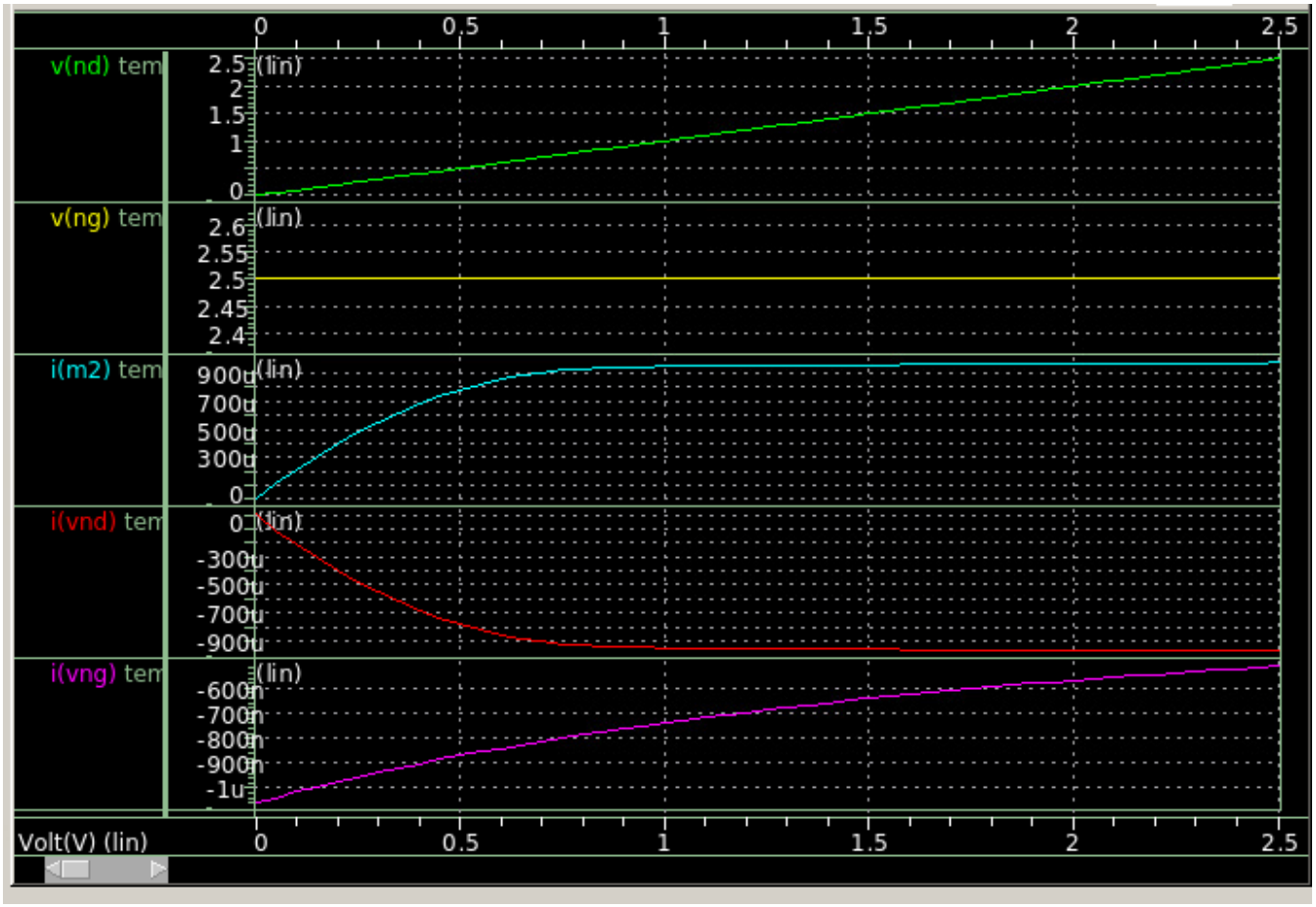
.option WDF=1
*.model  N25  nmos  level = 54
M2 ND NG 0 0 N25 W=10u L=1u
VND ND 0 2.5V
VNG NG 0 VDD

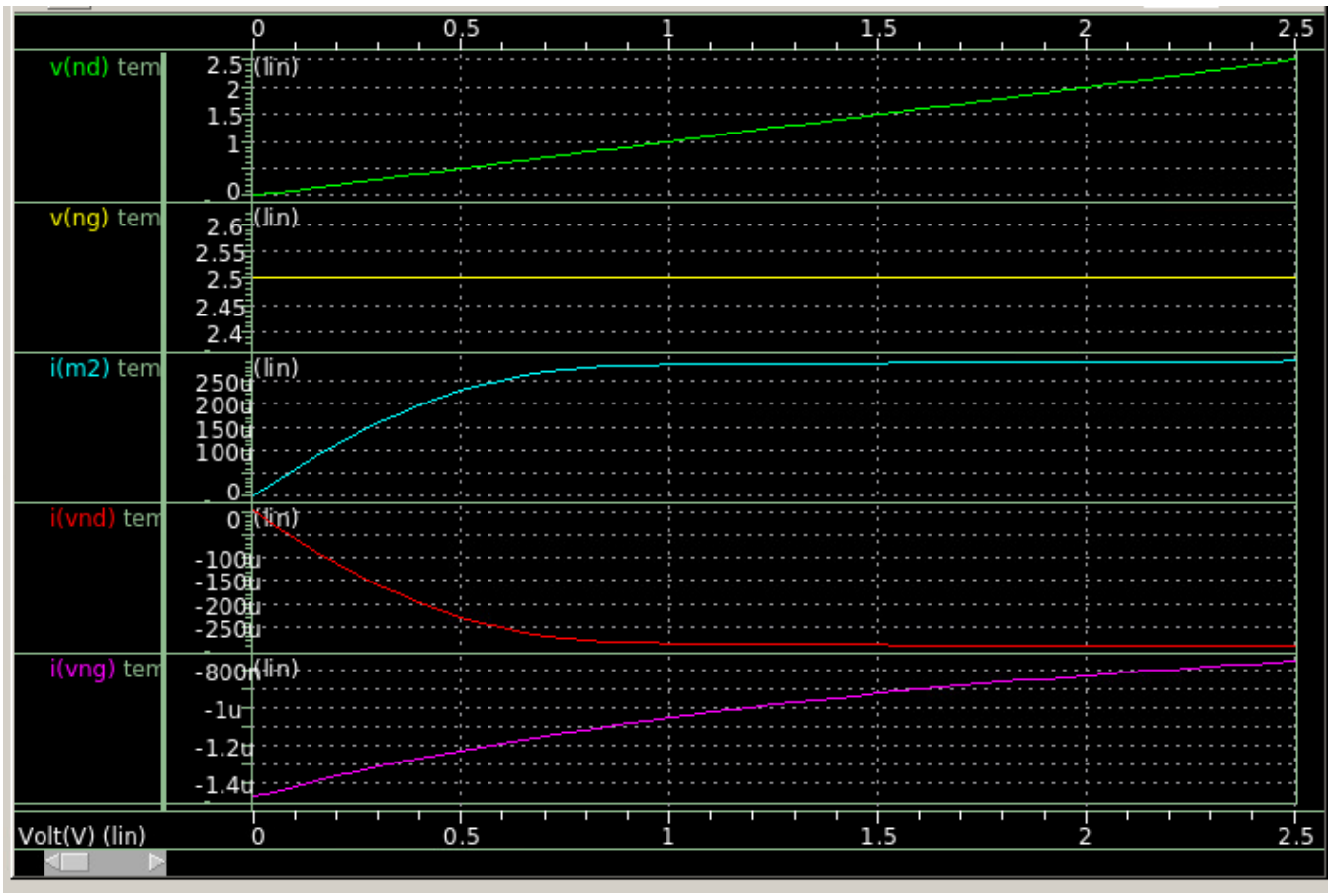
.temp -25 25 125

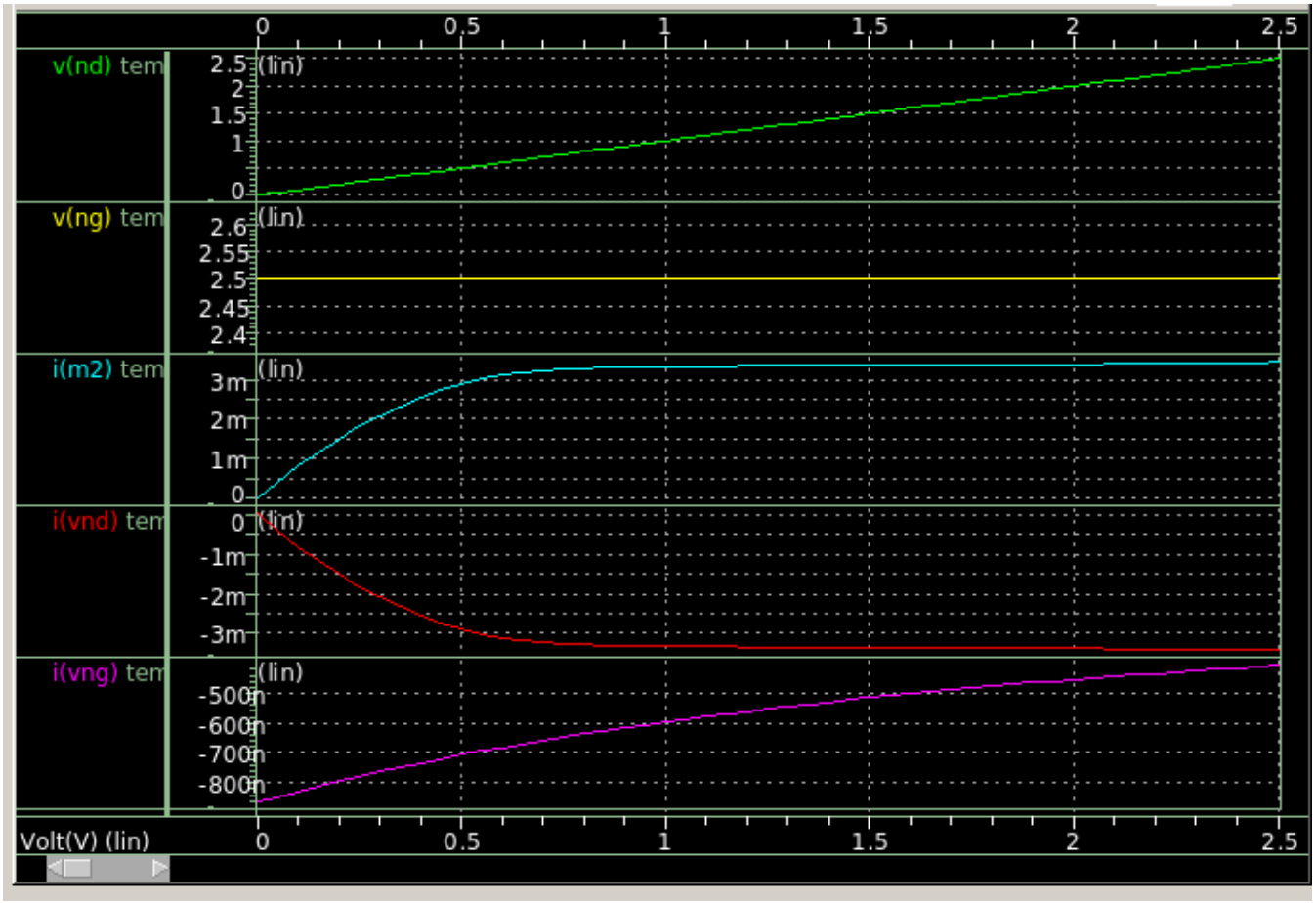
.DC VND 0 VDD 0.05
.probe DC I(M2)
.print DC I(M2)

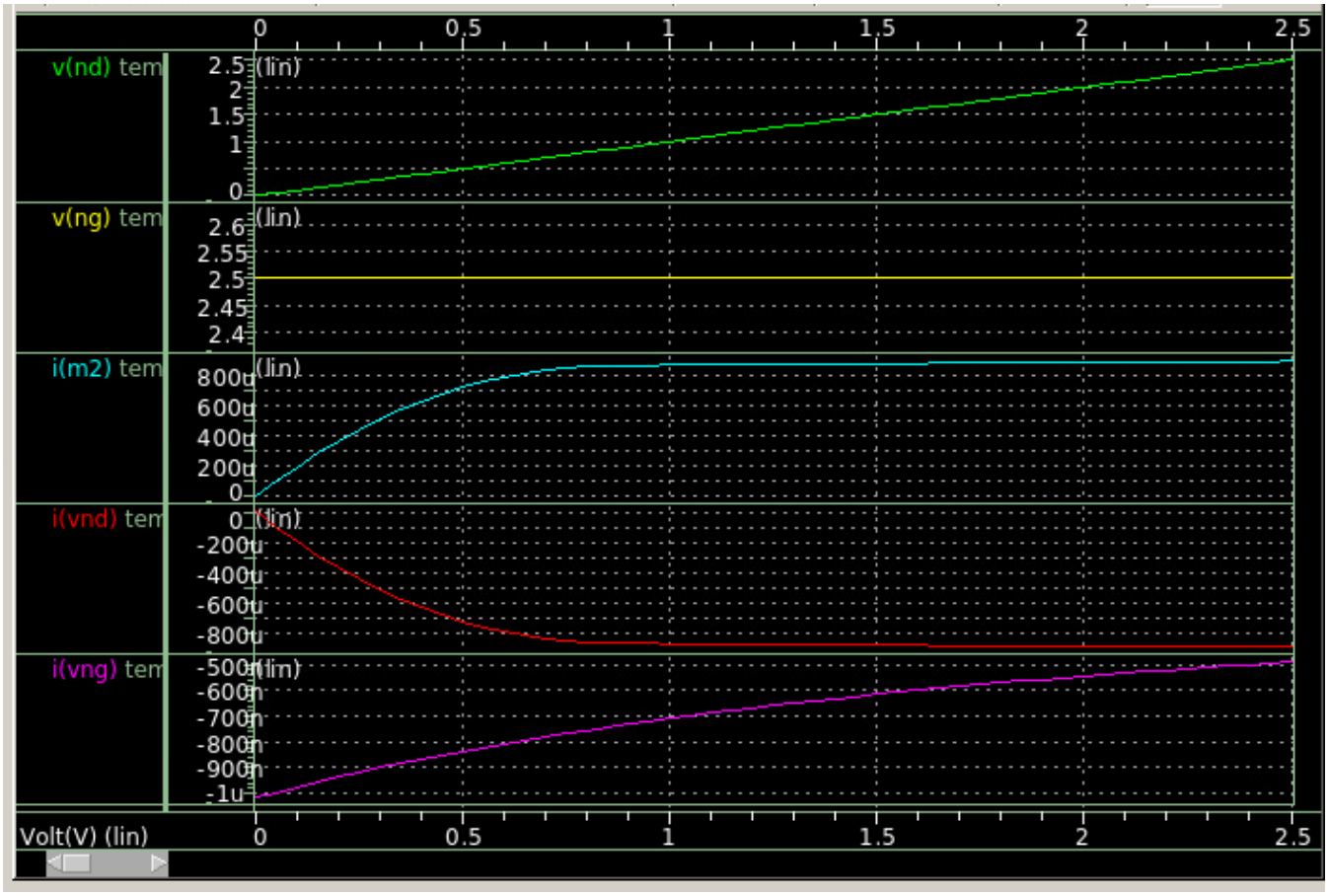
.alter
.lib 'reference40_models.lib' SS
.alter
.lib 'reference40_models.lib' FF
.end
```

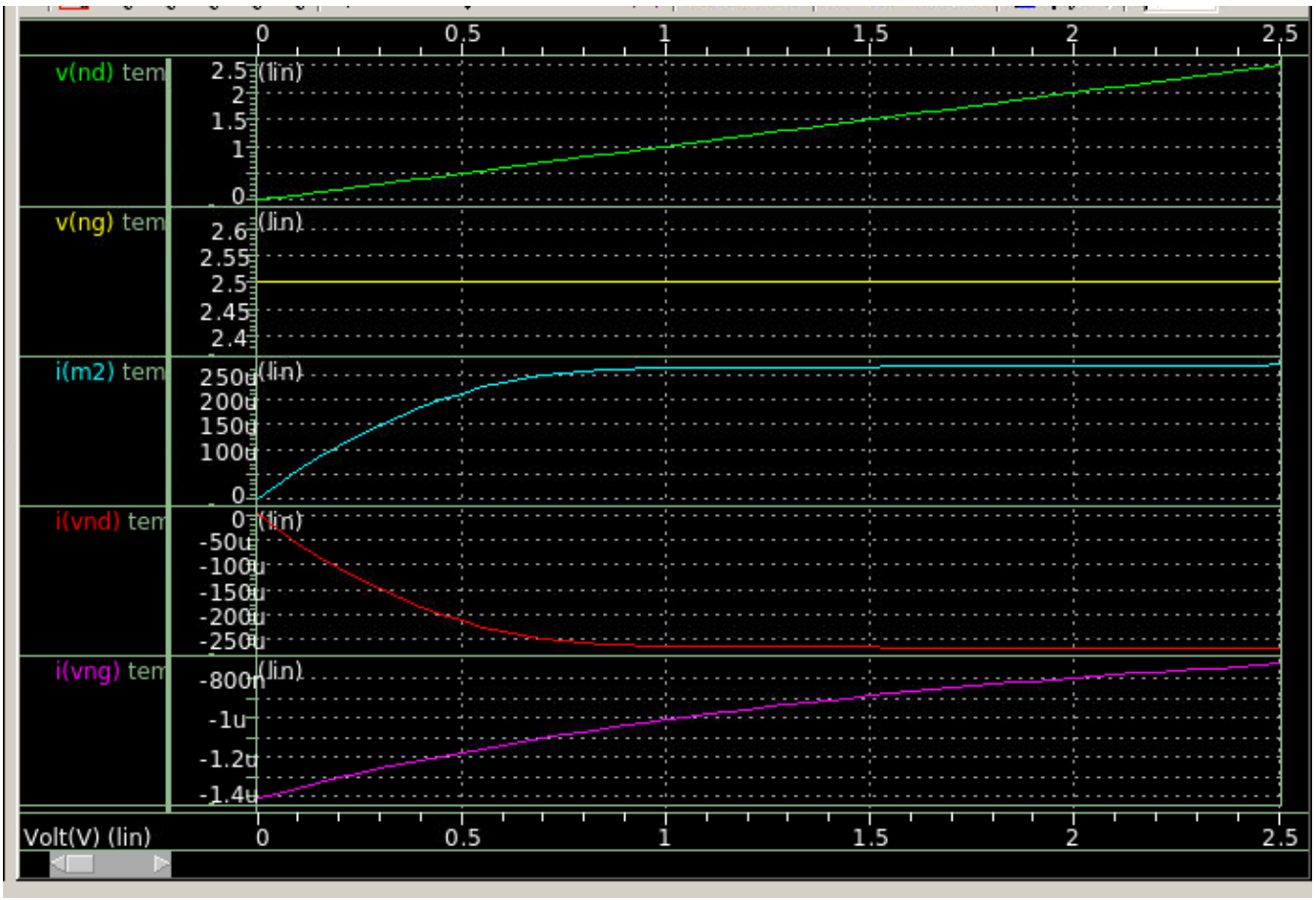

0
0.5
1
1.5
2
2.5
v(nd) tem
2.5
2
1.5
1
0
(lin)
v(ng) tem
2.6
2.55
2.5
2.45
2.4
(lin)
i(m2) tem
3.5m
2m
1m
0
(lin)
i(vnd) tem
0
-1.5m
-2.5m
-3.5m
(lin)
i(vng) tem
-500n
-600n
-700n
-800n
-900n
(lin)
0
0.5
1
1.5
2
2.5

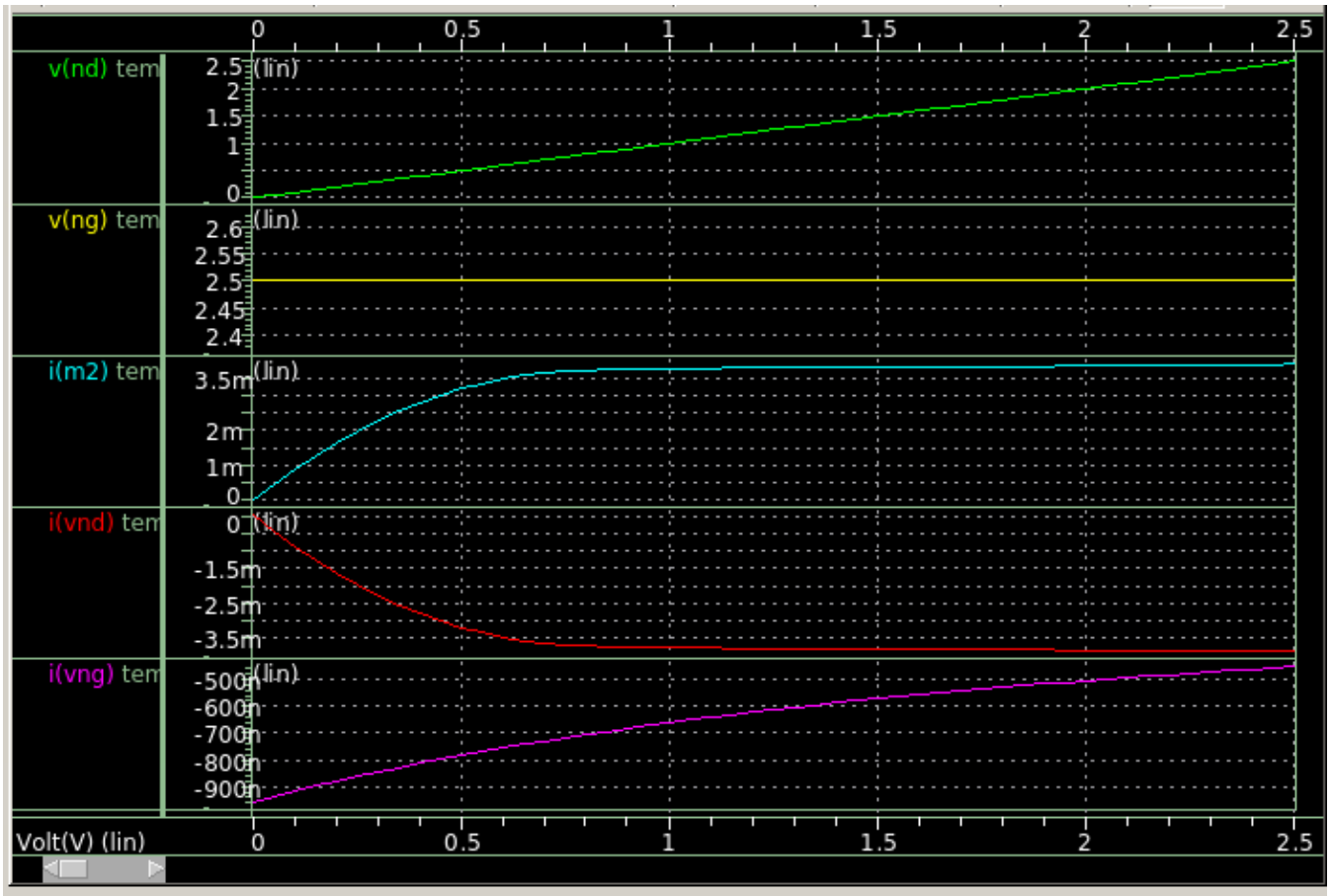

v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5
2.6
2.55
2.45
2.4
900u
700u
500u
300u
-300u
-500u
-700u
-900u
-600n
-700n
-800n
-900n
-1u

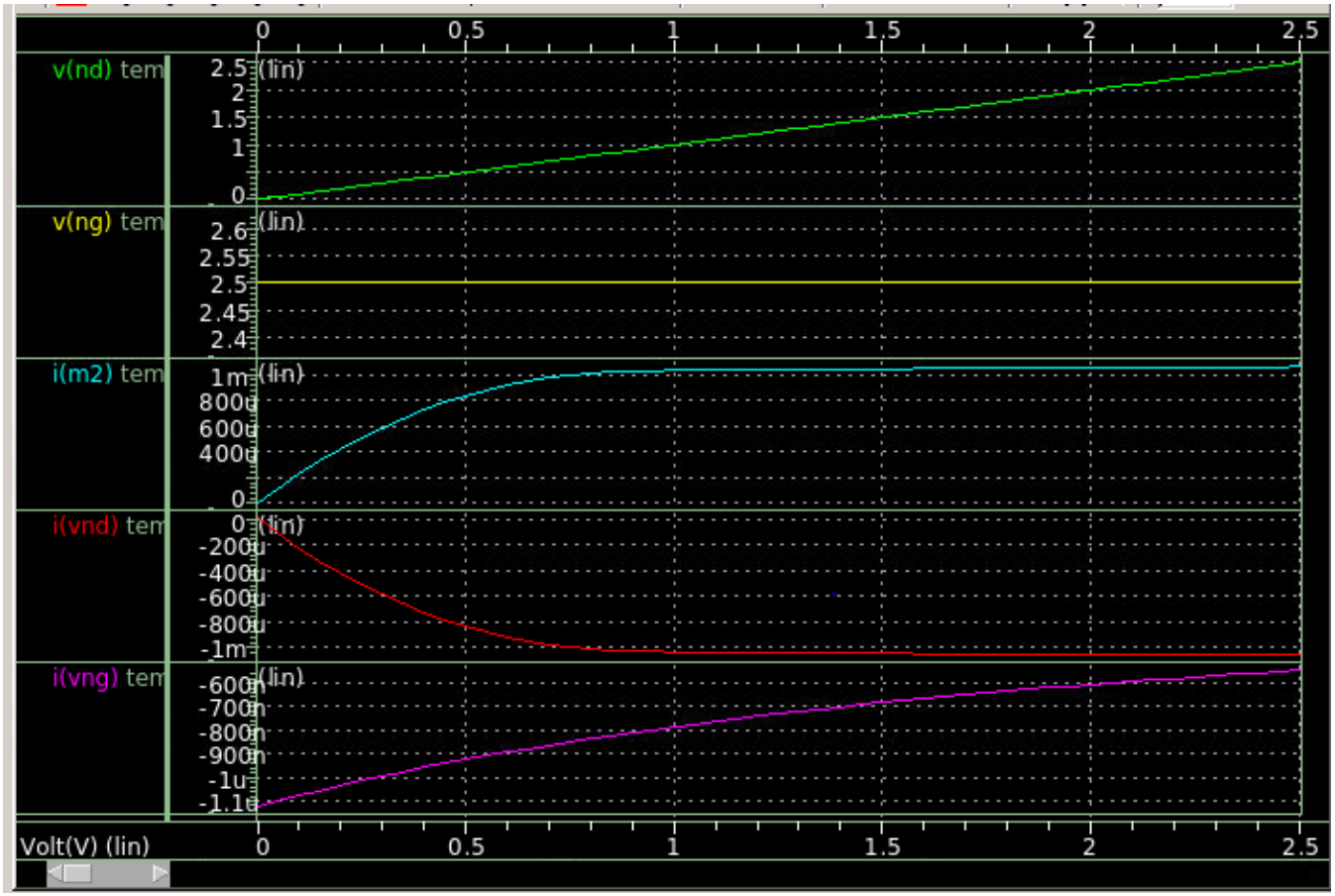

v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5
2.6
2.55
2.45
2.4
250u
200u
150u
100u
-100u
-150u
-200u
-250u
-800n
-1u
-1.2u
-1.4u

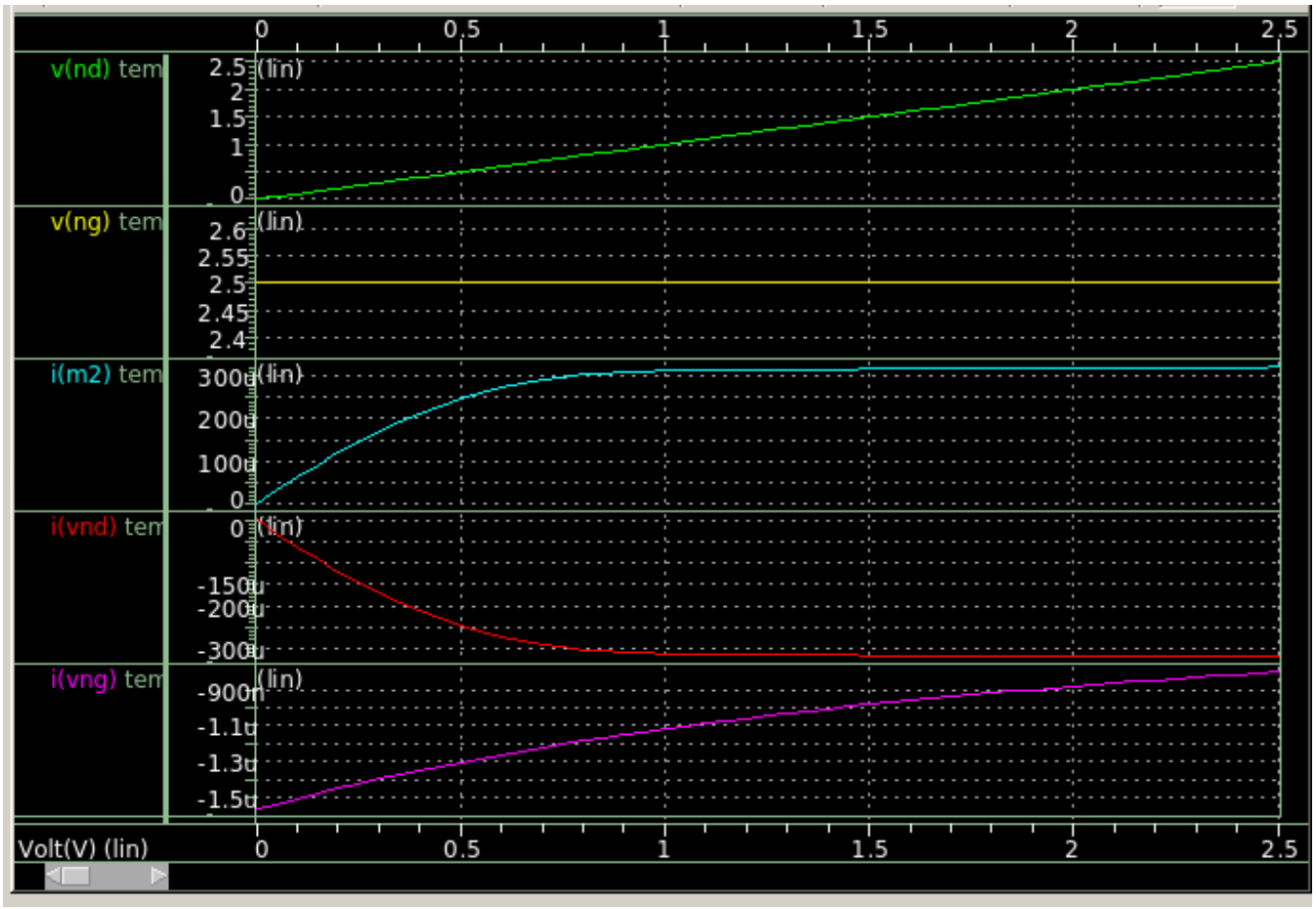

v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) tem
(lin)
v(ng) tem
(lin)
i(m2) tem
(lin)
i(vnd) tem
(lin)
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) tem
(lin)
v(ng) tem
(lin)
i(m2) tem
(lin)
i(vnd) tem
(lin)
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5


v(nd) tem
v(ng) tem
i(m2) tem
i(vnd) tem
i(vng) tem
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5

[b23n_sarif_727@prac3 experiment6]$ vi voltage_corner.sp

[b23n_sarif_727@prac3 experiment6]$ wv voltage_corner_wdf.sw0 &

[b23n_sarif_727@prac3 experiment6]$ wv voltage_corner_wdf.sw1 &

[b23n_sarif_727@prac3 experiment6]$ wv voltage_corner_wdf.sw2 &

```
* analyse the process corner analysis- SS FF TT

.option search=/data/pdk/pdk_cc/hspice

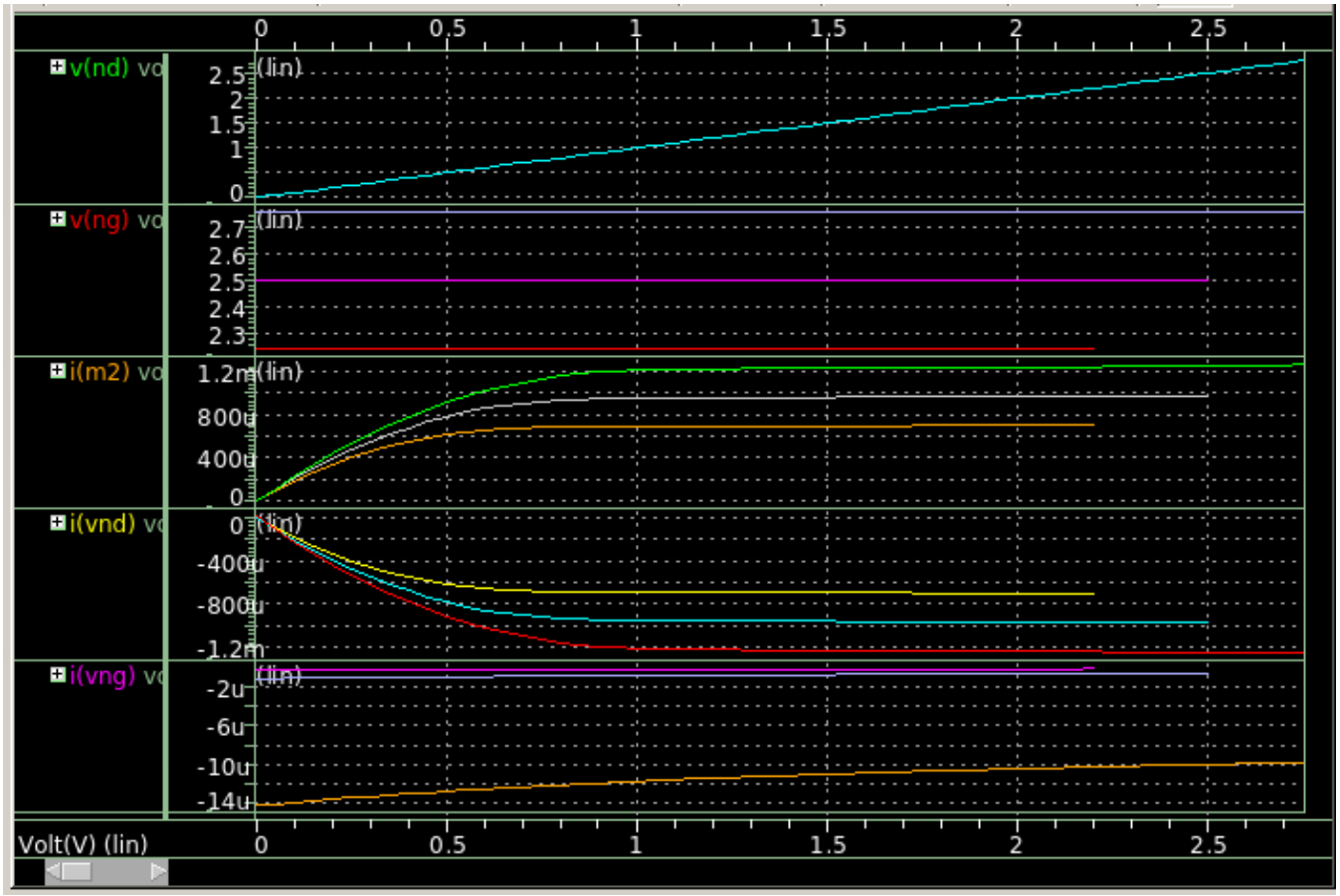
.lib 'reference40_models.lib' TT
.param VDD=2.5V

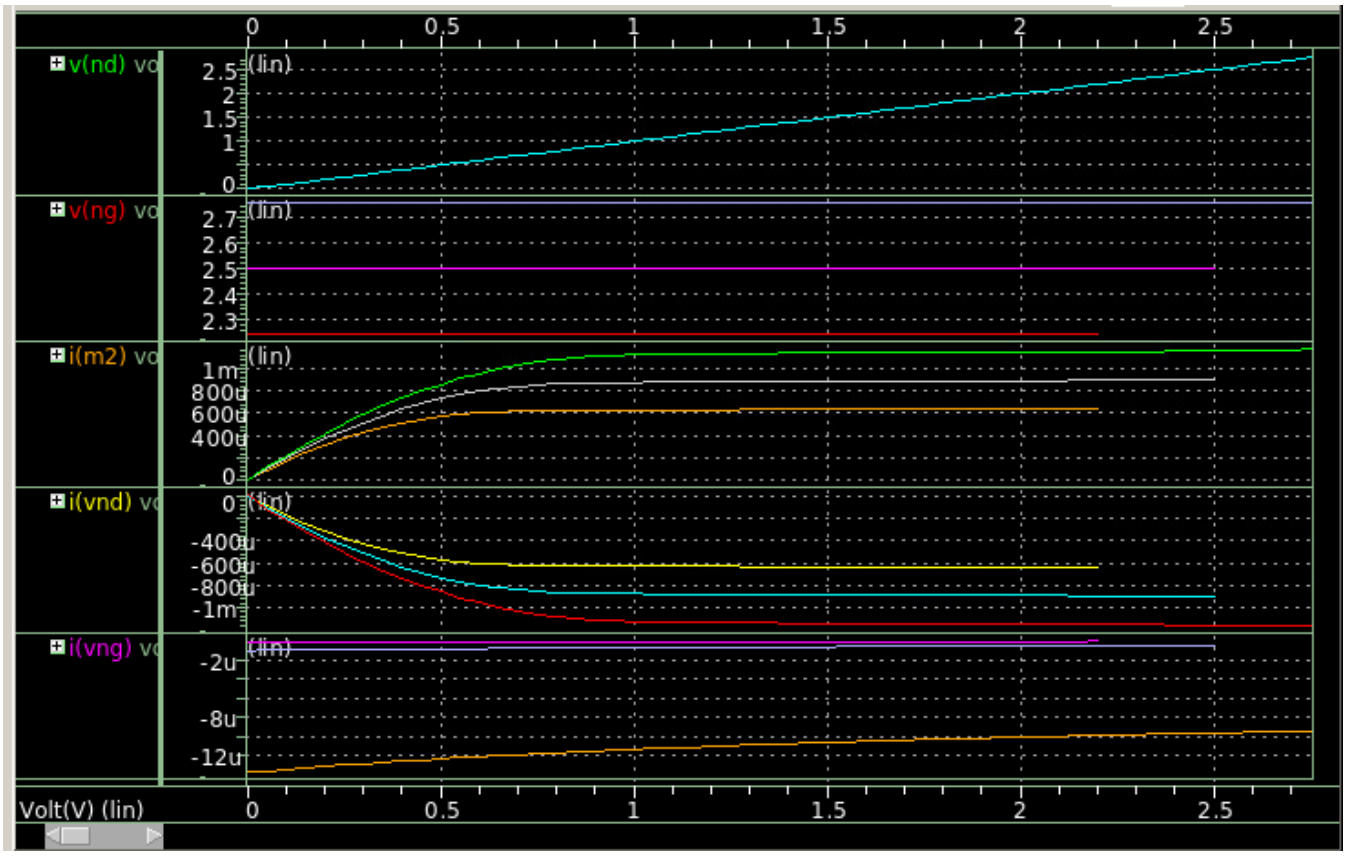
.option WDF=1
*.model  N25  nmos  level = 54
M2 ND NG 0 0 N25 W=10u L=1u
VND ND 0 0
VNG NG 0 VDD

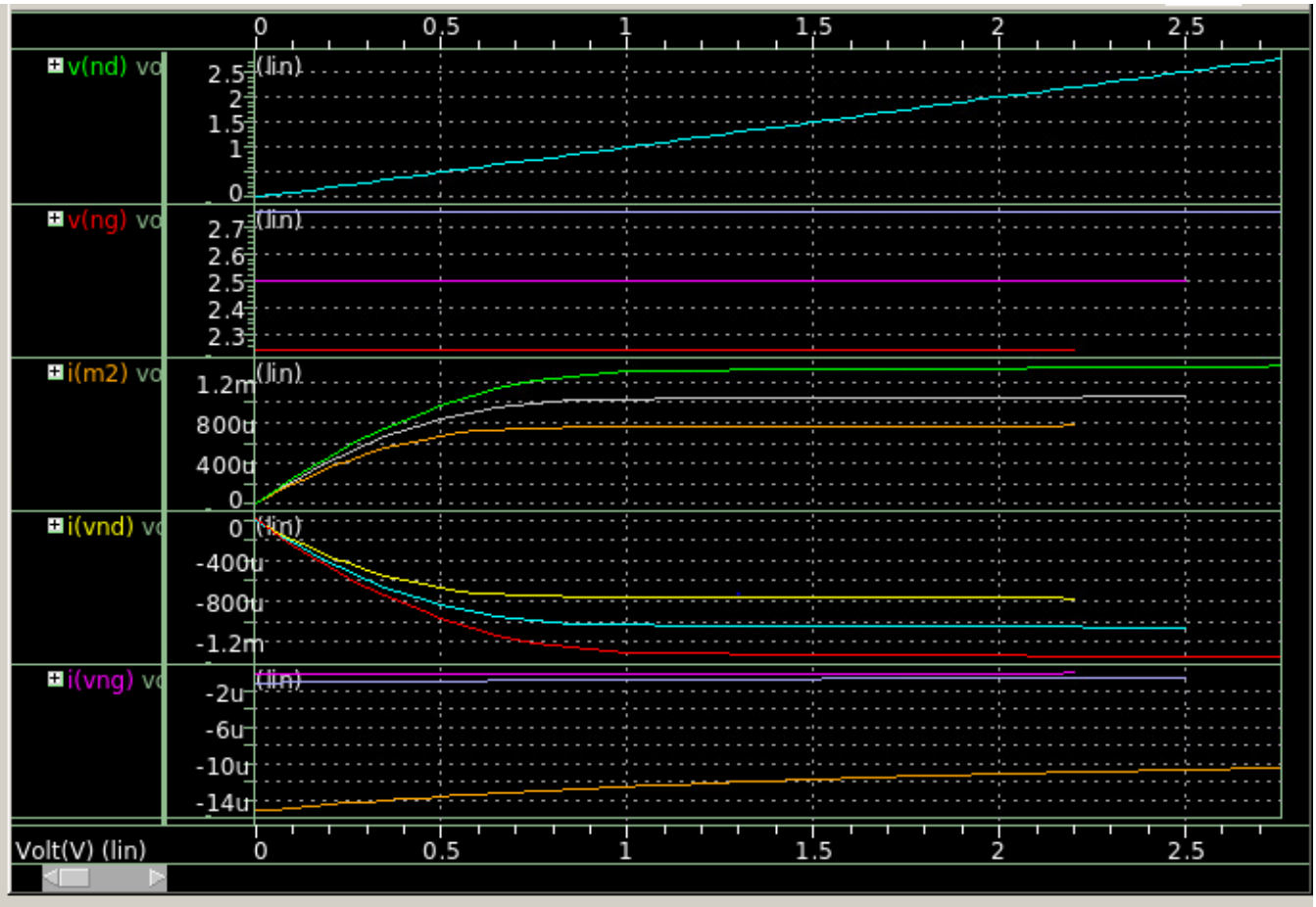
.DC VND 0 VDD 0.05 sweep DATA=supply_values
.DATA supply_values
VDD
2.24
2.5
2.75

.probe DC I(M2)
.print DC I(M2)
.ENDDATA

.alter
.lib 'reference40_models.lib' SS
.alter
.lib 'reference40_models.lib' FF
.end
```


v(nd) vo
v(ng) vo
i(m2) vo
i(vnd) vo
i(vng) vo
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5
2.5
2
1.5
1
0
2.7
2.6
2.5
2.4
2.3
1.2m
800u
400u
0
-400u
-800u
-1.2m
-2u
-6u
-10u
-14u


v(nd) vo
v(ng) vo
i(m2) vo
i(vnd) vo
i(vng) vo
(lin)
Volt(V) (lin)
0
0.5
1
1.5
2
2.5
2.5
2
1.5
1
0
2.7
2.6
2.5
2.4
2.3
1m
800u
600u
400u
0
-400u
-600u
-800u
-1m
-2u
-8u
-12u



Q2) Below is a netlist you need to give the value I(M1) for different PVT Corner

a) Take process corner as **SS, TT, FF**

b) Voltage corner as **2.24, 2.5, 2.75**

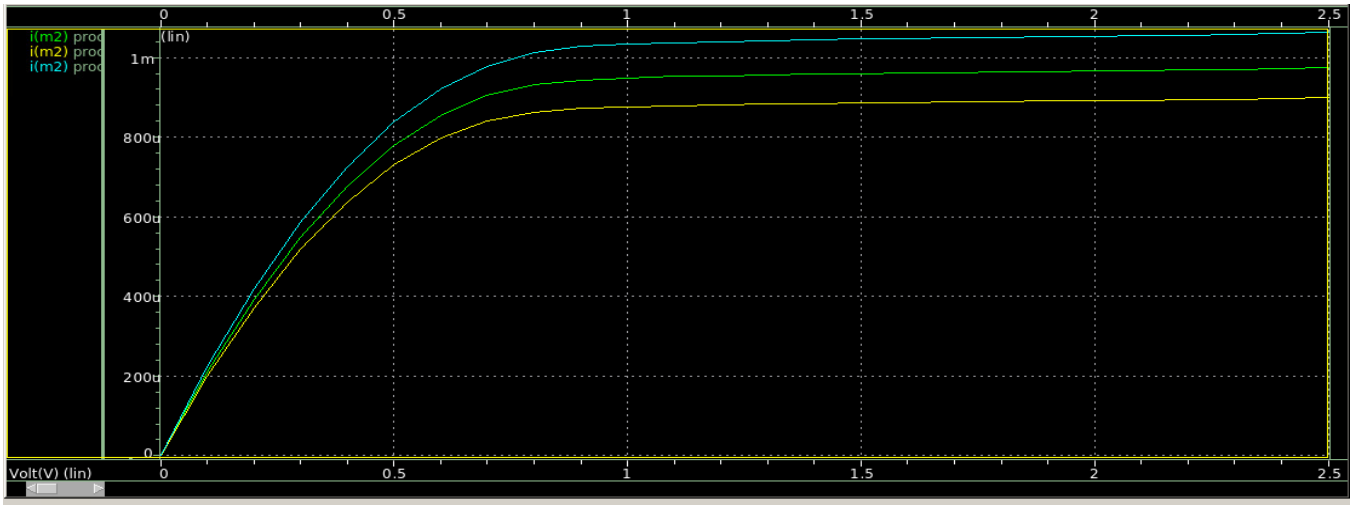
c) Temperature corner as **-25, 25 ,125**

Note: Make sure the voltage on VND and VNG are same

State the Best- and Worst-case PVT Corner

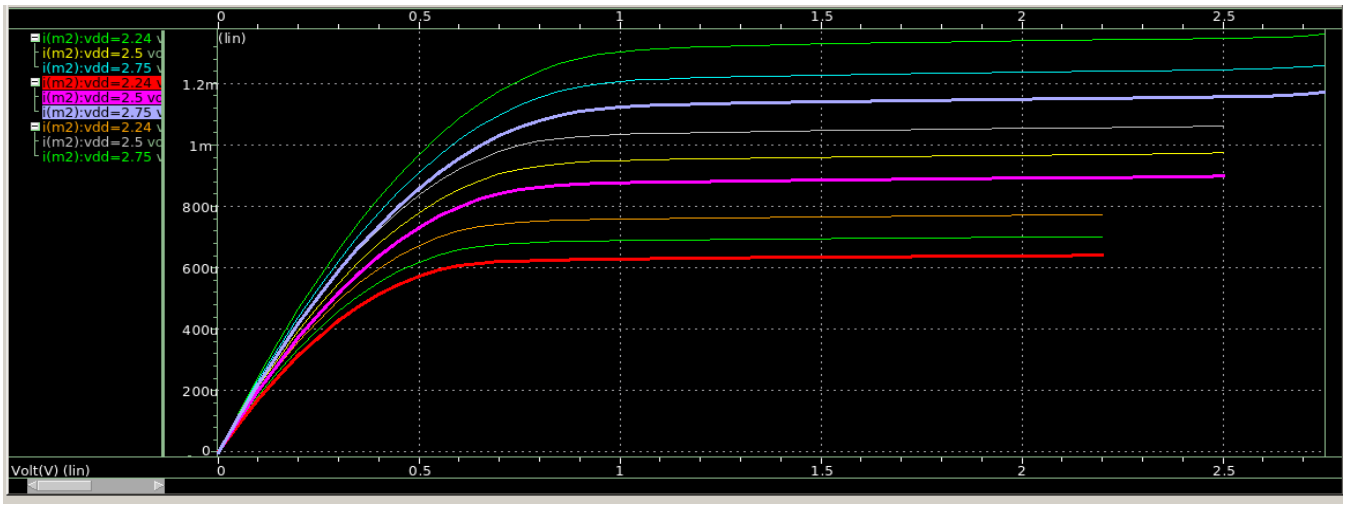
```
.option search='Add the path where you have hspice folder for
reference40nm'
.lib 'reference40_models.lib'
M2 ND NG 0 0 N25 L=1U W=10u
VND ND  0 2.5
VNG NG 0   2.5
.DC VND 0 2.5 2.5
.PRINT DC I(M1)
. PROBE DC I(M1)
. END
```

Solution: - (a) Process corner analysis at constant temp=25 and V=2.5V

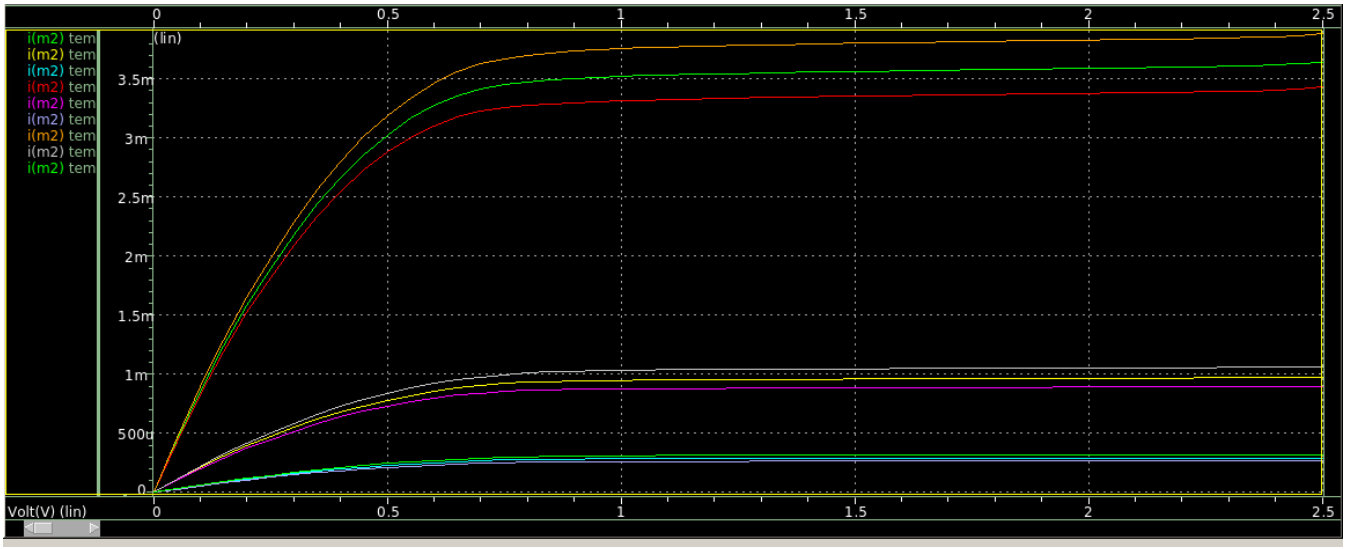


(This is the curve for process corner FF, TT, SS for NMOS (N25) Devices)

b) Voltage corner analysis at different 2.24, 2.5, 2.75, SS, TT, FF and at constant temp=25



c) Temperature corner analysis at -25, 25 ,125, SS, TT, FF and constant voltage=2.5V



d) PVT corner analysis of NMOS (N25)- by using the HSPICE tool

Design code-

```
* PVT corner analysis of NMOS N25
.option search=/data/pdk/pdk_cc/hspice

.lib 'reference40_models.lib' TT
.temp -25 25 125
.param VDD=2.5V

.option WDF=1
*.model  N25  nmos  level = 54
M2 ND NG 0 0 N25 W=10u L=1u
VND ND 0 0
VNG NG 0 VDD

.DC VND 0 VDD 0.05 sweep DATA=supply_values
.DATA supply_values
VDD
2.24
```

2.5

2.75

.probe DC I(M2)

.print DC I(M2)

.ENDDATA

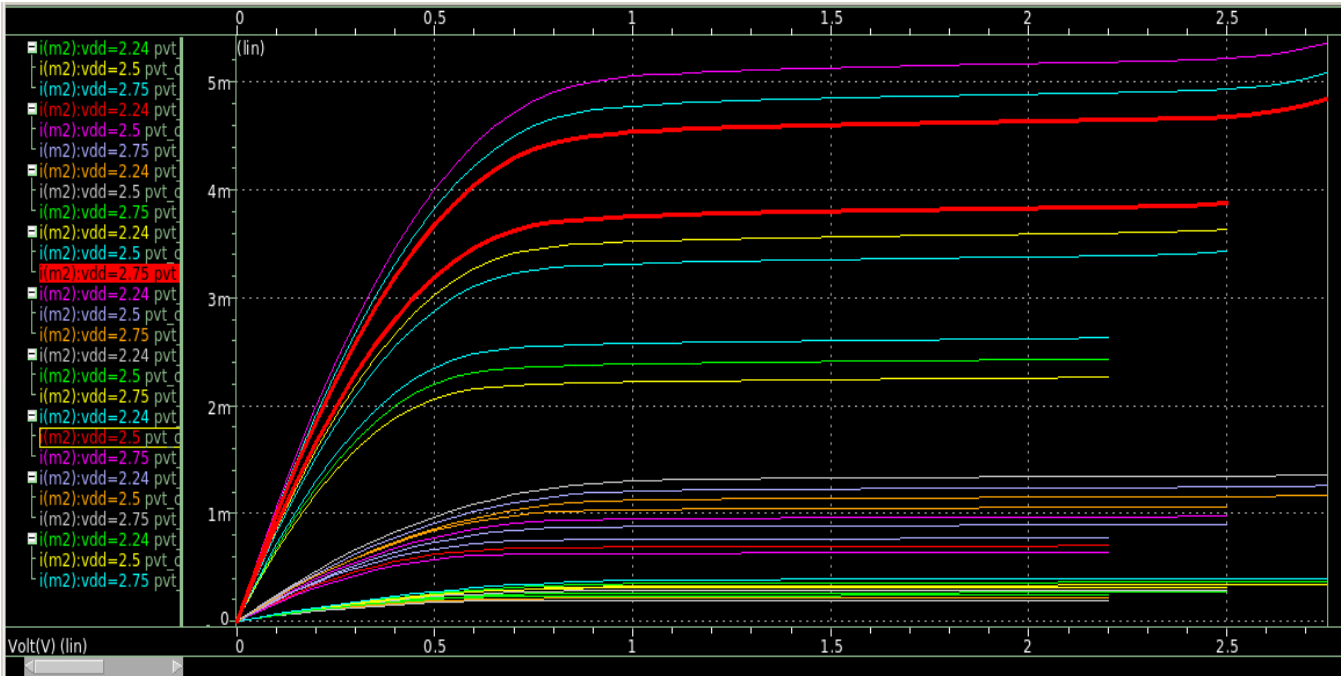
.alter

.lib 'reference40_models.lib' SS

.alter

.lib 'reference40_models.lib' FF

.end



- **Best-case (max current)- FF process, V=2.75V, T= -25 C**
- **Worst-case (min current)- SS Process, V=2.24V, T=125C**

Process corner (SS / TT / FF)-

- **FF(Fast)-** Higher mobility / lower threshold – higher current
- **SS(Slow)-** Lower mobility / higher threshold- lower current

Voltage corner – (2.24V, 2.5V, 2.75V)-

- In saturation region, current is directly proportional to $(V-V_{th})^2$ means higher voltage V produce higher current I. **voltage=2.75V gives higher current.**

Temperature corner – (-25, 25, 125)-

- **Mobility degrades with temperature** and reducing the current. So **lower temp gives larger current** and higher temp gives lower current.